

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\125-DopeRev 02.pvdb

Date of Analysis : Aug 13, 2026 4:37pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 609.601 mm ID	
Vessel Design Length, Tangent to Tangent	18134.00 mm
Distance of Bottom Tangent above Grade	15000.00 mm
Distance of Base above Grade	11000.00 mm
Specified Datum Line Distance	4000.00 mm
Internal Design Temperature	270 °C
Internal Design Pressure	2.215 MPa
External Design Temperature	200 °C
External Design Pressure	0.103 MPa
Maximum Allowable Working Pressure	4 MPa
External Max. Allowable Working Pressure	1 MPa
Shop Test Pressure	6.007 MPa
Field Test Pressure	2.896 MPa
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	+IT
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Base Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-2	...	...	...	3.0000	N/A	No
Skirt -1	...	...	...	3.0000	N/A	No
Bottom head	2.231	0.10	7.000	3.0000	No	No
Shell-1	2.229	0.10	6.000	3.0000	N/A	No
Body flange -1	2.215	0.10	6.000	3.0000	N/A	No
Body flange -2	2.215	0.10	6.000	3.0000	N/A	No
Shell-2	2.215	0.10	6.000	3.0000	N/A	No
Shell-3	2.215	0.10	6.000	3.0000	N/A	No
Body flange-3	2.215	0.10	6.000	3.0000	N/A	No
Body flange-4	2.215	0.10	4.000	3.0000	N/A	No
Shell-4	2.215	0.10	4.000	3.0000	N/A	No
Shell-5	2.215	0.10	4.000	3.0000	N/A	No
Body flange-5	2.215	0.10	4.000	3.0000	N/A	No
Body flange-6	2.215	0.10	4.000	3.0000	N/A	No
Shell-6	2.215	0.10	4.000	3.0000	N/A	No

Shell-7	2.215	0.10	4.000	3.0000	N/A	No
Body flange-7	2.215	0.10	4.000	3.0000	N/A	No
Body flange-8	2.215	0.10	4.000	3.0000	N/A	No
Shell-8	2.215	0.10	4.000	3.0000	N/A	No
Body flange-9	2.215	0.10	4.000	3.0000	N/A	No
Body flange-10	2.215	0.10	4.000	3.0000	N/A	No

Liquid Level: 1952.40 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.840

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-1000.0	3000.0	20.0	20.0	...	...	0.70	0.70
Skirt	0.0	1000.0	20.0	20.0	...	...	0.70	0.50
Ellipse	50.0	50.0	22.0	18.0	7.9	4.7	1.00	1.00
Cylinder	1850.0	1800.0	16.0	16.0	8.1	6.3	1.00	1.00
Body Flg	2040.8	186.3	90.0	88.0	81.1	81.1	1.00	1.00
Body Flg	2227.1	186.3	90.0	88.0	81.1	81.1	1.00	1.00
Cylinder	5227.1	3000.0	16.0	16.0	8.0	7.7	1.00	1.00
Cylinder	6607.1	1380.0	16.0	16.0	8.0	7.7	1.00	1.00
Body Flg	6797.9	186.3	98.0	88.0	81.1	81.1	1.00	1.00
Body Flg	7002.2	204.3	108.0	106.0	94.8	94.8	1.00	1.00
Cylinder	10002.2	3000.0	12.0	12.0	8.0	7.6	1.00	1.00
Cylinder	11302.2	1300.0	12.0	12.0	8.0	7.6	1.00	1.00
Body Flg	11521.0	214.3	118.0	116.0	96.9	96.9	1.00	1.00
Body Flg	11735.3	214.3	118.0	116.0	96.9	96.9	1.00	1.00
Cylinder	14735.3	3000.0	12.0	12.0	8.0	7.8	1.00	1.00
Cylinder	16435.3	1700.0	12.0	12.0	8.0	7.8	1.00	1.00
Body Flg	16654.1	214.3	118.0	116.0	96.9	96.9	1.00	1.00
Body Flg	16866.4	212.3	116.0	114.0	96.0	96.0	1.00	1.00
Cylinder	17834.5	968.1	12.0	12.0	8.0	5.5	1.00	1.00
Body Flg	18050.0	212.3	116.0	114.0	96.0	96.0	1.00	1.00
Body Flg	18134.0	84.0	90.0	88.0	78.1	78.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3_80	WRC-107/537	0.299	n/a	Passed
N6_100	WRC-297	0.506	Shell	Passed
N5_100	WRC-297	0.506	Shell	Passed
N1_80	WRC-297	0.705	Shell	Passed
N4_80	WRC-297	0.705	Shell	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	20475.5	11063.3	231635040	181349360
20	30	3500	18127.5	11010.7	173707008	148224896
30	40	4025	17248.4	10972.5	156011872	137228816
40	50	4950	17199.8	10903.3	155150336	136681712
50	60	5947.65	15459.7	10706.7	125675240	117176656
60	70	6133.95	15297.5	10620.1	122809072	115189288
70	80	7727.1	15135.3	10529.6	119973144	113218408
80	90	9917.1	12235.2	10026.1	78900712	82372376
90	100	10704.7	10901.2	9745.46	62881060	68680568
100	110	10900	10739	9523.84	60864496	66884932
110	120	12502.2	10562.2	9261.81	58687712	64965216
120	130	14652.2	7691.08	8414.31	31296648	38440276
130	140	15413.8	6446.91	6802.27	22074216	28514874
140	150	15628.1	6263.12	6374.97	20711792	27102370
150	160	17235.3	6079.34	5897.56	19388768	25786846

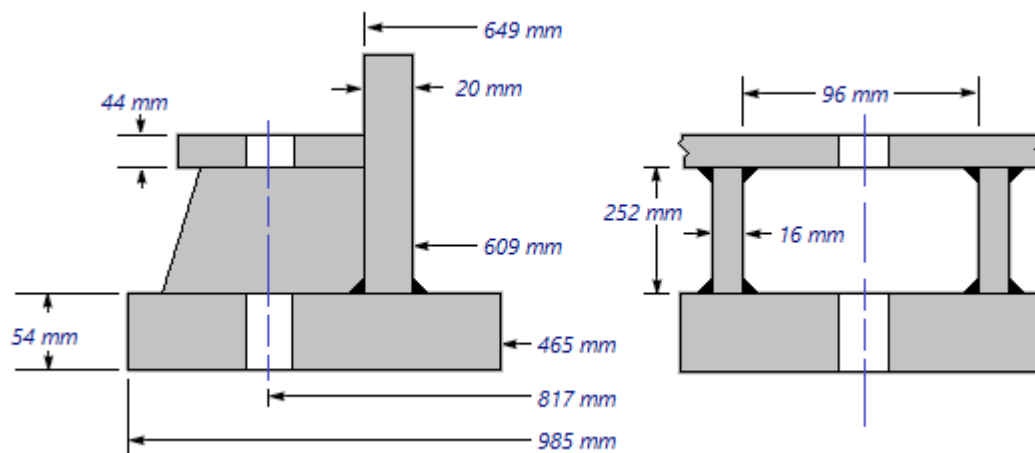
160	170	19585.3	3208.18	4525.64	5451829	10145697
170	180	20546.9	1581.2	3457.73	1372087	3341516
180	190	20760.2	1397.41	2804	1052802	2670303
190	200	21350.5	1215.04	2149.61	775380	2144267
200	210	21943.8	288.479	1629.13	46360.3	309176
210	220	22092	106.107	918.802	4458.28	38605.1

Abs Max of the all of the Stress Ratio's : 0.8081

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	54.0000	mm
Inside Diameter of Basing	465.0000	mm
Outside Diameter of Basing	985.0001	mm
Nominal Diameter of Bolts	48.0000	mm
Diameter of Bolt Circle	817.0000	mm
Number of Bolts	12	
Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	158.0000	mm
Height of Gussets	252.0000	mm
Distance between Gussets	96.0000	mm
Thickness of Top Plate or Ring	44.0000	mm
Radial Width of the Top Plate	168.0000	mm
Number of Gussets	24	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	40508.	N
Total Earthquake Shear on Support	31095.	N
Wind Moment on Support	453392416.	N-mm
Earthquake Moment on Support	403106752.	N-mm
Maximum Initial Bolt Load	172907.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	11864.8	kgm
Shop Test - Fabricated + water (full)	17121.1	kgm
Shipping - Fab. + rem. intls.+ shipping app.	13844.5	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	13844.5	kgm
Empty - Fab. + intls. + details + wghts.	13844.5	kgm



Operating - Empty + operating liquid (no ca)	14680.6	kgm
Field Test - Empty weight + test medium (full)	18374.3	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3_80	7	OK	...	OK	7.95	OK	Passed
N6_100		...	...	OK	8.02	OK	No Calc[*]
K1_50		...	...	OK	7.80	OK	No Calc[*]
K2_50		...	...	OK	7.80	OK	No Calc[*]
K6_50		...	...	OK	7.80	OK	No Calc[*]
N5_100		...	...	OK	8.02	OK	No Calc[*]
H3_150		...	...	...		OK	No Calc[*]
K3_50		...	...	OK	7.80	OK	No Calc[*]
H2_150		...	...	...		OK	No Calc[*]
N1_80	4	OK	...	OK	8.02	OK	Passed
H1_150	4	OK	...	...		OK	Passed
N4_80	4	OK	...	OK	8.02	OK	Passed
K5_50	4	OK	...	OK	7.80	OK	Passed
N2_100	4	OK	...	OK	8.26	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4 Nozzle : K550
 Minimum MAWP Shells/Flanges : 4 Element : Shell-8
 Minimum MAPnc Shells/Flanges : 5 Element : Body flange-10

Computed Vessel M.A.W.P. : 4 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N3_80	0.000	0.000	172.000
40	N6_100	1650.000	0.000	172.000
40	K1_50	1650.000	180.000	113.600
40	K2_50	200.000	180.000	113.600
40	K6_50	200.000	0.000	113.600
70	N5_100	2352.099	0.000	172.000
70	H3_150	3877.099	0.000	288.000
110	K3_50	7202.200	0.000	113.600
110	H2_150	8652.199	0.000	288.000
150	N1_80	12450.302	180.000	172.000
150	H1_150	13985.301	0.000	288.000
190	N4_80	17016.406	0.000	172.000
190	K5_50	17016.406	180.000	113.600
210	N2_100	0.000	0.000	284.300

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!



Checking Multiple Nozzles on Flat Head per ASME Sec. VIII Div. 1 UG-39

Comparing Nozzles on Element: Body flange-10

Note: No Nozzle pairs found on this element.

UG-39 Nozzle Diameter and Distance to Edge Checks:

Nozzle Description	Nozzle dia. mm	Head Dia. /2 mm	Distance from Edge mm	Nozzle dia./4 mm
N2_100	98.05	333.38	361.18	24.512

No Multiple Nozzle spacing violations detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1_50	50 mm	Actual	LWN	84.4	16.800	...	...	218	300
K2_50	50 mm	Actual	LWN	84.4	16.800	...	...	218	300
K6_50	50 mm	Actual	LWN	84.4	16.800	...	...	218	300
K3_50	50 mm	Actual	LWN	84.4	16.800	...	...	214	300
K5_50	50 mm	Actual	LWN	84.4	16.800	...	...	214	300
N3_80	80 mm	Actual	WNF	100.0	10.000	...	...	222	600
N1_80	80 mm	Actual	WNF	100.0	10.000	...	...	220	600
N4_80	80 mm	Actual	WNF	100.0	10.000	...	...	220	600
N6_100	100 mm	Actual	WNF	100.0	10.000	...	...	320	600
N5_100	100 mm	Actual	WNF	100.0	10.000	...	...	320	600
N2_100	100 mm	120	WNF	114.3	11.125	...	...	388	300
H3_150	150 mm	Actual	WNF	150.0	6.000	...	...	225	300
H2_150	150 mm	Actual	WNF	150.0	6.000	...	...	221	300
H1_150	150 mm	Actual	WNF	150.0	6.000	...	...	221	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grv Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grv Weld	Inside Weld
K1_50	SA-105	10.000	10.000	...	...	...
K2_50	SA-105	10.000	10.000	...	...	...
K6_50	SA-105	10.000	10.000	...	...	...
K3_50	SA-105	10.000	10.000	...	...	...
K5_50	SA-105	10.000	10.000	...	...	...
N3_80	SA-105	10.000	10.000	...	...	...
N1_80	SA-105	10.000	10.000	...	...	...
N4_80	SA-105	10.000	10.000	...	...	...
N6_100	SA-105	10.000	10.000	...	...	...
N5_100	SA-105	10.000	10.000	...	...	...
N2_100	SA-105	66.000	10.000	...	...	...
H3_150	SA-105	10.000	10.000	...	...	...
H2_150	SA-105	10.000	10.000	...	...	...
H1_150	SA-105	10.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K1_50	1650.000	180.0	200.00	0.00	Shell-1
K2_50	200.000	180.0	200.00	0.00	Shell-1
K6_50	200.000	0.0	200.00	0.00	Shell-1
K3_50	7202.200	0.0	200.00	0.00	Shell-4
K5_50	17016.404	180.0	200.00	0.00	Shell-8
N3_80	...	0.0	200.00	0.00	Bottom head
N1_80	12450.301	180.0	200.02	0.00	Shell-6
N4_80	17016.404	0.0	200.00	0.00	Shell-8
N6_100	1650.000	0.0	300.00	0.00	Shell-1

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N5_100	2352.099	0.0	300.00	0.00	Shell-2
N2_100	...	0.0	300.00	0.00	Body flange-10
H3_150	3877.099	0.0	200.00	0.00	Shell-2
H2_150	8652.199	0.0	200.00	0.00	Shell-4
H1_150	13985.301	0.0	200.00	0.00	Shell-6

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SKIRT: 20.0mm THK X 609.0mm ID X 3000.0mm	SA-516 70
1	SKIRT: 20.0mm THK X 609.0mm ID X 1000.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 22.0mm THK X 609.6mm ID X 50.0mm	SA-516 70
2	WELD NECK FLANGE: 90.0mm THK X 914.4mm OD	SA-266 2
1	CYLINDER: 16.0mm THK X 609.6mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 16.0mm THK X 609.6mm ID X 1380.0mm	SA-516 70
1	WELD NECK FLANGE: 98.0mm THK X 914.4mm OD	SA-266 2
1	WELD NECK FLANGE: 108.0mm THK X 914.4mm OD	SA-266 2
2	CYLINDER: 12.0mm THK X 609.6mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 12.0mm THK X 609.6mm ID X 1300.0mm	SA-516 70
3	WELD NECK FLANGE: 118.0mm THK X 914.4mm OD	SA-266 2
1	CYLINDER: 12.0mm THK X 609.6mm ID X 1700.0mm	SA-516 70
2	WELD NECK FLANGE: 116.0mm THK X 914.4mm OD	SA-266 2
1	CYLINDER: 12.0mm THK X 609.6mm ID X 968.1mm	SA-516 70
1	BLIND FLANGE: 90.0mm THK X 914.4mm OD	SA-266 2
1	INSULATION: 500mm X 80mm THK	
1	INSULATION: 202mm X 80mm THK	
1	INSULATION: 1800mm X 80mm THK	
9	INSULATION: 168mm X 80mm THK	
1	PACKING: 2350.000mm	...
3	INSULATION: 3000mm X 80mm THK	
1	PACKING: 1350.000mm	...
1	INSULATION: 1380mm X 80mm THK	
1	PACKING: 2480.000mm	...
1	PACKING: 1220.000mm	...
1	INSULATION: 1300mm X 80mm THK	
1	PACKING: 2100.000mm	...
1	PACKING: 1600.000mm	...
1	INSULATION: 1700mm X 80mm THK	
1	INSULATION: 968mm X 80mm THK	
1	INSULATION: 69mm X 80mm THK	
3	CLASS 300 GR 1.1, 150.0mm BLIND FLANGE(S)	SA-105
1	BASERING: 985.0mm OD X 465.0 ID X 54.0mm	SA-516 70



12	BASERING BOLTS: 48.0mm DIA	SA-193 B7
12	BASERING BOLT NUTS: 48.0mm DIA	...
1	BASERING TOP RING: 985.0mm OD X 649.0 ID X 44.0mm	...
24	BASERING GUSSET PLATES: 158.0mm X 158.0mm X 252.0mm X 16.0mm	...
4	GASKET: 680.0mm OD X 632.0mm ID	...
120	BODY FLANGE BOLTS: 38.1mm DIA	SA-193 B7
240	NUTS FOR BODY FLANGE BOLTS: 38.1mm DIA	...
1	GASKET: 680.0mm OD X 640.0mm ID	...
2	CYLINDRICAL SEGMENT 16.0mm THK X 760.7mm X 1800.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 393.7mm X 1800.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Body flange -[11]	B	-14	-41	-29	0.545	16.000	1.00	Yes
Body flange -[11]	B	-14	-43	-29	0.533	16.000	1.00	Yes
Body flange-3[11]	B	-14	-43	-29	0.533	16.000	1.00	Yes
Body flange-4[11]	B	-23	-39	-29	0.714	12.000	1.00	Yes
Body flange-5[11]	B	-23	-39	-29	0.714	12.000	1.00	Yes
Body flange-6[11]	B	-23	-39	-29	0.714	12.000	1.00	Yes
Body flange-7[11]	B	-40	-48	-29	0.714	12.000	1.00	Yes
Body flange-8[11]	B	-23	-39	-29	0.714	12.000	1.00	Yes
Body flange-9[11]	B	-23	-39	-29	0.714	12.000	1.00	Yes
Body flange-1[11]	B	-5	-13	-29	0.852	22.000	1.00	Yes
Bottom head [10]	B	-11	-34	-29	0.583	18.000	1.00	Yes
Bottom head [7]	B	-5	-41	-29	0.473	22.000	1.00	Yes
Shell-1 [8]	B	-14	-31	-29	0.691	16.000	1.00	Yes
Shell-2 [8]	B	-14	-31	-29	0.688	16.000	1.00	Yes
Shell-3 [8]	B	-14	-31	-29	0.688	16.000	1.00	Yes
Shell-4 [8]	B	-23	-23	-29	0.994	12.000	1.00	Yes
Shell-5 [8]	B	-23	-23	-29	0.994	12.000	1.00	Yes
Shell-6 [8]	B	-23	-23	-29	0.994	12.000	1.00	Yes
Shell-7 [8]	B	-23	-23	-29	0.994	12.000	1.00	Yes
Shell-8 [8]	B	-23	-23	-29	0.994	12.000	1.00	Yes
N380 [1]	B	-8	-104	-29	0.100	10.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104	-29	0.384	16.000	1.00	Yes
N6100 [1]	B	-14	-48	-29	0.384	16.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104	-29	0.384	16.000	1.00	Yes
K150 [1]	B	-14	-48	-29	0.384	16.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48	-29	0.386	16.000	1.00	Yes
K250 [1]	B	-14	-48	-29	0.386	16.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48	-29	0.386	16.000	1.00	Yes
K650 [1]	B	-14	-48	-29	0.386	16.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48	-29	0.384	16.000	1.00	Yes
N5100 [1]	B	-14	-48	-29	0.384	16.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104	-29	0.399	6.000	1.00	Yes
H3150 [1]	B	-8	-48	-29	0.399	6.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48	-29	0.554	12.000	1.00	Yes
K350 [1]	B	-23	-48	-29	0.554	12.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48	-29	0.554	12.000	1.00	Yes
H2150 [1]	B	-8	-48	-29	0.399	6.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48	-29	0.554	12.000	1.00	Yes
N180 [1]	B	-23	-48	-29	0.554	12.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104	-29	0.554	12.000	1.00	Yes
H1150 [1]	B	-8	-48	-29	0.399	6.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48	-29	0.554	12.000	1.00	Yes
N480 [1]	B	-23	-48	-29	0.554	12.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104	-29	0.554	12.000	1.00	Yes
K550 [1]	B	-23	-48	-29	0.554	12.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48	-29	0.118	9.735	1.00	Yes
N2100 [1]	B	-8	-104	-29	0.118	9.735	1.00	Yes
Nozzle Flg [5]	A	-18	-48	-29	1.000	64.000	1.00	Yes
Nozl. Bolting[22]		-48						
Bolting [21]		-48						
Bolting [21]		-48						
Bolting [21]		-48						
Bolting [21]		-48						
Bolting [21]		-48						

Warmest MDMT: -5 -13

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.



- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note	20	40	There is a high jump in the Joint Eff.
Note	10	20	The wind diameter multiplier is very big.
Note	20	30	The wind diameter multiplier is very big.
Note	30	40	The wind diameter multiplier is very big.
Note	40	50	The wind diameter multiplier is very big.
Note	50	60	The wind diameter multiplier is very big.
Note	60	70	The wind diameter multiplier is very big.
Note	70	80	The wind diameter multiplier is very big.
Note	80	90	The wind diameter multiplier is very big.
Note	90	100	The wind diameter multiplier is very big.
Note	100	110	The wind diameter multiplier is very big.
Note	110	120	The wind diameter multiplier is very big.
Note	120	130	The wind diameter multiplier is very big.
Note	130	140	The wind diameter multiplier is very big.
Note	140	150	The wind diameter multiplier is very big.
Note	150	160	The wind diameter multiplier is very big.
Note	160	170	The wind diameter multiplier is very big.
Note	170	180	The wind diameter multiplier is very big.
Note	180	190	The wind diameter multiplier is very big.
Note	190	200	The wind diameter multiplier is very big.
Note	200	210	The wind diameter multiplier is very big.
Note	210	220	The wind diameter multiplier is very big.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be



*computed under test conditions. Please see interpretation VIII-1-10-12 for more information.
PV Elite also computes weight due to test fluid and uses the weight in various calculations such as
saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The
static pressure is used in conjunction with the design and test pressures to determine a total
pressure in the determination of required thickness, MAWP etc. This ensures compliance with
UG-22 requirements.*

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	2.2148	MPa
Design Internal Temperature	270.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	Lethal	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16		IP+VE+EW	
Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	97.2	km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		11000	mm
Percent Wind for Hydrotest		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	mm
Distance Upwind of Crest	Lh	0	mm
Distance from Crest to the Vessel	x	0	mm
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	mm
Seismic Design Code		ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor		1.000	
Importance Factor		1.250	
Table Value Fa		1.600	
Table Value Fv		2.400	
Max. Mapped Res. Acceleration	[Ss]	0.147	
Max. Eff. Ground Acceleration	[S]	0.065	
Force Modification Factor R		2.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	2.500	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Moment Reduction Factor Tau		1.000	
Design Pressure + Static Head		Y	
Consider MAP New and Cold in Noz. Design		N	
Consider External Loads for Nozzle Des.		Y	
Use ASME VIII-1 Appendix 1-9		N	
Perform Blast Load Analysis		No	
Material Database Year	Current w/Addenda or Code Year		

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/21
Element From Node	10



Element To Node	20
Element Type	Skirt Support
Description	Skirt-2
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	649 mm
Diameter of Skirt at Base	649 mm
Skirt Thickness	20 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	270 °C
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	137.2 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	211.2 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No
Element From Node	10
Detail Type	Weight
Detail ID	Tailing lug
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	1961.2 N
Offset from Element Centerline	0 mm

Element#	2/21
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt -1
Distance "FROM" to "TO"	1000 mm
Skirt Outside Diameter	649 mm
Diameter of Skirt at Base	649 mm
Skirt Thickness	20 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	270 °C
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No
Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	500 mm
Height/Length of Insulation	500 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element#	3/21
Element From Node	30
Element To Node	40



Element Type	Elliptical
Description	Bottom head
Distance "FROM" to "TO"	50 mm
Inside Diameter	609.6 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	-152.4 mm
Height/Length of Liquid	202.4 mm
Liquid Density	0.0008387 kg/cm ³
Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	-152.4 mm
Height/Length of Insulation	202.4 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³
Element From Node	30
Detail Type	Nozzle
Detail ID	N380
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	80 mm
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	397.15 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	30
Detail Type	Weight
Detail ID	Tracing coil wt
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	2941.8 N
Offset from Element Centerline	0 mm
Element From Node	30
Detail Type	For./Mom.
Detail ID	Guide sup later
Dist. from "FROM" Node / Offset dist	-152 mm
Force in X Direction	-8667 N
Force in Y Direction	0 N
Force in Z Direction	-8667 N
Moment about X Axis	0 N-mm
Moment about Y Axis	0 N-mm
Moment about Z Axis	0 N-mm
Force/Moment Combination Method	SRSS



Element To Node	50	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	1800	mm
Inside Diameter	609.6	mm
Element Thickness	16	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	Yes	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1750	mm
Liquid Density	0.0008387	kg/cm³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1800	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N6100	
Dist. from "FROM" Node / Offset dist	1600	mm
Nozzle Diameter	100	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	294.43	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K150	
Dist. from "FROM" Node / Offset dist	1600	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K250	
Dist. from "FROM" Node / Offset dist	150	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N



Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	40
Detail Type	Nozzle
Detail ID	K650
Dist. from "FROM" Node / Offset dist	150 mm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element#	5/21
Element From Node	50
Element To Node	60
Element Type	Flange
Description	Body flange -1
Distance "FROM" to "TO"	186.3 mm
Flange Inside Diameter	577.6 mm
Element Thickness	88 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	90 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-266 2
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	133.2 MPa
Allowable Stress, Hydrotest	225 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	200 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	168.15 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element#	6/21
Element From Node	60
Element To Node	70
Element Type	Flange
Description	Body flange -2
Distance "FROM" to "TO"	186.3 mm
Flange Inside Diameter	577.6 mm
Element Thickness	88 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	90 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa



Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	168.15	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	7/21	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	609.6	mm
Element Thickness	16	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	137.2	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	211.2	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Packing	
Detail ID	Bed 3	
Dist. from "FROM" Node / Offset dist	650	mm
Height of Packed Section	2350	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.83999997	

Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	70	
Detail Type	Nozzle	
Detail ID	N5100	
Dist. from "FROM" Node / Offset dist	125	mm



Nozzle Diameter	100	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	294.43	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	70	
Detail Type	Nozzle	
Detail ID	H3150	
Dist. from "FROM" Node / Offset dist	1650	mm
Nozzle Diameter	150	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	547.33	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	70	
Detail Type	Weight	
Detail ID	H3 Davit	
Dist. from "FROM" Node / Offset dist	1650	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	300	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	N5Collector Wt	
Dist. from "FROM" Node / Offset dist	125	mm
Miscellaneous Weight	294.18	N
Offset from Element Centerline	0	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	N5Collector su	
Dist. from "FROM" Node / Offset dist	100	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	Bed3 Sup	
Dist. from "FROM" Node / Offset dist	550	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	Bed3 sup grid	
Dist. from "FROM" Node / Offset dist	600	mm
Miscellaneous Weight	686.42	N
Offset from Element Centerline	0	mm

Element#	8/21	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	1380	mm
Inside Diameter	609.6	mm
Element Thickness	16	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm



Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Packing	
Detail ID	Bed 3	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	1350	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.83999997	

Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1380	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	80	
Detail Type	Weight	
Detail ID	Bed3 limiter su	
Dist. from "FROM" Node / Offset dist	1360	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	80	
Detail Type	Weight	
Detail ID	Bed3 limiter	
Dist. from "FROM" Node / Offset dist	1370	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element#	9/21	
Element From Node	90	
Element To Node	100	
Element Type	Flange	
Description	Body flange-3	
Distance "FROM" to "TO"	186.3	mm
Flange Inside Diameter	577.6	mm
Element Thickness	88	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	98	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	133.2	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	200	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	



Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	90
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	168.15 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element#	10/21
Element From Node	100
Element To Node	110
Element Type	Flange
Description	Body flange-4
Distance "FROM" to "TO"	204.3 mm
Flange Inside Diameter	585.6 mm
Element Thickness	106 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	108 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	100
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	168.15 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element From Node	100
Detail Type	Weight
Detail ID	Dist sup
Dist. from "FROM" Node / Offset dist	29.205 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm

Element From Node	100
Detail Type	Weight
Detail ID	Distributor
Dist. from "FROM" Node / Offset dist	0.006977 mm
Miscellaneous Weight	294.18 N
Offset from Element Centerline	0 mm

Element#	11/21
Element From Node	110
Element To Node	120
Element Type	Cylinder
Description	Shell-4
Distance "FROM" to "TO"	3000 mm
Inside Diameter	609.6 mm
Element Thickness	12 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C



Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	137.2	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	211.2	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	110	
Detail Type	Packing	
Detail ID	Bed 2	
Dist. from "FROM" Node / Offset dist	520	mm
Height of Packed Section	2480	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.83999997	
Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	110	
Detail Type	Nozzle	
Detail ID	K350	
Dist. from "FROM" Node / Offset dist	200.01	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	99.439	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	110	
Detail Type	Nozzle	
Detail ID	H2150	
Dist. from "FROM" Node / Offset dist	1650	mm
Nozzle Diameter	150	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	542.51	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	110	
Detail Type	Weight	
Detail ID	H2 Davit	
Dist. from "FROM" Node / Offset dist	1650	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	300	mm
Element From Node	110	
Detail Type	Weight	



Detail ID	Bed2 sup	
Dist. from "FROM" Node / Offset dist	450.01	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Bed sup grid	
Dist. from "FROM" Node / Offset dist	500.01	mm
Miscellaneous Weight	686.42	N
Offset from Element Centerline	0	mm

Element#	12/21	
Element From Node	120	
Element To Node	130	
Element Type	Cylinder	
Description	Shell-5	
Distance "FROM" to "TO"	1300	mm
Inside Diameter	609.6	mm
Element Thickness	12	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	120	
Detail Type	Packing	
Detail ID	Bed 2	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	1220	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.83999997	

Element From Node	120	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1300	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	120	
Detail Type	Weight	
Detail ID	Clip weight	
Dist. from "FROM" Node / Offset dist	1050	mm
Miscellaneous Weight	4903	N
Offset from Element Centerline	0	mm

Element From Node	120	
Detail Type	Weight	
Detail ID	Bed2 limiter su	
Dist. from "FROM" Node / Offset dist	1230	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	120	
Detail Type	Weight	
Detail ID	Bed 2 limiter	
Dist. from "FROM" Node / Offset dist	1250	mm



Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	Misc wt 7%	
Dist. from "FROM" Node / Offset dist	997.81	mm
Miscellaneous Weight	7844.8	N
Offset from Element Centerline	0	mm

Element#	13/21	
Element From Node	130	
Element To Node	140	
Element Type	Flange	
Description	Body flange-5	
Distance "FROM" to "TO"	214.3	mm
Flange Inside Diameter	585.6	mm
Element Thickness	116	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	118	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	133.2	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	200	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	130	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	168.15	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element#	14/21	
Element From Node	140	
Element To Node	150	
Element Type	Flange	
Description	Body flange-6	
Distance "FROM" to "TO"	214.3	mm
Flange Inside Diameter	585.6	mm
Element Thickness	116	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	118	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	



Weld is pre-Heated	No
Element From Node	140
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	168.15 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³
Element From Node	140
Detail Type	Weight
Detail ID	Dist2 sup
Dist. from "FROM" Node / Offset dist	11.212 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm
Element From Node	140
Detail Type	Weight
Detail ID	Dist wt
Dist. from "FROM" Node / Offset dist	0.01163 mm
Miscellaneous Weight	294.18 N
Offset from Element Centerline	0 mm

Element#	15/21
Element From Node	150
Element To Node	160
Element Type	Cylinder
Description	Shell-6
Distance "FROM" to "TO"	3000 mm
Inside Diameter	609.6 mm
Element Thickness	12 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	137.2 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	211.2 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	150
Detail Type	Packing
Detail ID	Bed 1
Dist. from "FROM" Node / Offset dist	900 mm
Height of Packed Section	2100 mm
Density	0.000224 kg/cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.83999997

Element From Node	150
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm



Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element From Node	150	
Detail Type	Nozzle	
Detail ID	N180	
Dist. from "FROM" Node / Offset dist	715.01	mm
Nozzle Diameter	80	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	202.78	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	150	
Detail Type	Nozzle	
Detail ID	H1150	
Dist. from "FROM" Node / Offset dist	2250	mm
Nozzle Diameter	150	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	562.7	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	150	
Detail Type	Weight	
Detail ID	H1 Davit	
Dist. from "FROM" Node / Offset dist	2250	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	300	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	N1 pipe sup	
Dist. from "FROM" Node / Offset dist	700.01	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	N1 pipe	
Dist. from "FROM" Node / Offset dist	715.01	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	Bed1 sup	
Dist. from "FROM" Node / Offset dist	800.01	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	Bed 1 sup grid	
Dist. from "FROM" Node / Offset dist	850.01	mm
Miscellaneous Weight	686.42	N
Offset from Element Centerline	0	mm



Element Type	Cylinder
Description	Shell-7
Distance "FROM" to "TO"	1700 mm
Inside Diameter	609.6 mm
Element Thickness	12 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.2148 MPa
Design Temperature Internal Pressure	270 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.79
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	160
Detail Type	Packing
Detail ID	Bed 1
Dist. from "FROM" Node / Offset dist	0 mm
Height of Packed Section	1600 mm
Density	0.000224 kg/cm³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.83999997
Element From Node	160
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1700 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³
Element From Node	160
Detail Type	Weight
Detail ID	Bed1 limiter su
Dist. from "FROM" Node / Offset dist	1620 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm
Element From Node	160
Detail Type	Weight
Detail ID	Bed1 limiter
Dist. from "FROM" Node / Offset dist	1650 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm
Element From Node	160
Detail Type	Weight
Detail ID	Trunnion 1
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	980.6 N
Offset from Element Centerline	300 mm
Element From Node	160
Detail Type	Weight
Detail ID	Trunnion 2
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	980.6 N
Offset from Element Centerline	300 mm

Element#	17/21
Element From Node	170
Element To Node	180
Element Type	Flange
Description	Body flange-7



Distance "FROM" to "TO"	214.3	mm
Flange Inside Diameter	585.6	mm
Element Thickness	116	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	118	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	133.2	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	200	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	170	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	168.15	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element#	18/21	
Element From Node	180	
Element To Node	190	
Element Type	Flange	
Description	Body flange-8	
Distance "FROM" to "TO"	212.3	mm
Flange Inside Diameter	585.6	mm
Element Thickness	114	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	116	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	180	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	168.15	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element#	19/21	
Element From Node	190	
Element To Node	200	
Element Type	Cylinder	
Description	Shell-8	



Distance "FROM" to "TO"	968.14	mm
Inside Diameter	609.6	mm
Element Thickness	12	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	137.2	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	211.2	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	190	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	968.14	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	190	
Detail Type	Nozzle	
Detail ID	N480	
Dist. from "FROM" Node / Offset dist	150.02	mm
Nozzle Diameter	80	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	202.77	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	190	
Detail Type	Nozzle	
Detail ID	K550	
Dist. from "FROM" Node / Offset dist	150.02	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	99.439	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	190	
Detail Type	Weight	
Detail ID	N4 pipe sup	
Dist. from "FROM" Node / Offset dist	50.018	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	190	
Detail Type	Weight	
Detail ID	N4 pipe	



Dist. from "FROM" Node / Offset dist	0.01628	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	190	
Detail Type	Weight	
Detail ID	Demister sup	
Dist. from "FROM" Node / Offset dist	400.02	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	190	
Detail Type	Weight	
Detail ID	Demister	
Dist. from "FROM" Node / Offset dist	450.02	mm
Miscellaneous Weight	196.12	N
Offset from Element Centerline	0	mm

Element#	20/21	
Element From Node	200	
Element To Node	210	
Element Type	Flange	
Description	Body flange-9	
Distance "FROM" to "TO"	212.3	mm
Flange Inside Diameter	585.6	mm
Element Thickness	114	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	116	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	133.2	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	200	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	200	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	168.15	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element#	21/21	
Element From Node	210	
Element To Node	220	
Element Type	Flange	
Description	Body flange-10	
Distance "FROM" to "TO"	84	mm
Flange Outside Diameter	914.4	mm
Element Thickness	88	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	90	mm



External Corrosion Allowance	0	mm
Design Internal Pressure	2.2148	MPa
Design Temperature Internal Pressure	270	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.79	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	210	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	69.85	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	210	
Detail Type	Nozzle	
Detail ID	N2100	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	100	mm
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	522.38	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	210	
Detail Type	For./Mom.	
Detail ID	N2 F/M	
Dist. from "FROM" Node / Offset dist	0	mm
Force in X Direction	5497.9	N
Force in Y Direction	-8246.8	N
Force in Z Direction	5497.9	N
Moment about X Axis	1540080	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	1540080	N-mm
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Skirt-2	...	...	3000	...	3000
Skirt -1	...	...	4000	...	1000
Bottom head	...	...	4050	...	50
Shell-1	...	...	5850	...	1800
Body flange -1	...	...	6040.8	...	186.298
Body flange -2	...	...	6227.1	...	186.298
Shell-2	...	...	9227.09	...	3000
Shell-3	...	...	10607.1	...	1380
Body flange-3	...	...	10797.9	...	186.298
Body flange-4	...	...	11002.2	...	204.298
Shell-4	...	...	14002.2	...	3000
Shell-5	...	...	15302.2	...	1300
Body flange-5	...	...	15521	...	214.298
Body flange-6	...	...	15735.3	...	214.298
Shell-6	...	...	18735.3	...	3000
Shell-7	...	...	20435.3	...	1700
Body flange-7	...	...	20654.1	...	214.298
Body flange-8	...	...	20866.4	...	212.298
Shell-8	...	...	21834.5	...	968.14
Body flange-9	...	...	22050	...	212.298
Body flange-10	...	...	22134	...	84

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Warnings for Flange : FLANGE-1 :

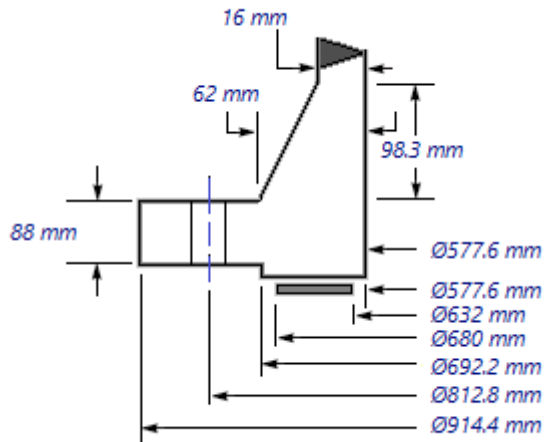
Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: FLANGE-1 :

Body flange -1

Flange Type	Integral Weld Neck		
Design Pressure	P	2.21	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	577.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	88.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	61.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	577.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	632.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		62100.00	N
Bending Moment on Flange		66296300.00	N-mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.21 \cdot (577.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.21) + Ca$$

$$= 7.9008 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133.2 \cdot 1.0 \cdot 13.0) / (291.8 + 0.6 \cdot 13.0)$$

$$= 5.780 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (288.8 + 0.6 \cdot 16.0)$$

$$= 7.399 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	583.600 mm
Corroded Large Hub,	g1Cor = g1-ci	58.974 mm
Corroded Small Hub,	g0Cor = go-ci	13.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	24.000 mm
Basic Gasket Width,	bo = N/2	12.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.729 mm
Gasket Reaction Diameter,	G = Go - 2 * b	662.542 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 662.5415^2 \cdot 2.215$$

$$= 763508.000 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.7293 \cdot \pi \cdot 662.5415 \cdot 3.0 \cdot 2.21$$

$$= 241428.969 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 583.6^2 \cdot 2.2148$$

$$= 592404.000 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 763508 - 592404$$

$$= 171104.000 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_o,1 + W_o,2 + H'p, 0) \\
 &= \max(763508 + 241429 + 0, 0) \\
 &= 1004937.000 \text{ N}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.95 \cdot 8.7293 \cdot \pi \cdot 662.542 + 0.0 \cdot 0.0 \cdot 0.0 \\
 &= 1252673.500 \text{ N}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((1004937 + 62100 + 4 \cdot 66296300 / 662.54) / 172, 1252674 / 172) \\
 &= 8530.700 \text{ mm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 38.1 + 6 \cdot 88.0 / (3.0 + 0.5) \\
 &= 227.057 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 812.8 \cdot \sin(\pi / 24) \\
 &= 106.092 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ Bs / (2a + t) }, 1) \\
 &= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 88.0) }, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	8530.700	21754.795	
Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	227.057

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 21754.795 \cdot 172.0 / (68.95 \cdot \pi (680.0 + 632.0)) \\
 &= 13.166 \text{ mm}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a \cdot A_b \\
 &= 172.0 \cdot 21754.7949 \\
 &= 3741439.00 \text{ N}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o - H \\
 &= 1004937 - 763508 \\
 &= 241428.95 \text{ N}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (812.8 - 662.5415) / 2 \\
 &= 75.1293 \text{ mm}
 \end{aligned}$$

Distance to Face Pressure Reaction [ht]:

$$\begin{aligned}
 &= 0.5 ((C - B) / 2 + h_g) \\
 &= 0.5 ((812.8 - 583.6) / 2 + 75.1293) \\
 &= 94.8646 \text{ mm}
 \end{aligned}$$

Distance to End Pressure Reaction [hd]:

$$\begin{aligned}
 &= (C - B - g_1) / 2 \\
 &= (812.8 - 583.6 - 58.974) / 2 \\
 &= 85.1130 \text{ mm}
 \end{aligned}$$



The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 165.4, Bb = 88.0, Cc = 98.298 \text{ and } Ddg = 35.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4)) \\ = 165.4 \cdot 88.0^3 \cdot (1/3 - 0.210 \cdot (88.0/165.4) \cdot (1 - 1/12 \cdot (88.0/165.4)^4)) \\ = 25062290.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4)) \\ = 98.3 \cdot 35.987^3 \cdot (1/3 - 0.105 \cdot (35.987/98.3) \cdot (1 - 1/192 \cdot (35.987/98.298)^4)) \\ = 1350985.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 1529.396 + 82.442 \\ = 26413274.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21 (t/AR) (1 - 1/12 (t/AR)^4)) \\ = 165.4 \cdot 88.0^3 (1/3 - 0.21 (88.0/165.4) (1 - 1/12 (88.0/6.512)^4)) \\ = 25062290.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2) \\ = \max(26413274, 25062290) \\ = 26413274.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V \\ = 0.0874 \cdot 1.364 \cdot 13.0^2 \cdot 87.102 \cdot 583.6 / 0.041 \\ = 24994664.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	592404.	85.1130	1.0000	50441732.
Face Pressure, Mt	171104.	94.8646	1.0000	16238300.
Gasket Load, Mg	241429.	75.1293	1.0000	18145732.
Gasket Seating, Matm	3741439.	75.1293	1.0000	281205504.
Moment Load, Moe	...	...	...	30273862.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD \\ = 4 \cdot 66296300 [24994664 / (0.384 \cdot 26413274 + 24994664)] \cdot (85.1 / (812.8 - 2 \cdot 85.1)) + \\ 62100 \cdot 85.113 \\ = 30273862.000 \text{ N-mm}$$

Total Moment for Operation, Mo	115099624. N-mm
Total Moment for Gasket seating, Mg	281205504. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	87.102 mm
Hub Ratio, h/h0 = HL / H0	1.129
Thickness Ratio, g1/g0 = (g1Cor/goCor)	4.536

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.641
Factor V per 4.16.5	0.041
Factor f per 4.16.5	1.541

Stress factor from Table 4.16.4 [K]:

$$= A / B \\ = 914.4 / 583.6 \\ = 1.567$$

Stress factor from Table 4.16.4 [Y]:



$$\begin{aligned}
 &= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1) \\
 &= (0.66845 + (5.7169(1.567^2 \cdot \log_{10}(1.567))) / (1.567^2 - 1)) / (1.567 - 1) \\
 &= 4.498
 \end{aligned}$$

Stress factor from Table 4.16.4 [T]:

$$\begin{aligned}
 &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1)) \\
 &= (1.567^2(1 + 8.55246 \cdot \log_{10}(1.567)) - 1) / ((1.0472 + 1.9448 \cdot 1.567^2) \cdot (1.567 - 1)) \\
 &= 1.682
 \end{aligned}$$

Stress factor from Table 4.16.4 [U]:

$$\begin{aligned}
 &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1)) \\
 &= (1.567^2(1 + 8.55246 \cdot \log_{10}(1.567)) - 1) / (1.36136(1.567^2 - 1) \cdot (1.567 - 1)) \\
 &= 4.943
 \end{aligned}$$

Stress factor from Table 4.16.4 [Z]:

$$\begin{aligned}
 &= (K^2 + 1) / (K^2 - 1) \\
 &= (1.567^2 + 1) / (1.567^2 - 1) \\
 &= 2.375
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned}
 &= U \cdot g_0^2 \cdot h_0 / V \\
 &= 4.943 \cdot 13.0^2 \cdot 87.102 / 0.041 \\
 &= 1776426.625 \text{ mm}^3
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned}
 &= F / h_0 \\
 &= 0.641 / 87.102 \\
 &= 0.007 \text{ 1/mm}
 \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned}
 &= (t \cdot e + 1) / T + t^3 / d \\
 &= (88.0 \cdot 0.007 + 1) / 1.682 + 88.0^3 / 1776427 \\
 &= 1.364
 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned}
 &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\
 &= (1.5412 \cdot 11510E+09 / 583.6) / (1.3635 \cdot 58.974^2) \\
 &= 64.08 \text{ MPa}
 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned}
 &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\
 &= (1.5412 \cdot 0.28121E+09 / 583.6) / (1.3635 \cdot 58.974^2) \\
 &= 156.55 \text{ MPa}
 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned}
 &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\
 &= (1.864 \cdot 11510E+09 / 583.6) / (1.3635 \cdot 88.0^2) \\
 &= 34.80 \text{ MPa}
 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned}
 &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\
 &= (1.864 \cdot 0.28121E+09 / 583.6) / (1.3635 \cdot 88.0^2) \\
 &= 85.03 \text{ MPa}
 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned}
 &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\
 &= (4.4981 \cdot 11510E+09 / (88.0^2 \cdot 583.6)) - 2.3746 \cdot 35 \\
 &= 31.87 \text{ MPa}
 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned}
 &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\
 &= (4.4981 \cdot 0.28121E+09 / (88.0^2 \cdot 583.6)) - 2.3746 \cdot 85 \\
 &= 77.87 \text{ MPa}
 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned}
 &= (SHo + \max(SRo, STo)) / 2 \\
 &= (64 + \max(35, 32)) / 2 \\
 &= 49.44 \text{ MPa}
 \end{aligned}$$



Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRA, STa)) / 2$$

$$= (157 + \max(85, 78)) / 2$$

$$= 120.79 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm1 + Fa + 4Me/G) / Ab$$

$$= 1467129 / 21754.7949$$

$$= 67.45 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (1252674 / 21754.7949)$$

$$= 57.59 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	64.08	199.80	156.55	207.00
Radial Flange	34.80	133.20	85.03	138.00
Tangential Flange	31.87	133.20	77.87	138.00
Maximum Average	49.44	133.20	120.79	138.00
Bolting	67.45	172.00	57.59	172.00

Minimum Required Flange Thickness 81.128 mm

Estimated M.A.W.P. (Operating) 6 MPa

Estimated Finished Weight of Flange at given Thk. 326.7 kgm

Estimated Unfinished Weight of Forging at given Thk 569.9 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Operating		stress ratio	Gasket Seating		stress ratio
		Actual	Allowed		Actual	Allowed	
Longitudinal Hub	Sh	356.49	457.15	0.780	356.49	499.99	0.713
Tangential Flange	St	178.02	342.86	0.519	178.02	374.99	0.475
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	410.23	685.72	0.598	410.23	749.99	0.547
Combined	Sr + St	372.41	685.72	0.543	372.41	749.99	0.497
Combined	Sh + Sto	585.06	685.72	0.853	585.06	749.99	0.780
Radial Flange	Sr	194.39	457.15	0.425	194.39	499.99	0.389
Maximum Stress Ratio				1.000			0.914



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\ = \max(1; 5.2751/3.436) \\ = 1.5352$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ = \max(0.25; 5.1118/3.8842) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 1.3175/1.3385) \\ = 0.9844$$

In this case, the hub ratio $h/h0$, 1.1285 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	393.2 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.202 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 632.0^2)$$

$$= 49461.3 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172.0 \cdot 49461.254 / (24 \cdot 906.45)$$

$$= 391.1 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(391.056; 507.5)$$

$$= 391.1 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(391.056; 290.0)$$

$$= 391.1 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(391.056; 393.191)$$

$$= 391.1 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin - S(Ag / (nb \cdot Abs))$$

$$:: 391.056 \geq 140.0(49461.254 / (24 \cdot 906.45))$$

$$:: 391.1 \geq 318.3 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin - O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 391.056 \geq (97.0 \cdot 49461.254 + \pi/4 \cdot 2.215 \cdot 632.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 391.1 \geq 360.7 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 391.056 \leq 380.0(49461.254 / (24 \cdot 906.45))$$

$$:: 391.1 \leq 864.0 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetamax} / \text{thetafmax})$$

$$:: 391.056 \leq 393.191(1.0/0.202)$$

$$:: 391.1 \leq 1948.8 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 391.056 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 2701945.0 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.25 \cdot 202713 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$$

$$= 0.536 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 0.11510E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.25 \cdot 187243 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$$

$$= 0.237 \text{ (should be } \leq 1)$$



Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.364 \cdot 202713 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.491$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.11510E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.364 \cdot 187243 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.218$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.545, Temperature Reduction per Fig. UCS 66.1 = 27 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -41 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	54.86	207.00	131.12	207.00
Radial Flange	37.17	138.00	88.84	138.00
Tangential Flange	29.20	138.00	69.80	138.00
Maximum Average	46.01	138.00	109.98	138.00
Bolting	67.45	172.00	57.59	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 76.479 mm

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Warnings for Flange : FLANGE-2 :

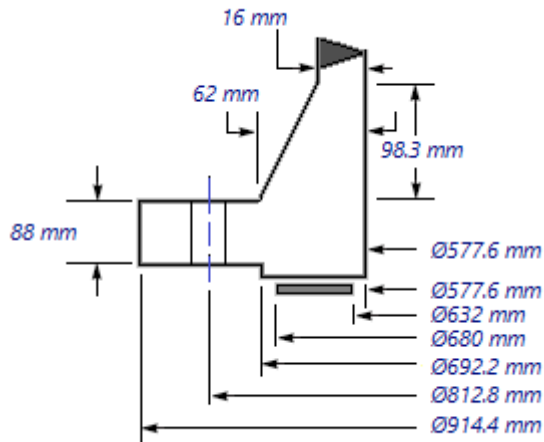
Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: FLANGE-2 :

Body flange -2

Flange Type	Integral Weld Neck		
Design Pressure	P	2.21	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	577.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	88.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	61.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	577.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	632.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		57710.00	N
Bending Moment on Flange		54796700.00	N-mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.21 \cdot (577.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.21) + Ca$$

$$= 7.9008 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133.2 \cdot 1.0 \cdot 13.0) / (291.8 + 0.6 \cdot 13.0)$$

$$= 5.780 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (288.8 + 0.6 \cdot 16.0)$$

$$= 7.399 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	583.600 mm
Corroded Large Hub,	g1Cor = g1-ci	58.974 mm
Corroded Small Hub,	g0Cor = go-ci	13.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	24.000 mm
Basic Gasket Width,	bo = N/2	12.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.729 mm
Gasket Reaction Diameter,	G = Go - 2 * b	662.542 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 662.5415^2 \cdot 2.215$$

$$= 763508.000 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.7293 \cdot \pi \cdot 662.5415 \cdot 3.0 \cdot 2.21$$

$$= 241428.969 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 583.6^2 \cdot 2.2148$$

$$= 592404.000 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 763508 - 592404$$

$$= 171104.000 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:



$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(763508 + 241429 + 0, 0)$$

$$= 1004937.000 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.7293 \cdot \pi \cdot 662.542 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1252673.500 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((1004937 + 57710 + 4 \cdot 54796700 / 662.54) / 172, 1252674 / 172)$$

$$= 8101.650 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 88.0 / (3.0 + 0.5)$$

$$= 227.057 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 812.8 \cdot \sin(\pi / 24)$$

$$= 106.092 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 88.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	8101.650	21754.795	
Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	227.057

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi (680.0 + 632.0))$$

$$= 13.166 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 1004937 - 763508$$

$$= 241428.95 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 662.5415) / 2$$

$$= 75.1293 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((812.8 - 583.6) / 2 + 75.1293)$$

$$= 94.8646 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (812.8 - 583.6 - 58.974) / 2$$

$$= 85.1130 \text{ mm}$$



The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 165.4, Bb = 88.0, Cc = 98.298 \text{ and } Ddg = 35.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4)) \\ = 165.4 \cdot 88.0^3 \cdot (1/3 - 0.210 \cdot (88.0/165.4) \cdot (1 - 1/12 \cdot (88.0/165.4)^4)) \\ = 25062290.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4)) \\ = 98.3 \cdot 35.987^3 \cdot (1/3 - 0.105 \cdot (35.987/98.3) \cdot (1 - 1/192 \cdot (35.987/98.298)^4)) \\ = 1350985.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 1529.396 + 82.442 \\ = 26413274.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21 (t/AR) (1 - 1/12 (t/AR)^4)) \\ = 165.4 \cdot 88.0^3 (1/3 - 0.21 (88.0/165.4) (1 - 1/12 (88.0/6.512)^4)) \\ = 25062290.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2) \\ = \max(26413274, 25062290) \\ = 26413274.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V \\ = 0.0874 \cdot 1.364 \cdot 13.0^2 \cdot 87.102 \cdot 583.6 / 0.041 \\ = 24994664.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	592404.	85.1130	1.0000	50441732.
Face Pressure, Mt	171104.	94.8646	1.0000	16238300.
Gasket Load, Mg	241429.	75.1293	1.0000	18145732.
Gasket Seating, Matm	3741439.	75.1293	1.0000	281205504.
Moment Load, Moe	...	...	...	25566020.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD \\ = 4 \cdot 54796700 [24994664 / (0.384 \cdot 26413274 + 24994664)] \cdot (85.1 / (812.8 - 2 \cdot 85.1)) + \\ 57710 \cdot 85.113 \\ = 25566020.000 \text{ N-mm}$$

Total Moment for Operation, Mo	110391784. N-mm
Total Moment for Gasket seating, Mg	281205504. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	87.102 mm
Hub Ratio, h/h0 = HL / H0	1.129
Thickness Ratio, g1/g0 = (g1Cor/goCor)	4.536

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.641
Factor V per 4.16.5	0.041
Factor f per 4.16.5	1.541

Stress factor from Table 4.16.4 [K]:

$$= A / B \\ = 914.4 / 583.6 \\ = 1.567$$

Stress factor from Table 4.16.4 [Y]:



$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.567^2 \cdot \log_{10}(1.567))) / (1.567^2 - 1)) / (1.567 - 1)$$

$$= 4.498$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.567^2(1 + 8.55246 \cdot \log_{10}(1.567)) - 1) / ((1.0472 + 1.9448 \cdot 1.567^2) \cdot (1.567 - 1))$$

$$= 1.682$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.567^2(1 + 8.55246 \cdot \log_{10}(1.567)) - 1) / (1.36136(1.567^2 - 1) \cdot (1.567 - 1))$$

$$= 4.943$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.567^2 + 1) / (1.567^2 - 1)$$

$$= 2.375$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 4.943 \cdot 13.0^2 \cdot 87.102 / 0.041$$

$$= 1776426.625 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.641 / 87.102$$

$$= 0.007 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (88.0 \cdot 0.007 + 1) / 1.682 + 88.0^3 / 1776427$$

$$= 1.364$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.5412 \cdot 1.11039\text{E}+09 / 583.6) / (1.3635 \cdot 58.974^2)$$

$$= 61.46 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.5412 \cdot 0.28121\text{E}+09 / 583.6) / (1.3635 \cdot 58.974^2)$$

$$= 156.55 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.864 \cdot 1.11039\text{E}+09 / 583.6) / (1.3635 \cdot 88.0^2)$$

$$= 33.38 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.864 \cdot 0.28121\text{E}+09 / 583.6) / (1.3635 \cdot 88.0^2)$$

$$= 85.03 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.4981 \cdot 1.11039\text{E}+09 / (88.0^2 \cdot 583.6)) - 2.3746 \cdot 33$$

$$= 30.57 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.4981 \cdot 0.28121\text{E}+09 / (88.0^2 \cdot 583.6)) - 2.3746 \cdot 85$$

$$= 77.87 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (61 + \max(33, 31)) / 2$$

$$= 47.42 \text{ MPa}$$



Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (157 + \max(85, 78)) / 2$$

$$= 120.79 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm1 + Fa + 4Me/G) / Ab$$

$$= 1393340 / 21754.7949$$

$$= 64.05 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (1252674 / 21754.7949)$$

$$= 57.59 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	61.46	199.80	156.55	207.00
Radial Flange	33.38	133.20	85.03	138.00
Tangential Flange	30.57	133.20	77.87	138.00
Maximum Average	47.42	133.20	120.79	138.00
Bolting	64.05	172.00	57.59	172.00

Minimum Required Flange Thickness 81.128 mm

Estimated M.A.W.P. (Operating) 6 MPa

Estimated Finished Weight of Flange at given Thk. 326.7 kgm

Estimated Unfinished Weight of Forging at given Thk 569.9 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Operating		stress ratio	Gasket Seating		stress ratio
		Actual	Allowed		Actual	Allowed	
Longitudinal Hub	Sh	356.49	457.15	0.780	356.49	499.99	0.713
Tangential Flange	St	178.02	342.86	0.519	178.02	374.99	0.475
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	410.23	685.72	0.598	410.23	749.99	0.547
Combined	Sr + St	372.41	685.72	0.543	372.41	749.99	0.497
Combined	Sh + Sto	585.06	685.72	0.853	585.06	749.99	0.780
Radial Flange	Sr	194.39	457.15	0.425	194.39	499.99	0.389
Maximum Stress Ratio				1.000			0.914



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\ = \max(1; 5.2751/3.436) \\ = 1.5352$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ = \max(0.25; 5.1118/3.8842) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 1.3175/1.3385) \\ = 0.9844$$

In this case, the hub ratio $h/h0$, 1.1285 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, $Sfmax$, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	393.2 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.202 deg

Gasket Contact Area [Ag]:

$$\begin{aligned} &= \pi/4 (GO.D.^2 - GI.D.^2) \\ &= \pi/4 (680.0^2 - 632.0^2) \\ &= 49461.3 \text{ mm}^2 \end{aligned}$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$\begin{aligned} &= SgT \cdot Ag / (nb \cdot Ab) \\ &= 172.0 \cdot 49461.254 / (24 \cdot 906.45) \\ &= 391.1 \text{ MPa} \end{aligned}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$\begin{aligned} &= \min(Sbsel; Sbmax) \\ &= \min(391.056; 507.5) \\ &= 391.1 \text{ MPa} \end{aligned}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$\begin{aligned} &= \max(Sbsel; Sbmin) \\ &= \max(391.056; 290.0) \\ &= 391.1 \text{ MPa} \end{aligned}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$\begin{aligned} &= \min(Sbsel; Sfmax) \\ &= \min(391.056; 393.191) \\ &= 391.1 \text{ MPa} \end{aligned}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\begin{aligned} &:: Sbsel \geq Sgmin \cdot S(Ag / (nb \cdot Abs)) \\ &:: 391.056 \geq 140.0(49461.254 / (24 \cdot 906.45)) \\ &:: 391.1 \geq 318.3 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\begin{aligned} &:: Sbsel \geq (Sgmin \cdot O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb) \\ &:: 391.056 \geq (97.0 \cdot 49461.254 + \pi/4 \cdot 2.215 \cdot 632.0^2) / (0.7 \cdot 906.45 \cdot 24) \\ &:: 391.1 \geq 360.7 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\begin{aligned} &:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs)) \\ &:: 391.056 \leq 380.0(49461.254 / (24 \cdot 906.45)) \\ &:: 391.1 \leq 864.0 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\begin{aligned} &:: Sbsel \leq Sfmax(\theta_{tagmax} / \theta_{tafmax}) \\ &:: 391.056 \leq 393.191(1.0/0.202) \\ &:: 391.1 \leq 1948.8 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$\begin{aligned} &= Sbsel \cdot K \cdot Abs \cdot \phi b \\ &= 391.056 \cdot 0.2 \cdot 906.45 \cdot 38.1 \\ &= 2701945.0 \text{ N-mm} \end{aligned}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$\begin{aligned} &= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki) \\ &= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.25 \cdot 202713 \cdot 13.0^2 \cdot 87.102 \cdot 0.3) \\ &= 0.536 \text{ (should be } \leq 1) \end{aligned}$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$\begin{aligned} &= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki) \\ &= 52.14 \cdot 0.11039E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.25 \cdot 187243 \cdot 13.0^2 \cdot 87.102 \cdot 0.3) \\ &= 0.228 \text{ (should be } \leq 1) \end{aligned}$$



Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.364 \cdot 202713 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.491$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.11039E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.364 \cdot 187243 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.209$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.533, Temperature Reduction per Fig. UCS 66.1 = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	52.56	207.00	131.12	207.00
Radial Flange	35.61	138.00	88.84	138.00
Tangential Flange	27.98	138.00	69.80	138.00
Maximum Average	44.09	138.00	109.98	138.00
Bolting	64.05	172.00	57.59	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 76.479 mm

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Warnings for Flange : FLANGE-3 :

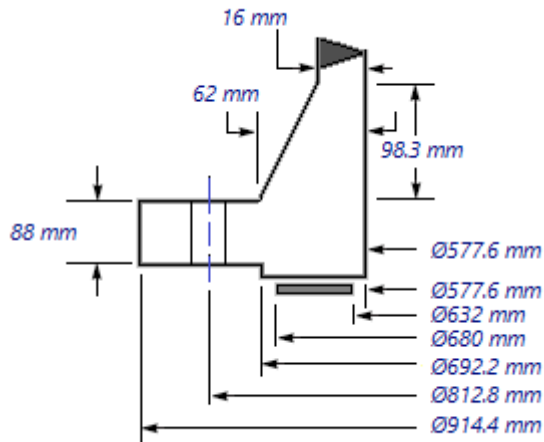
Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: FLANGE-3 :

Body flange-3

Flange Type	Integral Weld Neck		
Design Pressure	P	2.21	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	577.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	88.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	61.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	577.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	632.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		38010.80	N
Bending Moment on Flange		34100700.00	N-mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.21 \cdot (577.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.21) + Ca$$

$$= 7.9008 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133.2 \cdot 1.0 \cdot 13.0) / (291.8 + 0.6 \cdot 13.0)$$

$$= 5.780 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (288.8 + 0.6 \cdot 16.0)$$

$$= 7.399 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	583.600 mm
Corroded Large Hub,	g1Cor = g1-ci	58.974 mm
Corroded Small Hub,	g0Cor = go-ci	13.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	24.000 mm
Basic Gasket Width,	bo = N/2	12.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.729 mm
Gasket Reaction Diameter,	G = Go - 2 * b	662.542 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 662.5415^2 \cdot 2.215$$

$$= 763508.000 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.7293 \cdot \pi \cdot 662.5415 \cdot 3.0 \cdot 2.21$$

$$= 241428.969 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 583.6^2 \cdot 2.2148$$

$$= 592404.000 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 763508 - 592404$$

$$= 171104.000 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:



$$\begin{aligned}
 &= \max(W_o,1 + W_o,2 + H'p, 0) \\
 &= \max(763508 + 241429 + 0, 0) \\
 &= 1004937.000 \text{ N}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.95 \cdot 8.7293 \cdot \pi \cdot 662.542 + 0.0 \cdot 0.0 \cdot 0.0 \\
 &= 1252673.500 \text{ N}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((1004937 + 38011 + 4 \cdot 34100700 / 662.54) / 172, 1252674 / 172) \\
 &= 7283.736 \text{ mm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 38.1 + 6 \cdot 88.0 / (3.0 + 0.5) \\
 &= 227.057 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 812.8 \cdot \sin(\pi / 24) \\
 &= 106.092 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ Bs / (2a + t) }, 1) \\
 &= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 88.0) }, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	7283.736	21754.795	
Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	227.057

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 21754.795 \cdot 172.0 / (68.95 \cdot \pi (680.0 + 632.0)) \\
 &= 13.166 \text{ mm}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a \cdot A_b \\
 &= 172.0 \cdot 21754.7949 \\
 &= 3741439.00 \text{ N}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o - H \\
 &= 1004937 - 763508 \\
 &= 241428.95 \text{ N}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (812.8 - 662.5415) / 2 \\
 &= 75.1293 \text{ mm}
 \end{aligned}$$

Distance to Face Pressure Reaction [ht]:

$$\begin{aligned}
 &= 0.5 ((C - B) / 2 + h_g) \\
 &= 0.5 ((812.8 - 583.6) / 2 + 75.1293) \\
 &= 94.8646 \text{ mm}
 \end{aligned}$$

Distance to End Pressure Reaction [hd]:

$$\begin{aligned}
 &= (C - B - g_1) / 2 \\
 &= (812.8 - 583.6 - 58.974) / 2 \\
 &= 85.1130 \text{ mm}
 \end{aligned}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 165.4, Bb = 88.0, Cc = 98.298 \text{ and } Ddg = 35.987 \text{ mm}$$

Compute the value of [Kab]:

$$\begin{aligned} &= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4)) \\ &= 165.4 \cdot 88.0^3 \cdot (1/3 - 0.210 \cdot (88.0/165.4) \cdot (1 - 1/12 \cdot (88.0/165.4)^4)) \\ &= 25062290.000 \text{ mm}^4 \end{aligned}$$

Compute the value of [Kcd]:

$$\begin{aligned} &= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4)) \\ &= 98.3 \cdot 35.987^3 \cdot (1/3 - 0.105 \cdot (35.987/98.3) \cdot (1 - 1/192 \cdot (35.987/98.298)^4)) \\ &= 1350985.000 \text{ mm}^4 \end{aligned}$$

Flange Inertia Term [Ip,term1]:

$$\begin{aligned} &= Kab + Kcd = 1529.396 + 82.442 \\ &= 26413274.000 \text{ mm}^4 \end{aligned}$$

Flange Inertia Term [Ip,term2]:

$$\begin{aligned} &= Ar \cdot t^3 (1/3 - 0.21 (t/AR) (1 - 1/12 (t/AR)^4)) \\ &= 165.4 \cdot 88.0^3 (1/3 - 0.21 (88.0/165.4) (1 - 1/12 (88.0/6.512)^4)) \\ &= 25062290.000 \text{ mm}^4 \end{aligned}$$

Flange Inertia [Ip]:

$$\begin{aligned} &= \max(Ip,term1, Ip,term2) \\ &= \max(26413274, 25062290) \\ &= 26413274.0 \text{ mm}^4 \end{aligned}$$

Flange Moment of Inertia for Integral Types [I]:

$$\begin{aligned} &= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V \\ &= 0.0874 \cdot 1.364 \cdot 13.0^2 \cdot 87.102 \cdot 583.6 / 0.041 \\ &= 24994664.000 \text{ mm}^4 \end{aligned}$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	592404.	85.1130	1.0000	50441732.
Face Pressure, Mt	171104.	94.8646	1.0000	16238300.
Gasket Load, Mg	241429.	75.1293	1.0000	18145732.
Gasket Seating, Matm	3741439.	75.1293	1.0000	281205504.
Moment Load, Moe	...	...	...	16088630.

Moment Value, eq. 4.16.16, [Moe]:

$$\begin{aligned} &= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD \\ &= 4 \cdot 34100700 [24994664 / (0.384 \cdot 26413274 + 24994664)] \cdot (85.1 / (812.8 - 2 \cdot 85.1)) + \\ &\quad 38011 \cdot 85.113 \\ &= 16088630.000 \text{ N-mm} \end{aligned}$$

Total Moment for Operation, Mo	100914392. N-mm
Total Moment for Gasket seating, Mg	281205504. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	87.102 mm
Hub Ratio, h/h0 = HL / H0	1.129
Thickness Ratio, g1/g0 = (g1Cor/goCor)	4.536

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.641
Factor V per 4.16.5	0.041
Factor f per 4.16.5	1.541

Stress factor from Table 4.16.4 [K]:

$$\begin{aligned} &= A / B \\ &= 914.4 / 583.6 \\ &= 1.567 \end{aligned}$$

Stress factor from Table 4.16.4 [Y]:



$$\begin{aligned}
 &= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1) \\
 &= (0.66845 + (5.7169(1.567^2 \cdot \log_{10}(1.567))) / (1.567^2 - 1)) / (1.567 - 1) \\
 &= 4.498
 \end{aligned}$$

Stress factor from Table 4.16.4 [T]:

$$\begin{aligned}
 &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1)) \\
 &= (1.567^2(1 + 8.55246 \cdot \log_{10}(1.567)) - 1) / ((1.0472 + 1.9448 \cdot 1.567^2) \cdot (1.567 - 1)) \\
 &= 1.682
 \end{aligned}$$

Stress factor from Table 4.16.4 [U]:

$$\begin{aligned}
 &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1)) \\
 &= (1.567^2(1 + 8.55246 \cdot \log_{10}(1.567)) - 1) / (1.36136(1.567^2 - 1) \cdot (1.567 - 1)) \\
 &= 4.943
 \end{aligned}$$

Stress factor from Table 4.16.4 [Z]:

$$\begin{aligned}
 &= (K^2 + 1) / (K^2 - 1) \\
 &= (1.567^2 + 1) / (1.567^2 - 1) \\
 &= 2.375
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned}
 &= U \cdot g_0^2 \cdot h_0 / V \\
 &= 4.943 \cdot 13.0^2 \cdot 87.102 / 0.041 \\
 &= 1776426.625 \text{ mm}^3
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned}
 &= F / h_0 \\
 &= 0.641 / 87.102 \\
 &= 0.007 \text{ 1/mm}
 \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned}
 &= (t \cdot e + 1) / T + t^3 / d \\
 &= (88.0 \cdot 0.007 + 1) / 1.682 + 88.0^3 / 1776427 \\
 &= 1.364
 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned}
 &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\
 &= (1.5412 \cdot 10091E+09 / 583.6) / (1.3635 \cdot 58.974^2) \\
 &= 56.18 \text{ MPa}
 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned}
 &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\
 &= (1.5412 \cdot 0.28121E+09 / 583.6) / (1.3635 \cdot 58.974^2) \\
 &= 156.55 \text{ MPa}
 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned}
 &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\
 &= (1.864 \cdot 10091E+09 / 583.6) / (1.3635 \cdot 88.0^2) \\
 &= 30.51 \text{ MPa}
 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned}
 &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\
 &= (1.864 \cdot 0.28121E+09 / 583.6) / (1.3635 \cdot 88.0^2) \\
 &= 85.03 \text{ MPa}
 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned}
 &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\
 &= (4.4981 \cdot 10091E+09 / (88.0^2 \cdot 583.6)) - 2.3746 \cdot 31 \\
 &= 27.95 \text{ MPa}
 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned}
 &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\
 &= (4.4981 \cdot 0.28121E+09 / (88.0^2 \cdot 583.6)) - 2.3746 \cdot 85 \\
 &= 77.87 \text{ MPa}
 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned}
 &= (SHo + \max(SRo, STo)) / 2 \\
 &= (56 + \max(31, 28)) / 2 \\
 &= 43.35 \text{ MPa}
 \end{aligned}$$



Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (157 + \max(85, 78)) / 2$$

$$= 120.79 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm1 + Fa + 4Me/G) / Ab$$

$$= 1248743 / 21754.7949$$

$$= 57.41 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (1252674 / 21754.7949)$$

$$= 57.59 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	56.18	199.80	156.55	207.00
Radial Flange	30.51	133.20	85.03	138.00
Tangential Flange	27.95	133.20	77.87	138.00
Maximum Average	43.35	133.20	120.79	138.00
Bolting	57.41	172.00	57.59	172.00

Minimum Required Flange Thickness 81.128 mm

Estimated M.A.W.P. (Operating) 6 MPa

Estimated Finished Weight of Flange at given Thk. 326.7 kgm

Estimated Unfinished Weight of Forging at given Thk 569.9 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Operating		stress ratio	Gasket Seating		stress ratio
		Actual	Allowed		Actual	Allowed	
Longitudinal Hub	Sh	356.49	457.15	0.780	356.49	499.99	0.713
Tangential Flange	St	178.02	342.86	0.519	178.02	374.99	0.475
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	410.23	685.72	0.598	410.23	749.99	0.547
Combined	Sr + St	372.41	685.72	0.543	372.41	749.99	0.497
Combined	Sh + Sto	585.06	685.72	0.853	585.06	749.99	0.780
Radial Flange	Sr	194.39	457.15	0.425	194.39	499.99	0.389
Maximum Stress Ratio				1.000			0.914



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\ = \max(1; 5.2751/3.436) \\ = 1.5352$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ = \max(0.25; 5.1118/3.8842) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 1.3175/1.3385) \\ = 0.9844$$

In this case, the hub ratio $h/h0$, 1.1285 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	393.2 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.202 deg

Gasket Contact Area [Ag]:

$$\begin{aligned} &= \pi/4 (GO.D.^2 - GI.D.^2) \\ &= \pi/4 (680.0^2 - 632.0^2) \\ &= 49461.3 \text{ mm}^2 \end{aligned}$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$\begin{aligned} &= SgT \cdot Ag / (nb \cdot Ab) \\ &= 172.0 \cdot 49461.254 / (24 \cdot 906.45) \\ &= 391.1 \text{ MPa} \end{aligned}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$\begin{aligned} &= \min(Sbsel; Sbmax) \\ &= \min(391.056; 507.5) \\ &= 391.1 \text{ MPa} \end{aligned}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$\begin{aligned} &= \max(Sbsel; Sbmin) \\ &= \max(391.056; 290.0) \\ &= 391.1 \text{ MPa} \end{aligned}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$\begin{aligned} &= \min(Sbsel; Sfmax) \\ &= \min(391.056; 393.191) \\ &= 391.1 \text{ MPa} \end{aligned}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\begin{aligned} &:: Sbsel \geq Sgmin \cdot S(Ag / (nb \cdot Abs)) \\ &:: 391.056 \geq 140.0(49461.254 / (24 \cdot 906.45)) \\ &:: 391.1 \geq 318.3 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\begin{aligned} &:: Sbsel \geq (Sgmin \cdot O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb) \\ &:: 391.056 \geq (97.0 \cdot 49461.254 + \pi/4 \cdot 2.215 \cdot 632.0^2) / (0.7 \cdot 906.45 \cdot 24) \\ &:: 391.1 \geq 360.7 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\begin{aligned} &:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs)) \\ &:: 391.056 \leq 380.0(49461.254 / (24 \cdot 906.45)) \\ &:: 391.1 \leq 864.0 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\begin{aligned} &:: Sbsel \leq Sfmax(\text{thetamax} / \text{thetafmax}) \\ &:: 391.056 \leq 393.191(1.0/0.202) \\ &:: 391.1 \leq 1948.8 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$\begin{aligned} &= Sbsel \cdot K \cdot Abs \cdot \phi b \\ &= 391.056 \cdot 0.2 \cdot 906.45 \cdot 38.1 \\ &= 2701945.0 \text{ N-mm} \end{aligned}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$\begin{aligned} &= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki) \\ &= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.25 \cdot 202713 \cdot 13.0^2 \cdot 87.102 \cdot 0.3) \\ &= 0.536 \text{ (should be } \leq 1) \end{aligned}$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$\begin{aligned} &= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki) \\ &= 52.14 \cdot 0.10091E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.25 \cdot 187243 \cdot 13.0^2 \cdot 87.102 \cdot 0.3) \\ &= 0.208 \text{ (should be } \leq 1) \end{aligned}$$



Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.364 \cdot 202713 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.491$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.10091E+09 / 1.0 \cdot 1.0 \cdot 0.041 / (1.364 \cdot 187243 \cdot 13.0^2 \cdot 87.102 \cdot 0.3)$
 $= 0.191$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.533, Temperature Reduction per Fig. UCS 66.1 = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	47.96	207.00	131.12	207.00
Radial Flange	32.49	138.00	88.84	138.00
Tangential Flange	25.53	138.00	69.80	138.00
Maximum Average	40.22	138.00	109.98	138.00
Bolting	57.41	172.00	57.59	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 76.479 mm

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Warnings for Flange : FLANGE-4 :

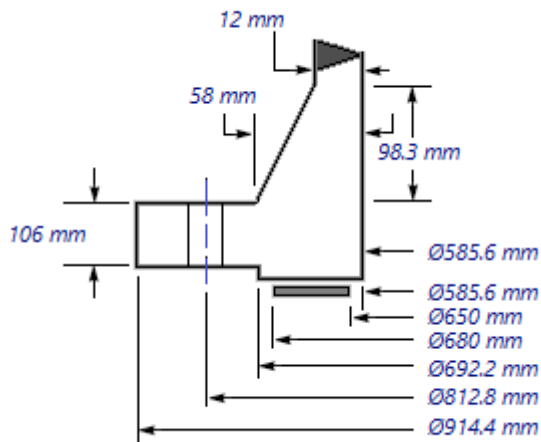
Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: FLANGE-4 :

Body flange-4

Flange Type	Integral Weld Neck		
Design Pressure	P	2.21	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	585.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	106.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	57.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	585.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	650.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		42450.00	N
Bending Moment on Flange		33097498.00	N-mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.21 \cdot (585.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.21) + Ca$$

$$= 7.9680 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133.2 \cdot 1.0 \cdot 9.0) / (295.8 + 0.6 \cdot 9.0)$$

$$= 3.980 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (292.8 + 0.6 \cdot 12.0)$$

$$= 5.520 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	591.600 mm
Corroded Large Hub,	g1Cor = g1-ci	54.974 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	15.000 mm
Basic Gasket Width,	bo = N/2	7.500 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.901 mm
Gasket Reaction Diameter,	G = Go - 2 * b	666.198 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 666.1978^2 \cdot 2.215$$

$$= 771958.438 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.9011 \cdot \pi \cdot 666.1978 \cdot 3.0 \cdot 2.21$$

$$= 191919.672 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 591.6^2 \cdot 2.2148$$

$$= 608756.688 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 771958 - 608757$$

$$= 163201.781 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\
 &= \max(771958 + 191920 + 0, 0) \\
 &= 963878.125 \text{ N}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.95 \cdot 6.9011 \cdot \pi \cdot 666.198 + 0.0 \cdot 0.0 \cdot 0.0 \\
 &= 995790.500 \text{ N}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((963878 + 42450 + 4 \cdot 33097498 / 666.2) / 172, 995791 / 172) \\
 &= 7006.375 \text{ mm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 38.1 + 6 \cdot 106.0 / (3.0 + 0.5) \\
 &= 257.914 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 812.8 \cdot \sin(\pi / 24) \\
 &= 106.092 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ Bs / (2a + t) }, 1) \\
 &= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 106.0) }, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	7006.375	21754.795	
Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	257.914

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 21754.795 \cdot 172.0 / (68.95 \cdot \pi (680.0 + 650.0)) \\
 &= 12.988 \text{ mm}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a \cdot A_b \\
 &= 172.0 \cdot 21754.7949 \\
 &= 3741439.00 \text{ N}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o - H \\
 &= 963878 - 771958 \\
 &= 191919.66 \text{ N}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (812.8 - 666.1978) / 2 \\
 &= 73.3011 \text{ mm}
 \end{aligned}$$

Distance to Face Pressure Reaction [ht]:

$$\begin{aligned}
 &= 0.5 ((C - B) / 2 + h_g) \\
 &= 0.5 ((812.8 - 591.6) / 2 + 73.3011) \\
 &= 91.9505 \text{ mm}
 \end{aligned}$$

Distance to End Pressure Reaction [hd]:

$$\begin{aligned}
 &= (C - B - g_1) / 2 \\
 &= (812.8 - 591.6 - 54.974) / 2 \\
 &= 83.1130 \text{ mm}
 \end{aligned}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 161.4, Bb = 106.0, Cc = 98.298 \text{ and } Ddg = 31.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 161.4 \cdot 106.0^3 \cdot (1/3 - 0.210 \cdot (106.0/161.4) \cdot (1 - 1/12 \cdot (106.0/161.4)^4))$$

$$= 37975636.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 31.987^3 \cdot (1/3 - 0.105 \cdot (31.987/98.3) \cdot (1 - 1/192 \cdot (31.987/98.298)^4))$$

$$= 962452.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 2317.418 + 58.732$$

$$= 38938088.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \cdot (1/3 - 0.21 \cdot (t/AR) \cdot (1 - 1/12 \cdot (t/AR)^4))$$

$$= 161.4 \cdot 106.0^3 \cdot (1/3 - 0.21 \cdot (106.0/161.4) \cdot (1 - 1/12 \cdot (106.0/161.4)^4))$$

$$= 37975636.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(38938088, 37975636)$$

$$= 38938088.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.954 \cdot 9.0^2 \cdot 72.968 \cdot 591.6 / 0.022$$

$$= 27344090.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	608757.	83.1130	1.0000	50616116.
Face Pressure, Mt	163202.	91.9505	1.0000	15012577.
Gasket Load, Mg	191920.	73.3011	1.0000	14073621.
Gasket Seating, Matm	3741439.	73.3011	1.0000	274362688.
Moment Load, Moe	...	...	...	14531458.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 33097498 [27344090 / (0.384 \cdot 38938088 + 27344090)] \cdot (83.1 / (812.8 - 2 \cdot 83.1)) + 42450 \cdot 83.113$$

$$= 14531458.000 \text{ N-mm}$$

Total Moment for Operation, Mo	94233768. N-mm
Total Moment for Gasket seating, Mg	274362688. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	72.968 mm
Hub Ratio, h/h0 = HL / H0	1.347
Thickness Ratio, g1/g0 = (g1Cor/goCor)	6.108

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.580
Factor V per 4.16.5	0.022
Factor f per 4.16.5	2.042

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 914.4 / 591.6$$

$$= 1.546$$

Stress factor from Table 4.16.4 [Y]:



$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.546^2 \cdot \log_{10}(1.546))) / (1.546^2 - 1)) / (1.546 - 1)$$

$$= 4.633$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.546^2(1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / ((1.0472 + 1.9448 \cdot 1.546^2) \cdot (1.546 - 1))$$

$$= 1.691$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.546^2(1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / (1.36136(1.546^2 - 1) \cdot (1.546 - 1))$$

$$= 5.091$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.546^2 + 1) / (1.546^2 - 1)$$

$$= 2.440$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 5.091 \cdot 9.0^2 \cdot 72.968 / 0.022$$

$$= 1378140.375 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.58 / 72.968$$

$$= 0.008 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (106.0 \cdot 0.008 + 1) / 1.691 + 106.0^3 / 1378140$$

$$= 1.954$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (2.0416 \cdot 94233768 / 591.6) / (1.9536 \cdot 54.974^2)$$

$$= 55.06 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (2.0416 \cdot 0.27436E+09 / 591.6) / (1.9536 \cdot 54.974^2)$$

$$= 160.32 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.1228 \cdot 94233768 / 591.6) / (1.9536 \cdot 106.0^2)$$

$$= 15.40 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.1228 \cdot 0.27436E+09 / 591.6) / (1.9536 \cdot 106.0^2)$$

$$= 44.83 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.6329 \cdot 94233768 / (106.0^2 \cdot 591.6)) - 2.4399 \cdot 15$$

$$= 28.09 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.6329 \cdot 0.27436E+09 / (106.0^2 \cdot 591.6)) - 2.4399 \cdot 45$$

$$= 81.77 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (55 + \max(15, 28)) / 2$$

$$= 41.57 \text{ MPa}$$



Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRA, STa)) / 2$$

$$= (160 + \max(45, 82)) / 2$$

$$= 121.04 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm1 + Fa + 4Me/G) / Ab$$

$$= 1204972 / 21754.7949$$

$$= 55.39 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (995791 / 21754.7949)$$

$$= 45.78 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	55.06	199.80	160.32	207.00
Radial Flange	15.40	133.20	44.83	138.00
Tangential Flange	28.09	133.20	81.77	138.00
Maximum Average	41.57	133.20	121.04	138.00
Bolting	55.39	172.00	45.78	172.00

Minimum Required Flange Thickness 94.767 mm

Estimated M.A.W.P. (Operating) 4 MPa

Estimated Finished Weight of Flange at given Thk. 370.2 kgm

Estimated Unfinished Weight of Forging at given Thk 613.3 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	403.56	457.15	0.883	403.56	499.99	0.807
Tangential Flange	St	200.76	342.86	0.586	200.76	374.99	0.535
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	393.54	685.72	0.574	393.54	749.99	0.525
Combined	Sr + St	310.83	685.72	0.453	310.83	749.99	0.414
Combined	Sh + Sto	632.13	685.72	0.922	632.13	749.99	0.843
Radial Flange	Sr	110.08	457.15	0.241	110.08	499.99	0.220
Maximum Stress Ratio				1.000			0.914



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\ = \max(1; 9.0327/4.3151) \\ = 2.0933$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ = \max(0.25; 8.5083/4.9044) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 2.2046/1.8595) \\ = 1.1856$$

In this case, the hub ratio $h/h0$, 1.3471 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	422.3 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.161 deg

Gasket Contact Area [Ag]:

$$\begin{aligned} &= \pi/4 (GO.D.^2 - GI.D.^2) \\ &= \pi/4 (680.0^2 - 650.0^2) \\ &= 31337.4 \text{ mm}^2 \end{aligned}$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$\begin{aligned} &= SgT \cdot Ag / (nb \cdot Ab) \\ &= 172.0 \cdot 31337.371 / (24 \cdot 906.45) \\ &= 247.8 \text{ MPa} \end{aligned}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$\begin{aligned} &= \min(Sbsel; Sbmax) \\ &= \min(247.763; 507.5) \\ &= 247.8 \text{ MPa} \end{aligned}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$\begin{aligned} &= \max(Sbsel; Sbmin) \\ &= \max(247.763; 290.0) \\ &= 290.0 \text{ MPa} \end{aligned}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$\begin{aligned} &= \min(Sbsel; Sfmax) \\ &= \min(290.0; 422.277) \\ &= 290.0 \text{ MPa} \end{aligned}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\begin{aligned} &:: Sbsel \geq Sgmin - S(Ag / (nb \cdot Abs)) \\ &:: 290.0 \geq 140.0(31337.371 / (24 \cdot 906.45)) \\ &:: 290.0 \geq 201.7 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\begin{aligned} &:: Sbsel \geq (Sgmin - O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb) \\ &:: 290.0 \geq (97.0 \cdot 31337.371 + \pi/4 \cdot 2.215 \cdot 650.0^2) / (0.7 \cdot 906.45 \cdot 24) \\ &:: 290.0 \geq 247.9 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\begin{aligned} &:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs)) \\ &:: 290.0 \leq 380.0(31337.371 / (24 \cdot 906.45)) \\ &:: 290.0 \leq 547.4 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\begin{aligned} &:: Sbsel \leq Sfmax(\theta_{tagmax} / \theta_{tafmax}) \\ &:: 290.0 \leq 422.277(1.0 / 0.161) \\ &:: 290.0 \leq 2615.4 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$\begin{aligned} &= Sbsel \cdot K \cdot Abs \cdot \phi b \\ &= 290.0 \cdot 0.2 \cdot 906.45 \cdot 38.1 \\ &= 2003714.6 \text{ N-mm} \end{aligned}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$\begin{aligned} &= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki) \\ &= 52.14 \cdot 0.27436E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.654 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3) \\ &= 0.528 \text{ (should be } \leq 1) \end{aligned}$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$\begin{aligned} &= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki) \\ &= 52.14 \cdot 94233768 / 1.0 \cdot 1.0 \cdot 0.022 / (1.654 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3) \\ &= 0.196 \text{ (should be } \leq 1) \end{aligned}$$



Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.27436E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.954 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.447$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 94233768 / 1.0 \cdot 1.0 \cdot 0.022 / (1.954 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.166$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.714, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	44.88	207.00	128.11	207.00
Radial Flange	16.92	138.00	48.29	138.00
Tangential Flange	25.72	138.00	73.44	138.00
Maximum Average	35.30	138.00	100.77	138.00
Bolting	55.39	172.00	45.78	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 81.864 mm

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Warnings for Flange : FLANGE-5 :

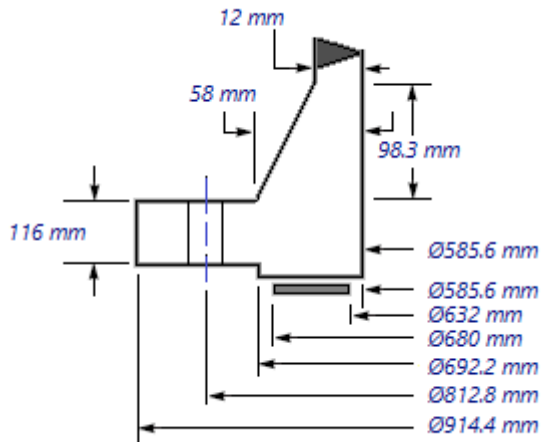
Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: FLANGE-5 :

Body flange-5

Flange Type	Integral Weld Neck		
Design Pressure	P	2.21	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	585.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	116.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	57.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	585.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	632.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		31880.00	N
Bending Moment on Flange		15933200.00	N-mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.21 \cdot (585.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.21) + C_a$$

$$= 7.9680 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133.2 \cdot 1.0 \cdot 9.0) / (295.8 + 0.6 \cdot 9.0)$$

$$= 3.980 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (292.8 + 0.6 \cdot 12.0)$$

$$= 5.520 \text{ MPa}$$

Corroded Flange ID,	$B_{cor} = B + 2 \cdot c_i$	591.600 mm
Corroded Large Hub,	$g1_{cor} = g1 - c_i$	54.974 mm
Corroded Small Hub,	$g0_{cor} = g0 - c_i$	9.000 mm
Code R Dimension,	$R = ((C - B_{cor}) / 2) - g1_{cor}$	55.626 mm
Gasket Contact Width,	$N = (G_o - G_i) / 2$	24.000 mm
Basic Gasket Width,	$b_o = N / 2$	12.000 mm
Effective Gasket Width,	$b = C_b \cdot \sqrt{b_o}$	8.729 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	662.542 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 662.5415^2 \cdot 2.215$$

$$= 763508.000 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.7293 \cdot \pi \cdot 662.5415 \cdot 3.0 \cdot 2.21$$

$$= 241428.969 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 591.6^2 \cdot 2.2148$$

$$= 608756.688 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 763508 - 608757$$

$$= 154751.344 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:



$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(763508 + 241429 + 0, 0)$$

$$= 1004937.000 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.7293 \cdot \pi \cdot 662.542 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1252673.500 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((1004937 + 31880 + 4 \cdot 15933200 / 662.54) / 172, 1252674 / 172)$$

$$= 7283.736 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 116.0 / (3.0 + 0.5)$$

$$= 275.057 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 812.8 \cdot \sin(\pi / 24)$$

$$= 106.092 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 116.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	7283.736	21754.795	
Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	275.057

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi (680.0 + 632.0))$$

$$= 13.166 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 1004937 - 763508$$

$$= 241428.95 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 662.5415) / 2$$

$$= 75.1293 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((812.8 - 591.6) / 2 + 75.1293)$$

$$= 92.8646 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (812.8 - 591.6 - 54.974) / 2$$

$$= 83.1130 \text{ mm}$$



The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 161.4, Bb = 116.0, Cc = 98.298 \text{ and } Ddg = 31.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 161.4 \cdot 116.0^3 \cdot (1/3 - 0.210 \cdot (116.0/161.4) \cdot (1 - 1/12 \cdot (116.0/161.4)^4))$$

$$= 46798196.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 31.987^3 \cdot (1/3 - 0.105 \cdot (31.987/98.3) \cdot (1 - 1/192 \cdot (31.987/98.298)^4))$$

$$= 962452.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 2855.804 + 58.732$$

$$= 47760648.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \cdot (1/3 - 0.21 \cdot (t/AR) \cdot (1 - 1/12 \cdot (t/AR)^4))$$

$$= 161.4 \cdot 116.0^3 \cdot (1/3 - 0.21 \cdot (116.0/161.4) \cdot (1 - 1/12 \cdot (116.0/161.4)^4))$$

$$= 46798196.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(47760648, 46798196)$$

$$= 47760648.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 2.269 \cdot 9.0^2 \cdot 72.968 \cdot 591.6 / 0.022$$

$$= 31758224.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	608757.	83.1130	1.0000	50616116.
Face Pressure, Mt	154751.	92.8646	1.0000	14376753.
Gasket Load, Mg	241429.	75.1293	1.0000	18145732.
Gasket Seating, Matm	3741439.	75.1293	1.0000	281205504.
Moment Load, Moe	...	...	...	7844060.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 15933200 [31758224 / (0.384 \cdot 47760648 + 31758224)] \cdot (83.1 / (812.8 - 2 \cdot 83.1)) +$$

$$31880 \cdot 83.113$$

$$= 7844059.500 \text{ N-mm}$$

Total Moment for Operation, Mo	90982664. N-mm
Total Moment for Gasket seating, Mg	281205504. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	72.968 mm
Hub Ratio, h/h0 = HL / H0	1.347
Thickness Ratio, g1/g0 = (g1Cor/goCor)	6.108

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.580
Factor V per 4.16.5	0.022
Factor f per 4.16.5	2.042

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 914.4 / 591.6$$

$$= 1.546$$

Stress factor from Table 4.16.4 [Y]:



$$\begin{aligned} &= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1) \\ &= (0.66845 + (5.7169(1.546^2 \cdot \log_{10}(1.546))) / (1.546^2 - 1)) / (1.546 - 1) \\ &= 4.633 \end{aligned}$$

Stress factor from Table 4.16.4 [T]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1)) \\ &= (1.546^2(1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / ((1.0472 + 1.9448 \cdot 1.546^2) \cdot (1.546 - 1)) \\ &= 1.691 \end{aligned}$$

Stress factor from Table 4.16.4 [U]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1)) \\ &= (1.546^2(1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / (1.36136(1.546^2 - 1) \cdot (1.546 - 1)) \\ &= 5.091 \end{aligned}$$

Stress factor from Table 4.16.4 [Z]:

$$\begin{aligned} &= (K^2 + 1) / (K^2 - 1) \\ &= (1.546^2 + 1) / (1.546^2 - 1) \\ &= 2.440 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 5.091 \cdot 9.0^2 \cdot 72.968 / 0.022 \\ &= 1378140.375 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.58 / 72.968 \\ &= 0.008 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3 / d \\ &= (116.0 \cdot 0.008 + 1) / 1.691 + 116.0^3 / 1378140 \\ &= 2.269 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (2.0416 \cdot 90982656 / 591.6) / (2.269 \cdot 54.974^2) \\ &= 45.77 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (2.0416 \cdot 0.28121\text{E}+09 / 591.6) / (2.269 \cdot 54.974^2) \\ &= 141.48 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.2287 \cdot 90982656 / 591.6) / (2.269 \cdot 116.0^2) \\ &= 11.22 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.2287 \cdot 0.28121\text{E}+09 / 591.6) / (2.269 \cdot 116.0^2) \\ &= 34.69 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (4.6329 \cdot 90982656 / (116.0^2 \cdot 591.6)) - 2.4399 \cdot 11 \\ &= 25.55 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (4.6329 \cdot 0.28121\text{E}+09 / (116.0^2 \cdot 591.6)) - 2.4399 \cdot 35 \\ &= 78.97 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (46 + \max(11, 26)) / 2 \\ &= 35.66 \text{ MPa} \end{aligned}$$



Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (141 + \max(35, 79)) / 2$
 $= 110.22 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (Wm1 + Fa + 4Me/G) / Ab$
 $= 1132973 / 21754.7949$
 $= 52.08 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (1252674 / 21754.7949)$
 $= 57.59 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	45.77	199.80	141.48	207.00
Radial Flange	11.22	133.20	34.69	138.00
Tangential Flange	25.55	133.20	78.97	138.00
Maximum Average	35.66	133.20	110.22	138.00
Bolting	52.08	172.00	57.59	172.00

Minimum Required Flange Thickness 96.901 mm

Estimated M.A.W.P. (Operating) 4 MPa

Estimated Finished Weight of Flange at given Thk. 400.2 kgm

Estimated Unfinished Weight of Forging at given Thk 643.4 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Seating		stress ratio
					Actual	Allowed	
Longitudinal Hub	Sh	403.56	457.15	0.883	403.56	499.99	0.807
Tangential Flange	St	219.71	342.86	0.641	219.71	374.99	0.586
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	412.50	685.72	0.602	412.50	749.99	0.550
Combined	Sr + St	316.21	685.72	0.461	316.21	749.99	0.422
Combined	Sh + Sto	632.13	685.72	0.922	632.13	749.99	0.843
Radial Flange	Sr	96.50	457.15	0.211	96.50	499.99	0.193
Maximum Stress Ratio				1.000			0.914



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\ = \max(1; 9.0327/4.3151) \\ = 2.0933$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ = \max(0.25; 8.5083/4.9044) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 2.2046/1.8595) \\ = 1.1856$$

In this case, the hub ratio $h/h0$, 1.3471 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	478.5 MPa
Target assembly gasket stress	SgT	208.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.161 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 632.0^2)$$

$$= 49461.3 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 49461.254 / (24 \cdot 906.45)$$

$$= 472.9 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(472.905; 507.5)$$

$$= 472.9 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(472.905; 290.0)$$

$$= 472.9 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(472.905; 478.511)$$

$$= 472.9 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin - S(Ag / (nb \cdot Abs))$$

$$:: 472.905 \geq 140.0(49461.254 / (24 \cdot 906.45))$$

$$:: 472.9 \geq 318.3 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin - O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 472.905 \geq (97.0 \cdot 49461.254 + \pi/4 \cdot 2.215 \cdot 632.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 472.9 \geq 360.7 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 472.905 \leq 380.0(49461.254 / (24 \cdot 906.45))$$

$$:: 472.9 \leq 864.0 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\theta_{tagmax} / \theta_{tafmax})$$

$$:: 472.905 \leq 478.511(1.0/0.161)$$

$$:: 472.9 \leq 2963.7 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 472.905 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 3267468.0 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.707 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$$

$$= 0.525 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 90982656 / 1.0 \cdot 1.0 \cdot 0.022 / (1.707 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$$

$$= 0.184 \text{ (should be } \leq 1)$$



Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (2.269 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
= 0.395 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 90982656 / 1.0 \cdot 1.0 \cdot 0.022 / (2.269 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
= 0.138 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.714, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	37.59	207.00	114.23	207.00
Radial Flange	12.41	138.00	37.70	138.00
Tangential Flange	23.54	138.00	71.52	138.00
Maximum Average	30.56	138.00	92.88	138.00
Bolting	52.08	172.00	57.59	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 83.109 mm

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Warnings for Flange : FLANGE-6 :

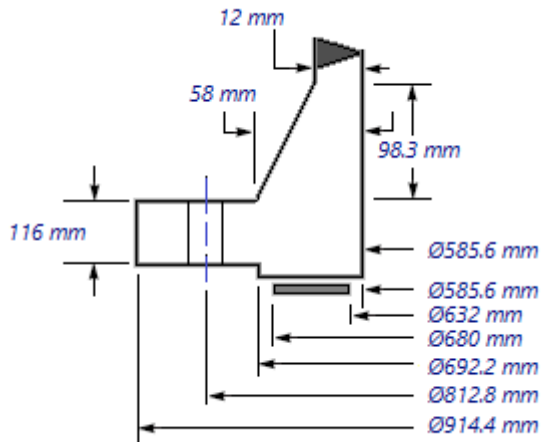
Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: FLANGE-6 :

Body flange-6

Flange Type	Integral Weld Neck		
Design Pressure	P	2.21	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	585.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	116.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	57.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	585.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	632.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		27490.00	N
Bending Moment on Flange		15107900.00	N-mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.21 \cdot (585.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.21) + C_a$$

$$= 7.9680 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133.2 \cdot 1.0 \cdot 9.0) / (295.8 + 0.6 \cdot 9.0)$$

$$= 3.980 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (292.8 + 0.6 \cdot 12.0)$$

$$= 5.520 \text{ MPa}$$

Corroded Flange ID,	$B_{cor} = B + 2 \cdot c_i$	591.600 mm
Corroded Large Hub,	$g1_{cor} = g1 - c_i$	54.974 mm
Corroded Small Hub,	$g0_{cor} = g0 - c_i$	9.000 mm
Code R Dimension,	$R = ((C - B_{cor}) / 2) - g1_{cor}$	55.626 mm
Gasket Contact Width,	$N = (G_o - G_i) / 2$	24.000 mm
Basic Gasket Width,	$b_o = N / 2$	12.000 mm
Effective Gasket Width,	$b = C_b \cdot \sqrt{b_o}$	8.729 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	662.542 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 662.5415^2 \cdot 2.215$$

$$= 763508.000 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.7293 \cdot \pi \cdot 662.5415 \cdot 3.0 \cdot 2.21$$

$$= 241428.969 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 591.6^2 \cdot 2.2148$$

$$= 608756.688 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 763508 - 608757$$

$$= 154751.344 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:



$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(763508 + 241429 + 0, 0)$$

$$= 1004937.000 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.7293 \cdot \pi \cdot 662.542 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1252673.500 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((1004937 + 27490 + 4 \cdot 15107900 / 662.54) / 172, 1252674 / 172)$$

$$= 7283.736 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 116.0 / (3.0 + 0.5)$$

$$= 275.057 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 812.8 \cdot \sin(\pi / 24)$$

$$= 106.092 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 116.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	7283.736	21754.795	
Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	275.057

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi (680.0 + 632.0))$$

$$= 13.166 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 1004937 - 763508$$

$$= 241428.95 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 662.5415) / 2$$

$$= 75.1293 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((812.8 - 591.6) / 2 + 75.1293)$$

$$= 92.8646 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (812.8 - 591.6 - 54.974) / 2$$

$$= 83.1130 \text{ mm}$$



The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 161.4, Bb = 116.0, Cc = 98.298 \text{ and } Ddg = 31.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 161.4 \cdot 116.0^3 \cdot (1/3 - 0.210 \cdot (116.0/161.4) \cdot (1 - 1/12 \cdot (116.0/161.4)^4))$$

$$= 46798196.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 31.987^3 \cdot (1/3 - 0.105 \cdot (31.987/98.3) \cdot (1 - 1/192 \cdot (31.987/98.298)^4))$$

$$= 962452.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 2855.804 + 58.732$$

$$= 47760648.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \cdot (1/3 - 0.21 \cdot (t/AR) \cdot (1 - 1/12 \cdot (t/AR)^4))$$

$$= 161.4 \cdot 116.0^3 \cdot (1/3 - 0.21 \cdot (116.0/161.4) \cdot (1 - 1/12 \cdot (116.0/161.4)^4))$$

$$= 46798196.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(47760648, 46798196)$$

$$= 47760648.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 2.269 \cdot 9.0^2 \cdot 72.968 \cdot 591.6 / 0.022$$

$$= 31758224.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	608757.	83.1130	1.0000	50616116.
Face Pressure, Mt	154751.	92.8646	1.0000	14376753.
Gasket Load, Mg	241429.	75.1293	1.0000	18145732.
Gasket Seating, Matm	3741439.	75.1293	1.0000	281205504.
Moment Load, Moe	...	...	...	7210042.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 15107900 [31758224 / (0.384 \cdot 47760648 + 31758224)] \cdot (83.1 / (812.8 - 2 \cdot 83.1)) +$$

$$27490 \cdot 83.113$$

$$= 7210042.500 \text{ N-mm}$$

Total Moment for Operation, Mo	90348648. N-mm
Total Moment for Gasket seating, Mg	281205504. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	72.968 mm
Hub Ratio, h/h0 = HL / H0	1.347
Thickness Ratio, g1/g0 = (g1Cor/goCor)	6.108

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.580
Factor V per 4.16.5	0.022
Factor f per 4.16.5	2.042

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 914.4 / 591.6$$

$$= 1.546$$

Stress factor from Table 4.16.4 [Y]:



$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.546^2 \cdot \log_{10}(1.546))) / (1.546^2 - 1)) / (1.546 - 1)$$

$$= 4.633$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.546^2(1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / ((1.0472 + 1.9448 \cdot 1.546^2) \cdot (1.546 - 1))$$

$$= 1.691$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.546^2(1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / (1.36136(1.546^2 - 1) \cdot (1.546 - 1))$$

$$= 5.091$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.546^2 + 1) / (1.546^2 - 1)$$

$$= 2.440$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 5.091 \cdot 9.0^2 \cdot 72.968 / 0.022$$

$$= 1378140.375 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.58 / 72.968$$

$$= 0.008 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (116.0 \cdot 0.008 + 1) / 1.691 + 116.0^3 / 1378140$$

$$= 2.269$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (2.0416 \cdot 90348648 / 591.6) / (2.269 \cdot 54.974^2)$$

$$= 45.46 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (2.0416 \cdot 0.28121\text{E}+09 / 591.6) / (2.269 \cdot 54.974^2)$$

$$= 141.48 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.2287 \cdot 90348648 / 591.6) / (2.269 \cdot 116.0^2)$$

$$= 11.14 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.2287 \cdot 0.28121\text{E}+09 / 591.6) / (2.269 \cdot 116.0^2)$$

$$= 34.69 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.6329 \cdot 90348648 / (116.0^2 \cdot 591.6)) - 2.4399 \cdot 11$$

$$= 25.37 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.6329 \cdot 0.28121\text{E}+09 / (116.0^2 \cdot 591.6)) - 2.4399 \cdot 35$$

$$= 78.97 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (45 + \max(11, 25)) / 2$$

$$= 35.41 \text{ MPa}$$



Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (141 + \max(35, 79)) / 2$$

$$= 110.22 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm1 + Fa + 4Me/G) / Ab$$

$$= 1123602 / 21754.7949$$

$$= 51.65 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (1252674 / 21754.7949)$$

$$= 57.59 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	45.46	199.80	141.48	207.00
Radial Flange	11.14	133.20	34.69	138.00
Tangential Flange	25.37	133.20	78.97	138.00
Maximum Average	35.41	133.20	110.22	138.00
Bolting	51.65	172.00	57.59	172.00

Minimum Required Flange Thickness 96.901 mm

Estimated M.A.W.P. (Operating) 4 MPa

Estimated Finished Weight of Flange at given Thk. 400.2 kgm

Estimated Unfinished Weight of Forging at given Thk 643.4 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	403.56	457.15	0.883	403.56	499.99	0.807
Tangential Flange	St	219.71	342.86	0.641	219.71	374.99	0.586
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	412.50	685.72	0.602	412.50	749.99	0.550
Combined	Sr + St	316.21	685.72	0.461	316.21	749.99	0.422
Combined	Sh + Sto	632.13	685.72	0.922	632.13	749.99	0.843
Radial Flange	Sr	96.50	457.15	0.211	96.50	499.99	0.193
Maximum Stress Ratio				1.000			0.914



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$\begin{aligned} &= \max(1; F1num / F1den) \\ &= \max(1; 9.0327/4.3151) \\ &= 2.0933 \end{aligned}$$

Hub Stress Correction Factor [f2]:

$$\begin{aligned} &= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ &= \max(0.25; 8.5083/4.9044) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 2.2046/1.8595) \\ &= 1.1856 \end{aligned}$$

In this case, the hub ratio $h/h0$, 1.3471 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	478.5 MPa
Target assembly gasket stress	SgT	208.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.161 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 632.0^2)$$

$$= 49461.3 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 49461.254 / (24 \cdot 906.45)$$

$$= 472.9 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(472.905; 507.5)$$

$$= 472.9 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(472.905; 290.0)$$

$$= 472.9 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(472.905; 478.511)$$

$$= 472.9 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin - S(Ag / (nb \cdot Abs))$$

$$:: 472.905 \geq 140.0(49461.254 / (24 \cdot 906.45))$$

$$:: 472.9 \geq 318.3 \text{ MPa Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin - O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 472.905 \geq (97.0 \cdot 49461.254 + \pi/4 \cdot 2.215 \cdot 632.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 472.9 \geq 360.7 \text{ MPa Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 472.905 \leq 380.0(49461.254 / (24 \cdot 906.45))$$

$$:: 472.9 \leq 864.0 \text{ MPa Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetamax} / \text{thetafmax})$$

$$:: 472.905 \leq 478.511(1.0/0.161)$$

$$:: 472.9 \leq 2963.7 \text{ MPa Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 472.905 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 3267468.0 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.707 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$$

$$= 0.525 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 90348648 / 1.0 \cdot 1.0 \cdot 0.022 / (1.707 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$$

$$= 0.182 \text{ (should be } \leq 1)$$



Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1 \cdot 0.022 / (2.269 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.395$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 90348648 / 1.0 \cdot 1 \cdot 0.022 / (2.269 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.137$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.714, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	37.33	207.00	114.23	207.00
Radial Flange	12.32	138.00	37.70	138.00
Tangential Flange	23.37	138.00	71.52	138.00
Maximum Average	30.35	138.00	92.88	138.00
Bolting	51.65	172.00	57.59	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 83.109 mm

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Warnings for Flange : FLANGE-7 :

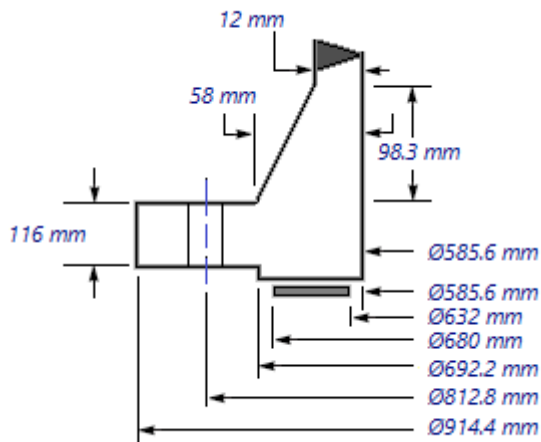
Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: FLANGE-7 :

Body flange-7

Flange Type	Integral Weld Neck		
Design Pressure	P	2.21	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	585.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	116.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	57.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	585.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	632.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		16940.00	N
Bending Moment on Flange		2682970.00	N-mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.21 \cdot (585.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.21) + C_a$$

$$= 7.9680 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133.2 \cdot 1.0 \cdot 9.0) / (295.8 + 0.6 \cdot 9.0)$$

$$= 3.980 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (292.8 + 0.6 \cdot 12.0)$$

$$= 5.520 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	591.600 mm
Corroded Large Hub,	g1Cor = g1-ci	54.974 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	24.000 mm
Basic Gasket Width,	bo = N/2	12.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.729 mm
Gasket Reaction Diameter,	G = Go - 2 * b	662.542 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 662.5415^2 \cdot 2.215$$

$$= 763508.000 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.7293 \cdot \pi \cdot 662.5415 \cdot 3.0 \cdot 2.21$$

$$= 241428.969 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 591.6^2 \cdot 2.2148$$

$$= 608756.688 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 763508 - 608757$$

$$= 154751.344 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_o,1 + W_o,2 + H'p, 0) \\
 &= \max(763508 + 241429 + 0, 0) \\
 &= 1004937.000 \text{ N}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.95 \cdot 8.7293 \cdot \pi \cdot 662.542 + 0.0 \cdot 0.0 \cdot 0.0 \\
 &= 1252673.500 \text{ N}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((1004937 + 16940 + 4 \cdot 2682970 / 662.54) / 172, 1252674 / 172) \\
 &= 7283.736 \text{ mm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 38.1 + 6 \cdot 116.0 / (3.0 + 0.5) \\
 &= 275.057 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 812.8 \cdot \sin(\pi / 24) \\
 &= 106.092 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ Bs / (2a + t) }, 1) \\
 &= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 116.0) }, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	7283.736	21754.795	
Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	275.057

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 21754.795 \cdot 172.0 / (68.95 \cdot \pi (680.0 + 632.0)) \\
 &= 13.166 \text{ mm}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a \cdot A_b \\
 &= 172.0 \cdot 21754.7949 \\
 &= 3741439.00 \text{ N}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o - H \\
 &= 1004937 - 763508 \\
 &= 241428.95 \text{ N}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (812.8 - 662.5415) / 2 \\
 &= 75.1293 \text{ mm}
 \end{aligned}$$

Distance to Face Pressure Reaction [ht]:

$$\begin{aligned}
 &= 0.5 ((C - B) / 2 + h_g) \\
 &= 0.5 ((812.8 - 591.6) / 2 + 75.1293) \\
 &= 92.8646 \text{ mm}
 \end{aligned}$$

Distance to End Pressure Reaction [hd]:

$$\begin{aligned}
 &= (C - B - g_1) / 2 \\
 &= (812.8 - 591.6 - 54.974) / 2 \\
 &= 83.1130 \text{ mm}
 \end{aligned}$$



The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 161.4, Bb = 116.0, Cc = 98.298 \text{ and } Ddg = 31.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 161.4 \cdot 116.0^3 \cdot (1/3 - 0.210 \cdot (116.0/161.4) \cdot (1 - 1/12 \cdot (116.0/161.4)^4))$$

$$= 46798196.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 31.987^3 \cdot (1/3 - 0.105 \cdot (31.987/98.3) \cdot (1 - 1/192 \cdot (31.987/98.298)^4))$$

$$= 962452.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 2855.804 + 58.732$$

$$= 47760648.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \cdot (1/3 - 0.21 \cdot (t/AR) \cdot (1 - 1/12 \cdot (t/AR)^4))$$

$$= 161.4 \cdot 116.0^3 \cdot (1/3 - 0.21 \cdot (116.0/161.4) \cdot (1 - 1/12 \cdot (116.0/161.4)^4))$$

$$= 46798196.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(47760648, 46798196)$$

$$= 47760648.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 2.269 \cdot 9.0^2 \cdot 72.968 \cdot 591.6 / 0.022$$

$$= 31758224.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	608757.	83.1130	1.0000	50616116.
Face Pressure, Mt	154751.	92.8646	1.0000	14376753.
Gasket Load, Mg	241429.	75.1293	1.0000	18145732.
Gasket Seating, Matm	3741439.	75.1293	1.0000	281205504.
Moment Load, Moe	...	...	...	2283005.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 2682970 [31758224 / (0.384 \cdot 47760648 + 31758224)] \cdot (83.1 / (812.8 - 2 \cdot 83.1)) + 16940 \cdot 83.113$$

$$= 2283005.000 \text{ N-mm}$$

Total Moment for Operation, Mo	85421608. N-mm
Total Moment for Gasket seating, Mg	281205504. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	72.968 mm
Hub Ratio, h/h0 = HL / H0	1.347
Thickness Ratio, g1/g0 = (g1Cor/goCor)	6.108

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.580
Factor V per 4.16.5	0.022
Factor f per 4.16.5	2.042

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 914.4 / 591.6$$

$$= 1.546$$

Stress factor from Table 4.16.4 [Y]:



$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.546^2 \cdot \log_{10}(1.546))) / (1.546^2 - 1)) / (1.546 - 1)$$

$$= 4.633$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.546^2(1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / ((1.0472 + 1.9448 \cdot 1.546^2) \cdot (1.546 - 1))$$

$$= 1.691$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.546^2(1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / (1.36136(1.546^2 - 1) \cdot (1.546 - 1))$$

$$= 5.091$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.546^2 + 1) / (1.546^2 - 1)$$

$$= 2.440$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 5.091 \cdot 9.0^2 \cdot 72.968 / 0.022$$

$$= 1378140.375 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.58 / 72.968$$

$$= 0.008 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (116.0 \cdot 0.008 + 1) / 1.691 + 116.0^3 / 1378140$$

$$= 2.269$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (2.0416 \cdot 85421608 / 591.6) / (2.269 \cdot 54.974^2)$$

$$= 42.98 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (2.0416 \cdot 0.28121E+09 / 591.6) / (2.269 \cdot 54.974^2)$$

$$= 141.48 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.2287 \cdot 85421608 / 591.6) / (2.269 \cdot 116.0^2)$$

$$= 10.54 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.2287 \cdot 0.28121E+09 / 591.6) / (2.269 \cdot 116.0^2)$$

$$= 34.69 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.6329 \cdot 85421608 / (116.0^2 \cdot 591.6)) - 2.4399 \cdot 11$$

$$= 23.99 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.6329 \cdot 0.28121E+09 / (116.0^2 \cdot 591.6)) - 2.4399 \cdot 35$$

$$= 78.97 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (43 + \max(11, 24)) / 2$$

$$= 33.48 \text{ MPa}$$



Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (141 + \max(35, 79)) / 2$$

$$= 110.22 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm1 + Fa + 4Me/G) / Ab$$

$$= 1038068 / 21754.7949$$

$$= 47.72 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (1252674 / 21754.7949)$$

$$= 57.59 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	42.98	199.80	141.48	207.00
Radial Flange	10.54	133.20	34.69	138.00
Tangential Flange	23.99	133.20	78.97	138.00
Maximum Average	33.48	133.20	110.22	138.00
Bolting	47.72	172.00	57.59	172.00

Minimum Required Flange Thickness 96.901 mm

Estimated M.A.W.P. (Operating) 4 MPa

Estimated Finished Weight of Flange at given Thk. 400.2 kgm

Estimated Unfinished Weight of Forging at given Thk 643.4 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	403.56	457.15	0.883	403.56	499.99	0.807
Tangential Flange	St	219.71	342.86	0.641	219.71	374.99	0.586
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	412.50	685.72	0.602	412.50	749.99	0.550
Combined	Sr + St	316.21	685.72	0.461	316.21	749.99	0.422
Combined	Sh + Sto	632.13	685.72	0.922	632.13	749.99	0.843
Radial Flange	Sr	96.50	457.15	0.211	96.50	499.99	0.193
Maximum Stress Ratio				1.000			0.914



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\ = \max(1; 9.0327/4.3151) \\ = 2.0933$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ = \max(0.25; 8.5083/4.9044) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 2.2046/1.8595) \\ = 1.1856$$

In this case, the hub ratio $h/h0$, 1.3471 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	478.5 MPa
Target assembly gasket stress	SgT	208.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.161 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 632.0^2)$$

$$= 49461.3 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 49461.254 / (24 \cdot 906.45)$$

$$= 472.9 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(472.905; 507.5)$$

$$= 472.9 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(472.905; 290.0)$$

$$= 472.9 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(472.905; 478.511)$$

$$= 472.9 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin - S(Ag / (nb \cdot Abs))$$

$$:: 472.905 \geq 140.0(49461.254 / (24 \cdot 906.45))$$

$$:: 472.9 \geq 318.3 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin - O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 472.905 \geq (97.0 \cdot 49461.254 + \pi/4 \cdot 2.215 \cdot 632.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 472.9 \geq 360.7 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 472.905 \leq 380.0(49461.254 / (24 \cdot 906.45))$$

$$:: 472.9 \leq 864.0 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetamax} / \text{thetafmax})$$

$$:: 472.905 \leq 478.511(1.0/0.161)$$

$$:: 472.9 \leq 2963.7 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 472.905 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 3267468.0 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.707 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$$

$$= 0.525 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 85421608 / 1.0 \cdot 1.0 \cdot 0.022 / (1.707 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$$

$$= 0.172 \text{ (should be } \leq 1)$$



Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.28121E+09 / 1.0 \cdot 1 \cdot 0.022 / (2.269 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.395$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 85421608 / 1.0 \cdot 1 \cdot 0.022 / (2.269 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.129$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.714, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	35.24	207.00	114.23	207.00
Radial Flange	11.63	138.00	37.70	138.00
Tangential Flange	22.07	138.00	71.52	138.00
Maximum Average	28.65	138.00	92.88	138.00
Bolting	47.72	172.00	57.59	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 83.109 mm

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Warnings for Flange : FLANGE-8 :

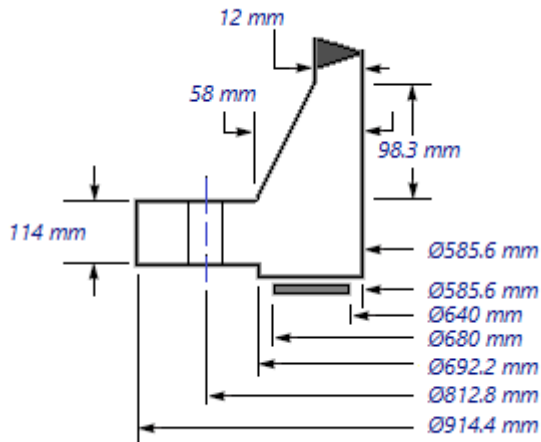
Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: FLANGE-8 :

Body flange-8

Flange Type	Integral Weld Neck		
Design Pressure	P	2.21	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	585.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	114.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	57.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	585.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	640.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		12550.00	N
Bending Moment on Flange		2153270.00	N-mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.21 \cdot (585.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.21) + C_a$$

$$= 7.9680 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133.2 \cdot 1.0 \cdot 9.0) / (295.8 + 0.6 \cdot 9.0)$$

$$= 3.980 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (292.8 + 0.6 \cdot 12.0)$$

$$= 5.520 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	591.600 mm
Corroded Large Hub,	g1Cor = g1-ci	54.974 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	20.000 mm
Basic Gasket Width,	bo = N/2	10.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.969 mm
Gasket Reaction Diameter,	G = Go - 2 * b	664.063 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 664.0626^2 \cdot 2.215$$

$$= 767018.000 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.9687 \cdot \pi \cdot 664.0626 \cdot 3.0 \cdot 2.21$$

$$= 220899.516 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 591.6^2 \cdot 2.2148$$

$$= 608756.688 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 767018 - 608757$$

$$= 158261.297 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_o,1 + W_o,2 + H'p, 0) \\
 &= \max(767018 + 220900 + 0, 0) \\
 &= 987917.500 \text{ N}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.95 \cdot 7.9687 \cdot \pi \cdot 664.063 + 0.0 \cdot 0.0 \cdot 0.0 \\
 &= 1146154.750 \text{ N}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((987918 + 12550 + 4 \cdot 2153270 / 664.06) / 172, 1146155 / 172) \\
 &= 6664.377 \text{ mm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 38.1 + 6 \cdot 114.0 / (3.0 + 0.5) \\
 &= 271.629 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 812.8 \cdot \sin(\pi / 24) \\
 &= 106.092 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ Bs / (2a + t) }, 1) \\
 &= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 114.0) }, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	6664.377	21754.795	
Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	271.629

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 21754.795 \cdot 172.0 / (68.95 \cdot \pi (680.0 + 640.0)) \\
 &= 13.086 \text{ mm}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a \cdot A_b \\
 &= 172.0 \cdot 21754.7949 \\
 &= 3741439.00 \text{ N}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o - H \\
 &= 987918 - 767018 \\
 &= 220899.50 \text{ N}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (812.8 - 664.0626) / 2 \\
 &= 74.3687 \text{ mm}
 \end{aligned}$$

Distance to Face Pressure Reaction [ht]:

$$\begin{aligned}
 &= 0.5((C - B) / 2 + h_g) \\
 &= 0.5((812.8 - 591.6) / 2 + 74.3687) \\
 &= 92.4844 \text{ mm}
 \end{aligned}$$

Distance to End Pressure Reaction [hd]:

$$\begin{aligned}
 &= (C - B - g_1) / 2 \\
 &= (812.8 - 591.6 - 54.974) / 2 \\
 &= 83.1130 \text{ mm}
 \end{aligned}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 161.4, Bb = 114.0, Cc = 98.298 \text{ and } Ddg = 31.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 161.4 \cdot 114.0^3 \cdot (1/3 - 0.210 \cdot (114.0/161.4) \cdot (1 - 1/12 \cdot (114.0/161.4)^4))$$

$$= 44974508.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 31.987^3 \cdot (1/3 - 0.105 \cdot (31.987/98.3) \cdot (1 - 1/192 \cdot (31.987/98.298)^4))$$

$$= 962452.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 2744.516 + 58.732$$

$$= 45936960.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \cdot (1/3 - 0.21 \cdot (t/AR) \cdot (1 - 1/12 \cdot (t/AR)^4))$$

$$= 161.4 \cdot 114.0^3 \cdot (1/3 - 0.21 \cdot (114.0/161.4) \cdot (1 - 1/12 \cdot (114.0/161.4)^4))$$

$$= 44974508.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(45936960, 44974508)$$

$$= 45936960.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 2.202 \cdot 9.0^2 \cdot 72.968 \cdot 591.6 / 0.022$$

$$= 30820798.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	608757.	83.1130	1.0000	50616116.
Face Pressure, Mt	158261.	92.4844	1.0000	14642629.
Gasket Load, Mg	220900.	74.3687	1.0000	16434668.
Gasket Seating, Matm	3741439.	74.3687	1.0000	278358752.
Moment Load, Moe	...	...	...	1747640.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 2153270 [30820798 / (0.384 \cdot 45936960 + 30820798)] \cdot (83.1 / (812.8 - 2 \cdot 83.1)) + 12550 \cdot 83.113$$

$$= 1747640.000 \text{ N-mm}$$

Total Moment for Operation, Mo	83441056. N-mm
Total Moment for Gasket seating, Mg	278358752. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	72.968 mm
Hub Ratio, h/h0 = HL / H0	1.347
Thickness Ratio, g1/g0 = (g1Cor/goCor)	6.108

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.580
Factor V per 4.16.5	0.022
Factor f per 4.16.5	2.042

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 914.4 / 591.6$$

$$= 1.546$$

Stress factor from Table 4.16.4 [Y]:



$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.546^2 \cdot \log_{10}(1.546))) / (1.546^2 - 1)) / (1.546 - 1)$$

$$= 4.633$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.546^2(1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / ((1.0472 + 1.9448 \cdot 1.546^2) \cdot (1.546 - 1))$$

$$= 1.691$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.546^2(1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / (1.36136(1.546^2 - 1) \cdot (1.546 - 1))$$

$$= 5.091$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.546^2 + 1) / (1.546^2 - 1)$$

$$= 2.440$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 5.091 \cdot 9.0^2 \cdot 72.968 / 0.022$$

$$= 1378140.375 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.58 / 72.968$$

$$= 0.008 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (114.0 \cdot 0.008 + 1) / 1.691 + 114.0^3 / 1378140$$

$$= 2.202$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (2.0416 \cdot 83441056 / 591.6) / (2.202 \cdot 54.974^2)$$

$$= 43.26 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (2.0416 \cdot 0.27836E+09 / 591.6) / (2.202 \cdot 54.974^2)$$

$$= 144.30 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.2075 \cdot 83441056 / 591.6) / (2.202 \cdot 114.0^2)$$

$$= 10.88 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.2075 \cdot 0.27836E+09 / 591.6) / (2.202 \cdot 114.0^2)$$

$$= 36.28 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.6329 \cdot 83441056 / (114.0^2 \cdot 591.6)) - 2.4399 \cdot 11$$

$$= 23.73 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.6329 \cdot 0.27836E+09 / (114.0^2 \cdot 591.6)) - 2.4399 \cdot 36$$

$$= 79.15 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (43 + \max(11, 24)) / 2$$

$$= 33.49 \text{ MPa}$$



Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRA, STa)) / 2$$

$$= (144 + \max(36, 79)) / 2$$

$$= 111.73 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm1 + Fa + 4Me/G) / Ab$$

$$= 1013433 / 21754.7949$$

$$= 46.59 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (1146155 / 21754.7949)$$

$$= 52.69 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	43.26	199.80	144.30	207.00
Radial Flange	10.88	133.20	36.28	138.00
Tangential Flange	23.73	133.20	79.15	138.00
Maximum Average	33.49	133.20	111.73	138.00
Bolting	46.59	172.00	52.69	172.00

Minimum Required Flange Thickness 96.012 mm

Estimated M.A.W.P. (Operating) 4 MPa

Estimated Finished Weight of Flange at given Thk. 394.2 kgm

Estimated Unfinished Weight of Forging at given Thk 637.4 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Operating		stress ratio	Gasket Seating		stress ratio
		Actual	Allowed		Actual	Allowed	
Longitudinal Hub	Sh	403.56	457.15	0.883	403.56	499.99	0.807
Tangential Flange	St	215.89	342.86	0.630	215.89	374.99	0.576
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	408.68	685.72	0.596	408.68	749.99	0.545
Combined	Sr + St	314.86	685.72	0.459	314.86	749.99	0.420
Combined	Sh + Sto	632.13	685.72	0.922	632.13	749.99	0.843
Radial Flange	Sr	98.97	457.15	0.216	98.97	499.99	0.198
Maximum Stress Ratio				1.000			0.914



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den) \\ = \max(1; 9.0327/4.3151) \\ = 2.0933$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ = \max(0.25; 8.5083/4.9044) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 2.2046/1.8595) \\ = 1.1856$$

In this case, the hub ratio $h/h0$, 1.3471 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	469.1 MPa
Target assembly gasket stress	SgT	208.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.161 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (680.0^2 - 640.0^2)$$

$$= 41469.0 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 41469.047 / (24 \cdot 906.45)$$

$$= 396.5 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(396.49; 507.5)$$

$$= 396.5 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(396.49; 290.0)$$

$$= 396.5 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(396.49; 469.135)$$

$$= 396.5 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin - S(Ag / (nb \cdot Abs))$$

$$:: 396.49 \geq 140.0(41469.047 / (24 \cdot 906.45))$$

$$:: 396.5 \geq 266.9 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin - O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 396.49 \geq (97.0 \cdot 41469.047 + \pi/4 \cdot 2.215 \cdot 640.0^2) / (0.7 \cdot 906.45 \cdot 24)$$

$$:: 396.5 \geq 310.9 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 396.49 \leq 380.0(41469.047 / (24 \cdot 906.45))$$

$$:: 396.5 \leq 724.4 \text{ MPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetamax} / \text{thetafmax})$$

$$:: 396.49 \leq 469.135(1.0/0.161)$$

$$:: 396.5 \leq 2905.6 \text{ MPa} \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 396.49 \cdot 0.2 \cdot 906.45 \cdot 38.1$$

$$= 2739493.5 \text{ N-mm}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 0.27836E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.685 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$$

$$= 0.526 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 83441056 / 1.0 \cdot 1.0 \cdot 0.022 / (1.685 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$$

$$= 0.170 \text{ (should be } \leq 1)$$



Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.27836E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (2.202 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.403$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 83441056 / 1.0 \cdot 1.0 \cdot 0.022 / (2.202 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.130$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.714, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	35.41	207.00	116.28	207.00
Radial Flange	11.99	138.00	39.37	138.00
Tangential Flange	21.79	138.00	71.56	138.00
Maximum Average	28.60	138.00	93.92	138.00
Bolting	46.59	172.00	52.69	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 82.601 mm

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Warnings for Flange : FLANGE-9 :

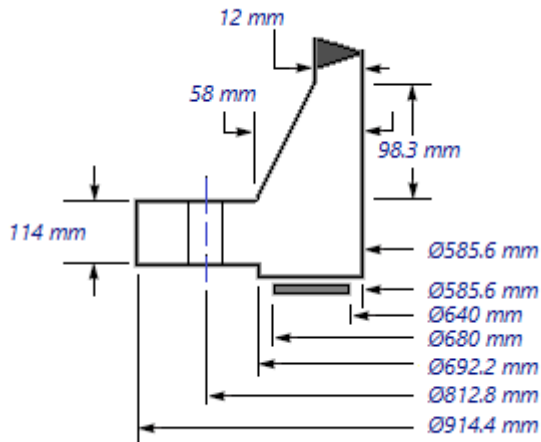
Note: Flange Hub Taper 0.4677 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: FLANGE-9 :

Body flange-9

Flange Type	Integral Weld Neck		
Design Pressure	P	2.21	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	585.600	mm
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	114.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	57.9740	mm
Length of Hub	h	98.2980	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	585.600	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	680.000	mm
Gasket Inside Diameter	Gi	640.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		6770.00	N
Bending Moment on Flange		286001.00	N-mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.21 \cdot (585.6/2 + 3.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 2.21) + C_a$$

$$= 7.9680 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133.2 \cdot 1.0 \cdot 9.0) / (295.8 + 0.6 \cdot 9.0)$$

$$= 3.980 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (292.8 + 0.6 \cdot 12.0)$$

$$= 5.520 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	591.600 mm
Corroded Large Hub,	g1Cor = g1-ci	54.974 mm
Corroded Small Hub,	g0Cor = go-ci	9.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	55.626 mm
Gasket Contact Width,	N = (Go - Gi)/2	20.000 mm
Basic Gasket Width,	bo = N/2	10.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.969 mm
Gasket Reaction Diameter,	G = Go - 2 * b	664.063 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 664.0626^2 \cdot 2.215$$

$$= 767018.000 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.9687 \cdot \pi \cdot 664.0626 \cdot 3.0 \cdot 2.21$$

$$= 220899.516 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 591.6^2 \cdot 2.2148$$

$$= 608756.688 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 767018 - 608757$$

$$= 158261.297 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\
 &= \max(767018 + 220900 + 0, 0) \\
 &= 987917.500 \text{ N}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.95 \cdot 7.9687 \cdot \pi \cdot 664.063 + 0.0 \cdot 0.0 \cdot 0.0 \\
 &= 1146154.750 \text{ N}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((987918 + 6770 + 4 \cdot 286001 / 664.06) / 172, 1146155 / 172) \\
 &= 6664.377 \text{ mm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 38.1 + 6 \cdot 114.0 / (3.0 + 0.5) \\
 &= 271.629 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 812.8 \cdot \sin(\pi / 24) \\
 &= 106.092 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ Bs / (2a + t) }, 1) \\
 &= \max(\sqrt{ 106.092 / (2 \cdot 38.1 + 114.0) }, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	6664.377	21754.795	
Radial Distance between Hub and Bolts:	50.800	55.626	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	271.629

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 21754.795 \cdot 172.0 / (68.95 \cdot \pi (680.0 + 640.0)) \\
 &= 13.086 \text{ mm}
 \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a \cdot A_b \\
 &= 172.0 \cdot 21754.7949 \\
 &= 3741439.00 \text{ N}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o - H \\
 &= 987918 - 767018 \\
 &= 220899.50 \text{ N}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (812.8 - 664.0626) / 2 \\
 &= 74.3687 \text{ mm}
 \end{aligned}$$

Distance to Face Pressure Reaction [ht]:

$$\begin{aligned}
 &= 0.5 ((C - B) / 2 + h_g) \\
 &= 0.5 ((812.8 - 591.6) / 2 + 74.3687) \\
 &= 92.4844 \text{ mm}
 \end{aligned}$$

Distance to End Pressure Reaction [hd]:

$$\begin{aligned}
 &= (C - B - g_1) / 2 \\
 &= (812.8 - 591.6 - 54.974) / 2 \\
 &= 83.1130 \text{ mm}
 \end{aligned}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 161.4, Bb = 114.0, Cc = 98.298 \text{ and } Ddg = 31.987 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 161.4 \cdot 114.0^3 \cdot (1/3 - 0.210 \cdot (114.0/161.4) \cdot (1 - 1/12 \cdot (114.0/161.4)^4))$$

$$= 44974508.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 98.3 \cdot 31.987^3 \cdot (1/3 - 0.105 \cdot (31.987/98.3) \cdot (1 - 1/192 \cdot (31.987/98.298)^4))$$

$$= 962452.125 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 2744.516 + 58.732$$

$$= 45936960.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21 (t/AR) (1 - 1/12 (t/AR)^4))$$

$$= 161.4 \cdot 114.0^3 (1/3 - 0.21 (114.0/161.4) (1 - 1/12 (114.0/161.4)^4))$$

$$= 44974508.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(45936960, 44974508)$$

$$= 45936960.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 2.202 \cdot 9.0^2 \cdot 72.968 \cdot 591.6 / 0.022$$

$$= 30820798.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	608757.	83.1130	1.0000	50616116.
Face Pressure, Mt	158261.	92.4844	1.0000	14642629.
Gasket Load, Mg	220900.	74.3687	1.0000	16434668.
Gasket Seating, Matm	3741439.	74.3687	1.0000	278358752.
Moment Load, Moe	...	...	...	656430.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 286001 [30820798 / (0.384 \cdot 45936960 + 30820798)] \cdot (83.1 / (812.8 - 2 \cdot 83.1)) +$$

$$6770 \cdot 83.113$$

$$= 656429.500 \text{ N-mm}$$

Total Moment for Operation, Mo	82349848. N-mm
Total Moment for Gasket seating, Mg	278358752. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor)	72.968 mm
Hub Ratio, h/h0 = HL / H0	1.347
Thickness Ratio, g1/g0 = (g1Cor/goCor)	6.108

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.580
Factor V per 4.16.5	0.022
Factor f per 4.16.5	2.042

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 914.4 / 591.6$$

$$= 1.546$$

Stress factor from Table 4.16.4 [Y]:



$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.546^2 \cdot \log_{10}(1.546))) / (1.546^2 - 1)) / (1.546 - 1)$$

$$= 4.633$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.546^2(1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / ((1.0472 + 1.9448 \cdot 1.546^2) \cdot (1.546 - 1))$$

$$= 1.691$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.546^2(1 + 8.55246 \cdot \log_{10}(1.546)) - 1) / (1.36136(1.546^2 - 1) \cdot (1.546 - 1))$$

$$= 5.091$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.546^2 + 1) / (1.546^2 - 1)$$

$$= 2.440$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 5.091 \cdot 9.0^2 \cdot 72.968 / 0.022$$

$$= 1378140.375 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.58 / 72.968$$

$$= 0.008 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (114.0 \cdot 0.008 + 1) / 1.691 + 114.0^3 / 1378140$$

$$= 2.202$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (2.0416 \cdot 82349840 / 591.6) / (2.202 \cdot 54.974^2)$$

$$= 42.69 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (2.0416 \cdot 0.27836E+09 / 591.6) / (2.202 \cdot 54.974^2)$$

$$= 144.30 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.2075 \cdot 82349840 / 591.6) / (2.202 \cdot 114.0^2)$$

$$= 10.73 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.2075 \cdot 0.27836E+09 / 591.6) / (2.202 \cdot 114.0^2)$$

$$= 36.28 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.6329 \cdot 82349840 / (114.0^2 \cdot 591.6)) - 2.4399 \cdot 11$$

$$= 23.42 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.6329 \cdot 0.27836E+09 / (114.0^2 \cdot 591.6)) - 2.4399 \cdot 36$$

$$= 79.15 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (43 + \max(11, 23)) / 2$$

$$= 33.05 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (144 + \max(36, 79)) / 2$
 $= 111.73 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (Wm1 + Fa + 4Me/G) / Ab$
 $= 996410 / 21754.7949$
 $= 45.81 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (1146155 / 21754.7949)$
 $= 52.69 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	42.69	199.80	144.30	207.00
Radial Flange	10.73	133.20	36.28	138.00
Tangential Flange	23.42	133.20	79.15	138.00
Maximum Average	33.05	133.20	111.73	138.00
Bolting	45.81	172.00	52.69	172.00

Minimum Required Flange Thickness 96.012 mm

Estimated M.A.W.P. (Operating) 4 MPa
 Estimated Finished Weight of Flange at given Thk. 394.2 kgm
 Estimated Unfinished Weight of Forging at given Thk 637.4 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Operating		stress ratio	Gasket Seating		stress ratio
		Actual	Allowed		Actual	Allowed	
Longitudinal Hub	Sh	403.56	457.15	0.883	403.56	499.99	0.807
Tangential Flange	St	215.89	342.86	0.630	215.89	374.99	0.576
Tangential Flange	Sto	228.57	228.57	1.000	228.57	250.00	0.914
Combined	Sh/f + St	408.68	685.72	0.596	408.68	749.99	0.545
Combined	Sr + St	314.86	685.72	0.459	314.86	749.99	0.420
Combined	Sh + Sto	632.13	685.72	0.922	632.13	749.99	0.843
Radial Flange	Sr	98.97	457.15	0.216	98.97	499.99	0.198
Maximum Stress Ratio				1.000			0.914

Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$\begin{aligned} &= \max(1; F1num / F1den) \\ &= \max(1; 9.0327/4.3151) \\ &= 2.0933 \end{aligned}$$

Hub Stress Correction Factor [f2]:

$$\begin{aligned} &= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3) \\ &= \max(0.25; 8.5083/4.9044) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 2.2046/1.8595) \\ &= 1.1856 \end{aligned}$$

In this case, the hub ratio $h/h0$, 1.3471 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sb_{max}	507.5 MPa
Minimum permissible bolt stress	Sb_{min}	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sf_{max}	469.1 MPa
Target assembly gasket stress	SgT	208.0 MPa
Maximum permissible gasket stress	Sg_{max}	380.0 MPa
Minimum gasket operating stress	Sg_{min-O}	97.0 MPa
Minimum gasket seating stress	Sg_{min-S}	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	$Phig$	0.700
Root Area of a Single Bolt	Abs	906.4498 mm ²
Max. permissible single flange rotation for gasket	$Thetag_{max}$	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	200.0 MPa
Single flange rotation at Sfmax	Thetafmax	0.161 deg

Gasket Contact Area [Ag]:

$$\begin{aligned} &= \pi/4 (GO.D.^2 - GI.D.^2) \\ &= \pi/4 (680.0^2 - 640.0^2) \\ &= 41469.0 \text{ mm}^2 \end{aligned}$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$\begin{aligned} &= SgT \cdot Ag / (nb \cdot Ab) \\ &= 208.0 \cdot 41469.047 / (24 \cdot 906.45) \\ &= 396.5 \text{ MPa} \end{aligned}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$\begin{aligned} &= \min(Sbsel; Sbmax) \\ &= \min(396.49; 507.5) \\ &= 396.5 \text{ MPa} \end{aligned}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$\begin{aligned} &= \max(Sbsel; Sbmin) \\ &= \max(396.49; 290.0) \\ &= 396.5 \text{ MPa} \end{aligned}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$\begin{aligned} &= \min(Sbsel; Sfmax) \\ &= \min(396.49; 469.135) \\ &= 396.5 \text{ MPa} \end{aligned}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\begin{aligned} &:: Sbsel \geq Sgmin - S(Ag / (nb \cdot Abs)) \\ &:: 396.49 \geq 140.0(41469.047 / (24 \cdot 906.45)) \\ &:: 396.5 \geq 266.9 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\begin{aligned} &:: Sbsel \geq (Sgmin - O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb) \\ &:: 396.49 \geq (97.0 \cdot 41469.047 + \pi/4 \cdot 2.215 \cdot 640.0^2) / (0.7 \cdot 906.45 \cdot 24) \\ &:: 396.5 \geq 310.9 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\begin{aligned} &:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs)) \\ &:: 396.49 \leq 380.0(41469.047 / (24 \cdot 906.45)) \\ &:: 396.5 \leq 724.4 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\begin{aligned} &:: Sbsel \leq Sfmax(\theta_{tagmax} / \theta_{tafmax}) \\ &:: 396.49 \leq 469.135(1.0/0.161) \\ &:: 396.5 \leq 2905.6 \text{ MPa} \text{ Check Passes} \end{aligned}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$\begin{aligned} &= Sbsel \cdot K \cdot Abs \cdot \phi_b \\ &= 396.49 \cdot 0.2 \cdot 906.45 \cdot 38.1 \\ &= 2739493.5 \text{ N-mm} \end{aligned}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 88 MPa
Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$\begin{aligned} &= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki) \\ &= 52.14 \cdot 0.27836E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (1.685 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3) \\ &= 0.526 \text{ (should be } \leq 1) \end{aligned}$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$\begin{aligned} &= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki) \\ &= 52.14 \cdot 82349840 / 1.0 \cdot 1.0 \cdot 0.022 / (1.685 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3) \\ &= 0.168 \text{ (should be } \leq 1) \end{aligned}$$



Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.27836E+09 / 1.0 \cdot 1.0 \cdot 0.022 / (2.202 \cdot 201334 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.403$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 82349840 / 1.0 \cdot 1.0 \cdot 0.022 / (2.202 \cdot 185988 \cdot 9.0^2 \cdot 72.968 \cdot 0.3)$
 $= 0.129$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.714, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	34.94	207.00	116.28	207.00
Radial Flange	11.83	138.00	39.37	138.00
Tangential Flange	21.50	138.00	71.56	138.00
Maximum Average	28.22	138.00	93.92	138.00
Bolting	45.81	172.00	52.69	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 82.601 mm

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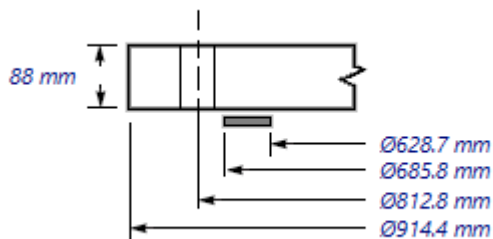


Flange Input Data Values

Description: FLANGE-10 :

Body flange-10

Flange Type		Blind	
Design Pressure	P	2.21	MPa
Design Temperature		270	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	914.400	mm
Flange Thickness	t	88.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	133.20	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Perimeter along the Center of the Bolts	L	2553.487	mm
Diameter of Bolt Circle	C	812.800	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	692.150	mm
Flange Face Inside Diameter	Fid	590.550	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	685.800	mm
Gasket Inside Diameter	Gi	628.650	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	28.575 mm
Basic Gasket Width,	$bo = N/2$	14.287 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	9.525 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	666.750 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 666.75^2 \cdot 2.215$
 $= 773238.625 \text{ N}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 9.525 \cdot \pi \cdot 666.75 \cdot 3.0 \cdot 2.21$$

$$= 265110.281 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(773239 + 265110 + 0, 0)$$

$$= 1038348.875 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 9.525 \cdot \pi \cdot 666.75 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 1375545.625 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((1038349 + 0 + 4 \cdot 0/666.75)/172, 1375546/172)$$

$$= 7998.183 \text{ mm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	7998.183	21754.795	
Radial Distance between Bolts and Edge:	38.100	50.800	
Circ. Spacing between the Bolts:	82.550	106.092	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 21754.795 \cdot 172.0 / (68.95 \cdot \pi (685.8 + 628.65))$$

$$= 13.142 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 21754.7949$$

$$= 3741439.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 1038348.88 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (812.8 - 666.75) / 2$$

$$= 73.0250 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 1038349 \cdot 73.025 \cdot 1.0 / (85.0^2 \cdot 666.75) + 0.3 \cdot 1.0 \cdot 2.21 \cdot 666.75^2 / 85.0^2$$

$$= 70.79 \text{ MPa}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 3741439 \cdot 73.025 \cdot 1.0 / (85.0^2 \cdot 666.75)$$

$$= 107.77 \text{ MPa}$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 1038349 / 21754.7949$$

$$= 47.73 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (1375546 / 21754.7949)$$



= 63.23 MPa

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	70.79	133.20	107.77	138.00
Bolting	47.73	172.00	63.23	172.00

Reqd. Blind Flange Thickness at Center 78.116 mm
 Reqd. Blind Flange Thickness at Gasket 78.116 mm

Estimated M.A.W.P. (Operating) 4 MPa
 Estimated Finished Weight of Flange at given Thk. 447.9 kgm
 Estimated Unfinished Weight of Forging at given Thk 447.9 kgm

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.852, Temperature Reduction per Fig. UCS 66.1 = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -13 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	66.05	138.00	100.55	138.00
Bolting	47.73	172.00	63.23	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 75.133 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-2	...	...	20	3	649	...	1
Skirt -1	...	...	20	3	649	...	2
Bottom head	2.2309	2.2309	22	3	609.6	137.2	3
Shell-1	2.2292	2.2292	16	3	609.6	137.2	4
Body flange -1	2.2148	2.2148	90	3	577.6	133.2	5
Body flange -2	2.2148	2.2148	90	3	577.6	133.2	6
Shell-2	2.2148	2.2148	16	3	609.6	137.2	7
Shell-3	2.2148	2.2148	16	3	609.6	137.2	8
Body flange-3	2.2148	2.2148	98	3	577.6	133.2	9
Body flange-4	2.2148	2.2148	108	3	585.6	133.2	10
Shell-4	2.2148	2.2148	12	3	609.6	137.2	11
Shell-5	2.2148	2.2148	12	3	609.6	137.2	12
Body flange-5	2.2148	2.2148	118	3	585.6	133.2	13
Body flange-6	2.2148	2.2148	118	3	585.6	133.2	14
Shell-6	2.2148	2.2148	12	3	609.6	137.2	15
Shell-7	2.2148	2.2148	12	3	609.6	137.2	16
Body flange-7	2.2148	2.2148	118	3	585.6	133.2	17
Body flange-8	2.2148	2.2148	116	3	585.6	133.2	18
Shell-8	2.2148	2.2148	12	3	609.6	137.2	19
Body flange-9	2.2148	2.2148	116	3	585.6	133.2	20
Body flange-10	2.2148	2.2148	90	3	914.4	133.2	21

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-2	...	...	No Calc	No Calc	20	No Calc	1
Skirt -1	...	...	No Calc	No Calc	20	No Calc	2
Bottom head	2.2148	2.2148	6.72347	8.1016	18	7.94885	3
Shell-1	2.2148	2.2148	5.63697	7.02276	16	8.0504	4
Body flange -1	2.2148	2.2148	5.77974	7.22736	88	81.1276	5
Body flange -2	2.2148	2.2148	5.77974	7.3905	88	81.1276	6
Shell-2	2.2148	2.2148	5.65138	7.02276	16	8.01743	7
Shell-3	2.2148	2.2148	5.65138	7.02276	16	8.01743	8
Body flange-3	2.2148	2.2148	5.77974	7.39933	88	81.1276	9
Body flange-4	2.2148	2.2148	3.9801	5.5199	106	94.7674	10
Shell-4	2.2148	2.2148	3.94248	5.30759	12	8.01743	11
Shell-5	2.2148	2.2148	3.94248	5.30759	12	8.01743	12
Body flange-5	2.2148	2.2148	3.9801	5.5199	116	96.901	13
Body flange-6	2.2148	2.2148	3.9801	5.5199	116	96.901	14
Shell-6	2.2148	2.2148	3.94248	5.30759	12	8.01743	15
Shell-7	2.2148	2.2148	3.94248	5.30759	12	8.01743	16
Body flange-7	2.2148	2.2148	3.9801	5.5199	116	96.901	17
Body flange-8	2.2148	2.2148	3.9801	5.5199	114	96.012	18
Shell-8	2.2148	2.2148	3.94248	5.30759	12	8.01743	19
Body flange-9	2.2148	2.2148	3.9801	5.5199	114	96.012	20
Body flange-10	2.2148	2.2148	4.16727	4.62751	88	78.1161	21

Minimum

4

5

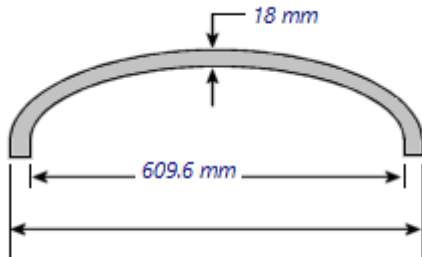
MAWP: 4 MPa, limited by: Shell-8.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom head

Elliptical Head Design Information:					
Design Pressure	P	2.231 MPa	Int. Design Temperature	270 °C	
Material		SA-516 70	Ext. Design Temperature	200 °C	
Length		50.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress	45.4 MPa	
Spec. Min. Thk	t,e, or sa	18.0000 mm	Spec. Nominal Thk.	22.000 mm	
Surface Area		565739.7 mm²	SG of Contents	0.840	
Element Volume		44.3 ltr	Weight of Contents	364.3 N	
Empty Weight		964.2 N	Operating Weight	1328.5 N	
			Comp. All. Ope. Stress	111.7 MPa	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.620 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.810 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.231 \cdot 615.601 \cdot 0.987) / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 2.231)$$

$$= 4.9489 + 3.0000 = 7.9489 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.016 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 137.2 \cdot 1.0 \cdot 15.0) / (0.987 \cdot 615.601 + 0.2 \cdot 15.0)$$

$$= 7 - 0 = 7 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 18.0) / (1.0 \cdot 609.601 + 0.2 \cdot 18.0)$$

$$= 8 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.231 \cdot (0.987 \cdot 615.601 + 0.2 \cdot 15.0)) / (2 \cdot 1.0 \cdot 15.0)$$

$$= 45.414 \text{ MPa}$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (2.231 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.231) + 3.0$
 $= 8.054 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.015 MPa
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (137.2 \cdot 1.0 \cdot 19.0) / (307.8005 + 0.6 \cdot 19.0)$
 $= 8 - 0 = 8 \text{ MPa}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (615.601 / (2 \cdot 155.4))^2) / 6$
 $= 0.987192$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 14.394 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 18.0$, $t_r = 8.741$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.583$, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 22.0$, $t_r = 8.984$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.473$, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -41 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.959 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.473	Temperature Difference	36 °C
UCS-66(c) red.	No	Computed Min. Temp.	-41 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.620 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.810 mm

EXTERNAL PRESSURE RESULTS for: Bottom head

Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B



15.000	645.60	43.04	0.0032270	100.70
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$$MAEP = B/(K0*Do/t) = 100.7003/(0.9 * 43.0401) = 2.5997 \text{ MPa}$$

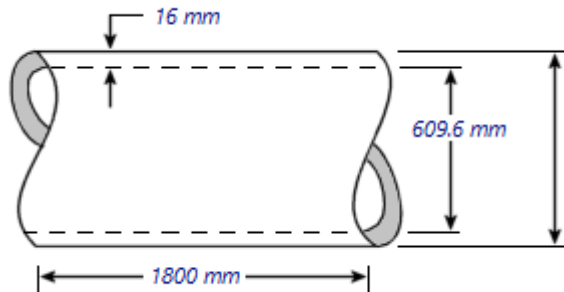
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.699	645.60	380.06	0.0003654	35.38

$$MAEP = B/(K0*Do/t) = 35.3773/(0.9 * 380.0567) = 0.1034 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	2.229 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		1800.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		54.1 MPa
Spec. Min. Thk t,e, or sa		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		3628168.2 mm²	SG of Contents		0.840
Element Volume		525.5 ltr	Weight of Contents		4205.6 N
Empty Weight		4301.9 N	Operating Weight		8507.5 N
			Comp. All. Ope. Stress		109.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.229 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.229)$$

$$= 5.0504 + 3.0000 = 8.0504 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.014 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 13.0) / (307.8005 + 0.6 \cdot 13.0)$$

$$= 6 - 0 = 6 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (304.8005 + 0.6 \cdot 16.0)$$

$$= 7 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.229 \cdot (307.8005 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 54.118 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 2.558 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 8.98, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.691$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.957 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.691	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	641.60	1900.80	49.35	2.9626	0.0011991	81.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*81.8269)/(3*49.3539) = 2.2106 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.281	641.60	1900.80	195.54	2.9626	0.0001567	15.17

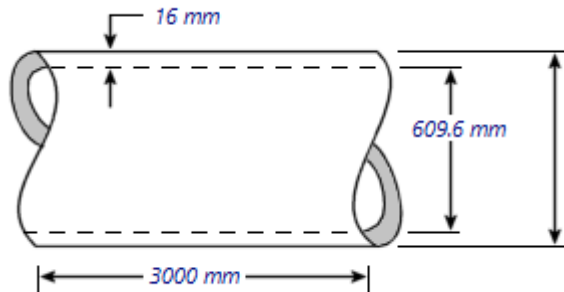
$$MAEP = (4*B)/(3*(Do/t)) = (4*15.1675)/(3*195.5382) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	641.60	101609.57	49.35	50.0000	0.0004538	43.93

$$MAEP = (4*B)/(3*(Do/t)) = (4*43.9328)/(3*49.3539) = 1.1869 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	2.215 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		53.8 MPa
Spec. Min. Thk t _e or s _a		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		6046947.0 mm ²	SG of Contents		0.000
Element Volume		875.8 ltr	Weight of Contents		0.0 N
Empty Weight		7169.8 N	Operating Weight		7169.8 N
			Comp. All. Ope. Stress		109.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.215 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.215)$$

$$= 5.0174 + 3.0000 = 8.0174 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 13.0) / (307.8005 + 0.6 \cdot 13.0)$$

$$= 6 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (304.8005 + 0.6 \cdot 16.0)$$

$$= 7 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.215 \cdot (307.8005 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 53.769 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.558 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 16.0, t_r = 8.947, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.688, Temp. Reduction = 17 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.942 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.688	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	641.60	4380.00	49.35	6.8267	0.0005235	50.68

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.6796)/(3*49.3539) = 1.3691 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.718	641.60	4380.00	136.00	6.8267	0.0001090	10.55

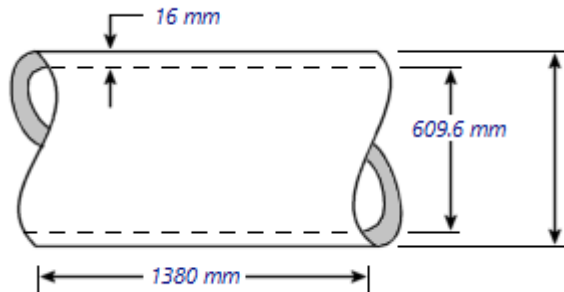
$$MAEP = (4*B)/(3*(Do/t)) = (4*10.5494)/(3*135.9958) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	641.60	183309.41	49.35	50.0000	0.0004538	43.93

$$MAEP = (4*B)/(3*(Do/t)) = (4*43.9328)/(3*49.3539) = 1.1869 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	2.215 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		1380.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		53.8 MPa
Spec. Min. Thk t _e or s _a		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		2781595.5 mm ²	SG of Contents		0.000
Element Volume		402.8 ltr	Weight of Contents		0.0 N
Empty Weight		3298.1 N	Operating Weight		3298.1 N
			Comp. All. Ope. Stress		109.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.215 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.215)$$

$$= 5.0174 + 3.0000 = 8.0174 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 13.0) / (307.8005 + 0.6 \cdot 13.0)$$

$$= 6 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (304.8005 + 0.6 \cdot 16.0)$$

$$= 7 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.215 \cdot (307.8005 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 53.769 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.558 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 8.947, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.688, Temp. Reduction = 17 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.942 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.688	Temperature Difference	17 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	641.60	4380.00	49.35	6.8267	0.0005235	50.68

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.6796)/(3*49.3539) = 1.3691 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.718	641.60	4380.00	136.00	6.8267	0.0001090	10.55

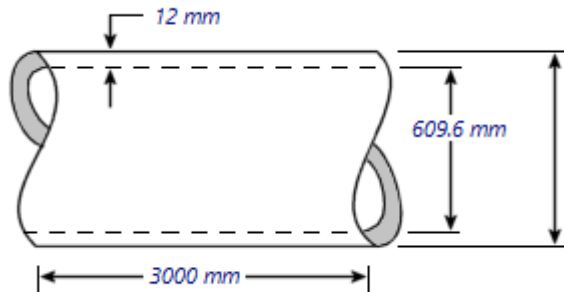
$$MAEP = (4*B)/(3*(Do/t)) = (4*10.5494)/(3*135.9958) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	641.60	183309.41	49.35	50.0000	0.0004538	43.93

$$MAEP = (4*B)/(3*(Do/t)) = (4*43.9328)/(3*49.3539) = 1.1869 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	2.215 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		77.1 MPa
Spec. Min. Thk t _e or s _a		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		5971548.5 mm ²	SG of Contents		0.000
Element Volume		875.8 ltr	Weight of Contents		0.0 N
Empty Weight		5343.0 N	Operating Weight		5343.0 N
			Comp. All. Ope. Stress		103.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.215 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.215)$$

$$= 5.0174 + 3.0000 = 8.0174 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 9.0) / (307.8005 + 0.6 \cdot 9.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (304.8005 + 0.6 \cdot 12.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.215 \cdot (307.8005 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 77.075 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.930 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 12.0, t_r = 8.947, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.994, Temp. Reduction = 0 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.942 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.994	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-4

Cylindrical Shell From 110 to 120 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	4300.00	70.40	6.7866	0.0002950	28.56

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.5569)/(3*70.4001) = 0.5409 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.647	633.60	4300.00	136.36	6.7866	0.0001093	10.58

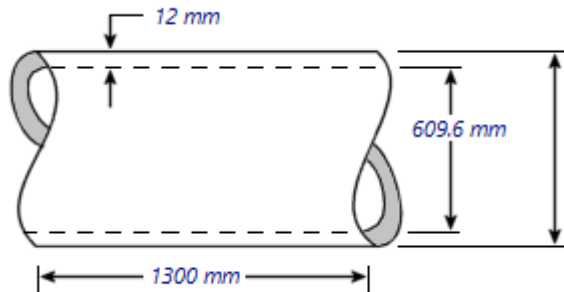
$$MAEP = (4*B)/(3*(Do/t)) = (4*10.5771)/(3*136.3574) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	163814.86	70.40	50.0000	0.0002292	22.19

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.1929)/(3*70.4001) = 0.4203 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	2.215 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		1300.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		77.1 MPa
Spec. Min. Thk t _e or s _a		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		2587671.2 mm ²	SG of Contents		0.000
Element Volume		379.5 ltr	Weight of Contents		0.0 N
Empty Weight		2315.3 N	Operating Weight		2315.3 N
			Comp. All. Ope. Stress		103.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.215 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.215)$$

$$= 5.0174 + 3.0000 = 8.0174 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 9.0) / (307.8005 + 0.6 \cdot 9.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (304.8005 + 0.6 \cdot 12.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.215 \cdot (307.8005 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 77.075 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.930 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 12.0, t_r = 8.947, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.994, Temp. Reduction = 0 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.942 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.994	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-5

Cylindrical Shell From 120 to 130 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	4300.00	70.40	6.7866	0.0002950	28.56

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.5569)/(3*70.4001) = 0.5409 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.647	633.60	4300.00	136.36	6.7866	0.0001093	10.58

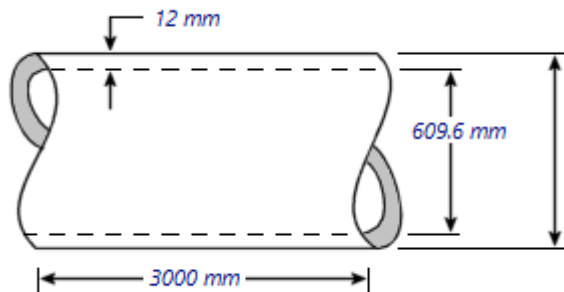
$$MAEP = (4*B)/(3*(Do/t)) = (4*10.5771)/(3*136.3574) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	163814.86	70.40	50.0000	0.0002292	22.19

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.1929)/(3*70.4001) = 0.4203 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	2.215 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		77.1 MPa
Spec. Min. Thk t _e or s _a		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		5971548.5 mm ²	SG of Contents		0.000
Element Volume		875.8 ltr	Weight of Contents		0.0 N
Empty Weight		5343.0 N	Operating Weight		5343.0 N
			Comp. All. Ope. Stress		103.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.215 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.215)$$

$$= 5.0174 + 3.0000 = 8.0174 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 9.0) / (307.8005 + 0.6 \cdot 9.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (304.8005 + 0.6 \cdot 12.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.215 \cdot (307.8005 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 77.075 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.930 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 8.947, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.994, Temp. Reduction = 0 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.942 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.994	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-6

Cylindrical Shell From 150 to 160 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	4700.00	70.40	7.4179	0.0002770	26.82

$$MAEP = (4*B)/(3*(Do/t)) = (4*26.8195)/(3*70.4001) = 0.5079 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.838	633.60	4700.00	130.96	7.4179	0.0001049	10.16

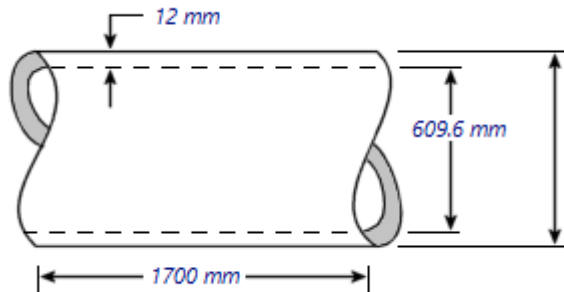
$$MAEP = (4*B)/(3*(Do/t)) = (4*10.1581)/(3*130.955) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	173385.11	70.40	50.0000	0.0002292	22.19

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.1929)/(3*70.4001) = 0.4203 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	2.215 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		1700.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		77.1 MPa
Spec. Min. Thk t _e or s _a		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		3383878.0 mm ²	SG of Contents		0.000
Element Volume		496.3 ltr	Weight of Contents		0.0 N
Empty Weight		3027.7 N	Operating Weight		3027.7 N
			Comp. All. Ope. Stress		103.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.215 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.215)$$

$$= 5.0174 + 3.0000 = 8.0174 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 9.0) / (307.8005 + 0.6 \cdot 9.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (304.8005 + 0.6 \cdot 12.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.215 \cdot (307.8005 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 77.075 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.930 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 12.0, t_r = 8.947, c = 3.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.994, Temp. Reduction = 0 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.942 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.994	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-7

Cylindrical Shell From 160 to 170 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	4700.00	70.40	7.4179	0.0002770	26.82

$$MAEP = (4*B)/(3*(Do/t)) = (4*26.8195)/(3*70.4001) = 0.5079 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.838	633.60	4700.00	130.96	7.4179	0.0001049	10.16

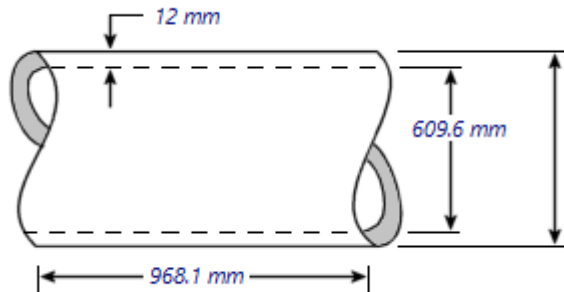
$$MAEP = (4*B)/(3*(Do/t)) = (4*10.1581)/(3*130.955) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	173385.11	70.40	50.0000	0.0002292	22.19

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.1929)/(3*70.4001) = 0.4203 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-8

Cylindrical Shell Design Information:					
Design Pressure	P	2.215 MPa	Int. Design Temperature		270 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		968.140 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	137.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	609.601 mm	Actual Stress		77.1 MPa
Spec. Min. Thk t _e or s _a		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		1927098.4 mm ²	SG of Contents		0.000
Element Volume		282.6 ltr	Weight of Contents		0.0 N
Empty Weight		1724.3 N	Operating Weight		1724.3 N
			Comp. All. Ope. Stress		103.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.215 \cdot 307.8005) / (137.2 \cdot 1.0 - 0.6 \cdot 2.215)$$

$$= 5.0174 + 3.0000 = 8.0174 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.2 \cdot 1.0 \cdot 9.0) / (307.8005 + 0.6 \cdot 9.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (304.8005 + 0.6 \cdot 12.0)$$

$$= 5 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.215 \cdot (307.8005 + 0.6 \cdot 9.0)) / (1.0 \cdot 9.0)$$

$$= 77.075 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.930 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 8.947, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.994, Temp. Reduction = 0 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.942 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.994	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-23 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-8

Cylindrical Shell From 190 to 200 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193613 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	968.14	70.40	1.5280	0.0014771	86.34

$$MAEP = (4*B)/(3*(Do/t)) = (4*86.3398)/(3*70.4001) = 1.6352 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.457	633.60	968.14	257.87	1.5280	0.0002066	20.00

$$MAEP = (4*B)/(3*(Do/t)) = (4*20.0027)/(3*257.8679) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.000	633.60	37962.85	70.40	50.0000	0.0002292	22.19

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.1929)/(3*70.4001) = 0.4203 \text{ MPa}$$



Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.155 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2.896 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6.007 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	4.362 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.167 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.167 MPa
App 27-4	= M.A.W.P.	3.942 MPa

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	102.4	234.0	0.438	6.01
Shell-1	118.2	234.0	0.505	6.01
Shell-2	118.2	234.0	0.505	6.01
Shell-3	118.2	234.0	0.505	6.01
Shell-4	156.3	234.0	0.668	6.01
Shell-5	156.3	234.0	0.668	6.01
Shell-6	156.3	234.0	0.668	6.01
Shell-7	156.3	234.0	0.668	6.01
Shell-8	156.3	234.0	0.668	6.01

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_80	Nozzle	138.00	133.20	1.036
N6_100	Nozzle	138.00	133.20	1.036
K1_50	Nozzle	138.00	133.20	1.036
K2_50	Nozzle	138.00	133.20	1.036
K6_50	Nozzle	138.00	133.20	1.036
N5_100	Nozzle	138.00	133.20	1.036
H3_150	Nozzle	138.00	133.20	1.036
K3_50	Nozzle	138.00	133.20	1.036
H2_150	Nozzle	138.00	133.20	1.036
N1_80	Nozzle	138.00	133.20	1.036
H1_150	Nozzle	138.00	133.20	1.036
N4_80	Nozzle	138.00	133.20	1.036
K5_50	Nozzle	138.00	133.20	1.036
N2_100	Nozzle	138.00	133.20	1.036

Minimum

1.036

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	137.20	1.006
Shell-1	138.00	137.20	1.006
Body flange -1	138.00	133.20	1.036
Body flange -2	138.00	133.20	1.036
Shell-2	138.00	137.20	1.006
Shell-3	138.00	137.20	1.006
Body flange-3	138.00	133.20	1.036
Body flange-4	138.00	133.20	1.036
Shell-4	138.00	137.20	1.006
Shell-5	138.00	137.20	1.006
Body flange-5	138.00	133.20	1.036
Body flange-6	138.00	133.20	1.036
Shell-6	138.00	137.20	1.006



Shell-7	138.00	137.20	1.006
Body flange-7	138.00	133.20	1.036
Body flange-8	138.00	133.20	1.036
Shell-8	138.00	137.20	1.006
Body flange-9	138.00	133.20	1.036
Body flange-10	138.00	133.20	1.036

Minimum 1.006

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_80	27.66	225.00	0.123
N6_100	27.66	225.00	0.123
K1_50	12.70	225.00	0.056
K2_50	12.70	225.00	0.056
K6_50	12.70	225.00	0.056
N5_100	27.66	225.00	0.123
H3_150	72.75	225.00	0.323
K3_50	12.70	225.00	0.056
H2_150	72.75	225.00	0.323
N1_80	27.66	225.00	0.123
H1_150	72.75	225.00	0.323
N4_80	27.66	225.00	0.123
K5_50	12.70	225.00	0.056
N2_100	28.50	225.00	0.127

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5.155 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2.896 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6.015 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	4.362 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.167 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.167 MPa
App 27-4	= M.A.W.P.	3.942 MPa

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 2.215 • 1.006
 = 2.896 MPa

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	62.6	234.0	0.268	3.08
Shell-1	74.6	234.0	0.319	3.07
Shell-2	74.1	234.0	0.317	3.05
Shell-3	73.4	234.0	0.314	3.02
Shell-4	104.6	234.0	0.447	3.01
Shell-5	103.6	234.0	0.443	2.98
Shell-6	103.0	234.0	0.440	2.96
Shell-7	101.9	234.0	0.436	2.93
Shell-8	101.2	234.0	0.433	2.91

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_80	Nozzle	138.00	133.20	1.036
N6_100	Nozzle	138.00	133.20	1.036
K1_50	Nozzle	138.00	133.20	1.036
K2_50	Nozzle	138.00	133.20	1.036
K6_50	Nozzle	138.00	133.20	1.036
N5_100	Nozzle	138.00	133.20	1.036
H3_150	Nozzle	138.00	133.20	1.036
K3_50	Nozzle	138.00	133.20	1.036
H2_150	Nozzle	138.00	133.20	1.036
N1_80	Nozzle	138.00	133.20	1.036
H1_150	Nozzle	138.00	133.20	1.036
N4_80	Nozzle	138.00	133.20	1.036
K5_50	Nozzle	138.00	133.20	1.036
N2_100	Nozzle	138.00	133.20	1.036

Minimum

1.036

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	137.20	1.006
Shell-1	138.00	137.20	1.006
Body flange -1	138.00	133.20	1.036
Body flange -2	138.00	133.20	1.036
Shell-2	138.00	137.20	1.006
Shell-3	138.00	137.20	1.006
Body flange-3	138.00	133.20	1.036
Body flange-4	138.00	133.20	1.036



Shell-4	138.00	137.20	1.006
Shell-5	138.00	137.20	1.006
Body flange-5	138.00	133.20	1.036
Body flange-6	138.00	133.20	1.036
Shell-6	138.00	137.20	1.006
Shell-7	138.00	137.20	1.006
Body flange-7	138.00	133.20	1.036
Body flange-8	138.00	133.20	1.036
Shell-8	138.00	137.20	1.006
Body flange-9	138.00	133.20	1.036
Body flange-10	138.00	133.20	1.036

Minimum

1.006

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_80	14.15	225.00	0.063
N6_100	14.14	225.00	0.063
K1_50	6.49	225.00	0.029
K2_50	6.49	225.00	0.029
K6_50	6.49	225.00	0.029
N5_100	14.04	225.00	0.062
H3_150	36.93	225.00	0.164
K3_50	6.35	225.00	0.028
H2_150	36.36	225.00	0.162
N1_80	13.61	225.00	0.060
H1_150	35.80	225.00	0.159
N4_80	13.38	225.00	0.059
K5_50	6.14	225.00	0.027
N2_100	13.72	225.00	0.061

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	4000	...	...	No Calc	No Calc
20	30	4000	...	...	No Calc	No Calc
30	40	No Calc	645.601	15	0.003227	100.7
40	50	1900.8	641.601	13	0.0011991	81.8269
50	60	No Calc	...	85	No Calc	No Calc
60	70	No Calc	...	85	No Calc	No Calc
70	80	4380	641.601	13	0.00052351	50.6796
80	90	4380	641.601	13	0.00052351	50.6796
90	100	No Calc	...	85	No Calc	No Calc
100	110	No Calc	...	103	No Calc	No Calc
110	120	4300	633.601	9	0.00029499	28.5569
120	130	4300	633.601	9	0.00029499	28.5569
130	140	No Calc	...	113	No Calc	No Calc
140	150	No Calc	...	113	No Calc	No Calc
150	160	4700	633.601	9	0.00027704	26.8195
160	170	4700	633.601	9	0.00027704	26.8195
170	180	No Calc	...	113	No Calc	No Calc
180	190	No Calc	...	111	No Calc	No Calc
190	200	968.14	633.601	9	0.0014771	86.3398
200	210	No Calc	...	111	No Calc	No Calc
210	220	No Calc	...	85	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	18	4.6987	0.10342	2.59965
40	50	16	6.28121	0.10342	2.21061
50	60	88	81.1276	0.10342	No Calc
60	70	88	81.1276	0.10342	No Calc
70	80	16	7.7178	0.10342	1.36915
80	90	16	7.7178	0.10342	1.36915
90	100	88	81.1276	0.10342	No Calc
100	110	106	94.7674	0.10342	No Calc
110	120	12	7.64662	0.10342	0.54085
120	130	12	7.64662	0.10342	0.54085
130	140	116	96.901	0.10342	No Calc
140	150	116	96.901	0.10342	No Calc
150	160	12	7.83831	0.10342	0.50794
160	170	12	7.83831	0.10342	0.50794
170	180	116	96.901	0.10342	No Calc
180	190	114	96.012	0.10342	No Calc
190	200	12	5.45708	0.10342	1.63522
200	210	114	96.012	0.10342	No Calc
210	220	88	78.1304	0.10342	No Calc

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
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10	20	4000	No Calc	No Calc	No Calc
20	30	4000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	1900.8	101610	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	4380	183309	No Calc	No Calc
80	90	4380	183309	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc
110	120	4300	163815	No Calc	No Calc
120	130	4300	163815	No Calc	No Calc
130	140	No Calc	No Calc	No Calc	No Calc
140	150	No Calc	No Calc	No Calc	No Calc
150	160	4700	173385	No Calc	No Calc
160	170	4700	173385	No Calc	No Calc
170	180	No Calc	No Calc	No Calc	No Calc
180	190	No Calc	No Calc	No Calc	No Calc
190	200	968.14	37962.8	No Calc	No Calc
200	210	No Calc	No Calc	No Calc	No Calc
210	220	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	1432.33	...	1298.21	...	...
20	30	306.31	...	261.604	...	...
30	40	98.3269	44.2546	84.9187	45.4278	...
40	50	438.702	525.45	358.154	535.845	...
50	60	332.834	49.8404	324.848	50.3784	...
60	70	332.834	49.8404	324.848	50.3784	...
70	80	731.169	875.75	596.924	893.074	...
80	90	336.338	402.845	274.585	410.814	...
90	100	357.305	51.9807	348.98	52.5187	...
100	110	376.208	56.1724	367.338	56.7178	...
110	120	544.87	875.75	410.625	893.074	...
120	130	236.11	379.492	177.937	386.999	...
130	140	406.23	58.9217	396.93	59.4671	...
140	150	406.23	58.9217	396.93	59.4671	...
150	160	544.87	875.75	410.625	893.074	...
160	170	308.76	496.259	232.687	506.076	...
170	180	406.23	58.9217	396.93	59.4671	...
180	190	400.226	58.3718	391.012	58.9172	...
190	200	175.837	282.616	132.514	288.207	...
200	210	400.226	58.3718	391.012	58.9172	...
210	220	458.073	...	458.073	...	...
Total		9030	5259.51	8035	5358.82	0

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Wght	200	...	...	...	Tailing lug
20	Ins1	32.2841	...	750	...	Ins: 20
30	Liqd	37.1532	...	-76.2001	...	Liquid 30
30	Ins1	22.9734	...	-51.2001	...	Ins: 20
30	Noz1	40.5012	...	-293.4	...	N3_80
30	Wght	300	...	...	...	Tracing coil wt
30	Forc	...	...	-152	...	Guide sup later
40	Liqd	428.878	...	875	...	Liquid 40
40	Ins1	115.043	...	900	...	Ins: 20
40	Noz1	30.0256	454.8	1600	...	N6_100
40	Noz1	10.2512	-404.8	1600	...	K1_50
40	Noz1	10.2512	-404.8	150	...	K2_50
40	Noz1	10.2512	404.8	150	...	K6_50
50	Ins1	17.2306	...	84.074	...	Ins: 20
60	Ins1	17.2306	...	84.074	...	Ins: 20
70	Pack	153.817	...	1825	...	Bed 3
70	Ins1	191.738	...	1500	...	Ins: 20
70	Noz1	30.0256	454.8	125.004	...	N5_100
70	Noz1	55.8155	404.8	1650	...	H3_150
70	Wght	100	300	1650	...	H3 Davit
70	Wght	30	...	125.004	...	N5_Collector Wt
70	Wght	10	...	100.004	...	N5_Collector su
70	Wght	10	...	550.004	...	Bed3 Sup
70	Wght	70	...	600.004	...	Bed3 sup grid
80	Pack	88.3632	...	675	...	Bed 3
80	Ins1	88.1996	...	690	...	Ins: 20
80	Wght	10	...	1360	...	Bed3 limiter su
80	Wght	10	...	1370	...	Bed3 limiter
90	Ins1	18.4974	...	84.074	...	Ins: 20
100	Ins1	16.3823	...	84.074	...	Ins: 20
100	Wght	10	...	29.2052	...	Dist sup
100	Wght	30	...	0.0069769	...	Distributor
110	Pack	162.326	...	1760	...	Bed 2



110	Ins1	189.613	...	1500	...	Ins: 20
110	Noz1	10.1406	404.8	200.008	...	K3_50
110	Noz1	55.3246	404.8	1650.01	...	H2_150
110	Wght	100	300	1650.01	...	H2 Davit
110	Wght	10	...	450.009	...	Bed2 sup
110	Wght	70	...	500.008	...	Bed sup grid
120	Pack	79.8541	...	610	...	Bed 2
120	Ins1	82.1655	...	650	...	Ins: 20
120	Wght	500	...	1050.01	...	Clip weight
120	Wght	10	...	1230.01	...	Bed2 limiter su
120	Wght	10	...	1250.01	...	Bed 2 limiter
120	Wght	800	...	997.809	...	Misc wt 7%
130	Ins1	16.2974	...	84.074	...	Ins: 20
140	Ins1	16.2974	...	84.074	...	Ins: 20
140	Wght	10	...	11.2121	...	Dist2 sup
140	Wght	30	...	0.011628	...	Dist wt
150	Pack	137.454	...	1950	...	Bed 1
150	Ins1	189.613	...	1500	...	Ins: 20
150	Noz1	20.6788	-404.813	715.013	...	N1_80
150	Noz1	57.3837	404.8	2250.01	...	H1_150
150	Wght	100	300	2250.01	...	H1 Davit
150	Wght	10	...	700.014	...	N1 pipe sup
150	Wght	10	...	715.014	...	N1 pipe
150	Wght	10	...	800.014	...	Bed1 sup
150	Wght	70	...	850.014	...	Bed 1 sup grid
160	Pack	104.727	...	800	...	Bed 1
160	Ins1	107.447	...	850	...	Ins: 20
160	Wght	10	...	1620.01	...	Bed1 limiter su
160	Wght	10	...	1650.01	...	Bed1 limiter
160	Wght	100	300	1500.01	...	Trunnion 1
160	Wght	100	-300	1500.01	...	Trunnion 2
170	Ins1	16.2974	...	84.074	...	Ins: 20
180	Ins1	16.3119	...	84.074	...	Ins: 20
190	Ins1	61.1905	...	484.07	...	Ins: 20
190	Noz1	20.6782	404.8	150.018	...	N4_80
190	Noz1	10.1406	-404.8	150.018	...	K5_50
190	Wght	10	...	50.0179	...	N4 pipe sup
190	Wght	10	...	0.016279	...	N4 pipe
190	Wght	10	...	400.017	...	Demister sup
190	Wght	20	...	450.017	...	Demister
200	Ins1	16.3119	...	84.074	...	Ins: 20
210	Ins1	22.1083	...	34.925	...	Ins: 20
210	Noz1	53.2713	...	234	...	N2_100
210	Forc	...	...	...	...	N2 F/M

Total Weight of Each Detail Type:

Packing	726.5
Liquid	466.0
Insulation	1253.2
Nozzles	414.7
Weights	2790.0

Sum of the Detail Weights 5650.5 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	9030.0	9030.0	9030.0	9030.0	9030.0	9030.0
Nozzles	414.7	414.7	414.7	414.7	414.7	414.7
Wld Weights	2420.0	2420.0	2420.0	2420.0	2420.0	2420.0
Packing	...	...	726.5	726.5	726.5	726.5
Insulation	...	...	1253.2	1253.2	1253.2	1253.2
Ope Weights	...	...	...	...	...	370.0
Ope. Liquid	...	...	...	...	...	466.0
Test Liquid	...	5256.3	...	...	...	...
Totals	11864.8	17121.1	13844.5	13844.5	13844.5	14680.6



Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	11864.8 kgm
Shop Test Wt.	- Fabricated weight + water (full)	17121.1 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	13844.5 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	13844.5 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	13844.5 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	14680.6 kgm
Field Test Wt.	- Empty weight + test medium (full)	18374.3 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	6042.7 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	8217.3 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	6207.1 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	6116681	5739690
20	30	2038893	1913230
30	40	565740	498579
40	50	3628168	3447212
50	60	869854	341682
60	70	869854	341682
70	80	6046947	5745354
80	90	2781595	2642863
90	100	892836	356199
100	110	914254	379530
110	120	5971548	5745354
120	130	2587671	2489653
130	140	942980	397927
140	150	942980	397927
150	160	5971548	5745354
160	170	3383878	3255701
170	180	942980	397927
180	190	937235	394248
190	200	1927098	1854102
200	210	937235	394248
210	220	915233	...

Total	50185216.0	42478468.0 mm ²
Total	50.2	42.5 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	1632.33	1632.33	1632.33	...	1632.33
20	30	338.594	338.594	338.594	...	338.594
30	40	461.801	498.955	506.029	...	461.801
40	50	614.524	1043.4	1139.65	93253.1	614.524
50	60	350.065	350.065	399.875	...	350.065
60	70	350.065	350.065	399.875	...	350.065
70	80	1282.57	1382.57	2003.96	649909	1382.57
80	90	522.901	532.901	837.137	...	532.901
90	100	375.802	375.802	427.751	...	375.802
100	110	402.59	432.59	458.728	...	432.59
110	120	1072.27	1142.27	1785.16	554267	1142.27
120	130	1708.13	1718.13	2007.54	...	1718.13
130	140	422.527	422.527	481.413	...	422.527
140	150	432.527	462.527	491.413	...	462.527



150	160	1070	1150	1807.76	440055	1150
160	170	730.934	740.934	1122.16	...	740.934
170	180	422.527	422.527	481.413	...	422.527
180	190	416.537	416.537	474.874	...	416.537
190	200	287.846	317.846	570.29	41845.6	317.846
200	210	416.537	416.537	474.874	...	416.537
210	220	533.453	533.453	533.453	...	533.453

Empty Support Force + the Sum of the Y forces 144006.20 N
 Operating Support Force + the Sum of the Y forces 152204.33 N
 Hydro Support Force + the Sum of the Y forces 188424.97 N

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	14214.5	14680.6	18374.3
20	30	12582.2	13048.2	16742
30	40	12243.6	12709.6	16403.4
40	50	11781.8	12210.7	15897.3
50	60	11167.3	11167.3	14757.7
60	70	10817.2	10817.2	14357.8
70	80	10467.2	10467.2	13957.9
80	90	9084.59	9084.59	11954
90	100	8551.69	8551.69	11116.8
100	110	8175.88	8175.88	10689.1
110	120	7743.29	7743.29	10230.4
120	130	6601.02	6601.02	8445.19
130	140	4882.89	4882.89	6437.65
140	150	4460.36	4460.36	5956.24
150	160	3997.83	3997.83	5464.83
160	170	2847.83	2847.83	3657.07
170	180	2106.9	2106.9	2534.9
180	190	1684.37	1684.37	2053.49
190	200	1267.84	1267.84	1578.62
200	210	949.99	949.99	1008.33
210	220	533.453	533.453	533.453

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	1779328	1779328	1779328
20	30	1779328	1779328	1779328
30	40	1779328	1779328	1779328
40	50	1779328	1779328	1779328
50	60	1686075	1686075	1686075
60	70	1686075	1686075	1686075
70	80	1686075	1686075	1686075
80	90	1036167	1036167	1036167
90	100	1036167	1036167	1036167
100	110	1036167	1036167	1036167
110	120	1036167	1036167	1036167
120	130	481900	481900	481900
130	140	481900	481900	481900
140	150	481900	481900	481900
150	160	481900	481900	481900
160	170	41845.6	41845.6	41845.6
170	180	41845.6	41845.6	41845.6
180	190	41845.6	41845.6	41845.6
190	200	41845.6	41845.6	41845.6



200	210	...	...	...
210	220	...	...	...



Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	UG-44	Max Pressure 50%	Max Pressure DNV
N3_80	8.23	10.20	270	600	GR 1.1	5.320	7.020	8.227	7.020
N6_100	8.23	10.20	270	600	GR 1.1	4.621	7.719	8.227	7.719
K1_50	4.12	5.10	270	300	GR 1.1	...	...	...	...
K2_50	4.12	5.10	270	300	GR 1.1	...	...	...	...
K6_50	4.12	5.10	270	300	GR 1.1	...	...	...	...
N5_100	8.23	10.20	270	600	GR 1.1	4.621	7.719	8.227	7.719
H3_150	4.12	5.10	270	300	GR 1.1	...	...	...	...
K3_50	4.12	5.10	270	300	GR 1.1	...	...	...	...
H2_150	4.12	5.10	270	300	GR 1.1	...	...	...	...
N1_80	8.23	10.20	270	600	GR 1.1	5.320	7.020	8.227	7.020
H1_150	4.12	5.10	270	300	GR 1.1	...	...	...	...
N4_80	8.23	10.20	270	600	GR 1.1	5.320	7.020	8.227	7.020
K5_50	4.12	5.10	270	300	GR 1.1	...	...	...	...
N2_100	4.12	5.10	270	300	GR 1.1	...	...	...	...
Min Rating	4.116	5.100, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N3_80	111.125	0.50	1267201	5998	3.942	7.020	Pass
N6_100	135.001	0.50	1979995	7497	3.942	7.719	Pass
N5_100	135.001	0.50	1979995	7497	3.942	7.719	Pass
N1_80	111.125	0.50	1267201	5998	3.942	7.020	Pass
N4_80	111.125	0.50	1267201	5998	3.942	7.020	Pass

Evaluating Flange on Nozzle [N380]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1267201) + 4 \cdot \text{abs}(5998)111.125 \leq \pi \cdot 111.125^3 \cdot ((8.227 - 3.942) + .500 \cdot 8.227)$$

$$= 0.22942E+08 \leq 0.36213E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N380 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8.227, 0.500 \cdot 8.227 + 8.227 - 22942268.0 / (\pi \cdot 111.125^3))$$

$$= \min(8.227, 7.020)$$

$$= 7.020 \text{ MPa}$$

Evaluating Flange on Nozzle [N6100]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1979995) + 4 \cdot \text{abs}(7497)135.001 \leq \pi \cdot 135.001^3 \cdot ((8.227 - 3.942) + .500 \cdot 8.227)$$

$$= 0.35730E+08 \leq 0.64930E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N6100 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8.227, 0.500 \cdot 8.227 + 8.227 - 35730028.0 / (\pi \cdot 135.001^3))$$

$$= \min(8.227, 7.719)$$

$$= 7.719 \text{ MPa}$$

Evaluating Flange on Nozzle [N5100]:



$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1979995) + 4 \cdot \text{abs}(7497)135.001 \leq \pi \cdot 135.001^3 [(8.227 - 3.942) + .500 \cdot 8.227] \\ &= 0.35730E+08 \leq 0.64930E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N5100 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(8.227, 0.500 \cdot 8.227 + 8.227 - 35730028.0 / (\pi \cdot 135.001^3)) \\ &= \min(8.227, 7.719) \\ &= 7.719 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N180]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1267201) + 4 \cdot \text{abs}(5998)111.125 \leq \pi \cdot 111.125^3 [(8.227 - 3.942) + .500 \cdot 8.227] \\ &= 0.22942E+08 \leq 0.36213E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N180 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(8.227, 0.500 \cdot 8.227 + 8.227 - 22942268.0 / (\pi \cdot 111.125^3)) \\ &= \min(8.227, 7.020) \\ &= 7.020 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N480]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1267201) + 4 \cdot \text{abs}(5998)111.125 \leq \pi \cdot 111.125^3 [(8.227 - 3.942) + .500 \cdot 8.227] \\ &= 0.22942E+08 \leq 0.36213E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N480 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(8.227, 0.500 \cdot 8.227 + 8.227 - 22942268.0 / (\pi \cdot 111.125^3)) \\ &= \min(8.227, 7.020) \\ &= 7.020 \text{ MPa} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 0.6858 Hz.

The Natural Frequency for the Vessel (Ope...) is 0.6856 Hz.

The Natural Frequency for the Vessel (Filled) is 0.6246 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	12257	-1863817
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	7775.19	2178002

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	12257	-1863817
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	7775.19	2178002

User Input Forces and Moments:

From Node	Distance From	- - - Fx	Forces - - - Fy	- - - Fz	- - - Mx	Moments - - - My	- - - Mz
30	-152.00	-8667.		-8667.			
210		5498.	-8247.	5498.	0.154E+07		0.154E+07



Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	11000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Dynamic Gust-Effect Factor, Operating Case [Gf]:
$$= 0.925[(1+1.7 \cdot I_{zbar} \cdot \sqrt{(gQ^2 \cdot Q^2 + gR^2 \cdot R^2)}) / (1+1.7 \cdot gV \cdot I_{zbar})]$$
$$= 0.925[(1+1.7 \cdot 0.191 \cdot \sqrt{(3.4^2 \cdot 0.919^2 + 4.099^2 \cdot 0.928^2)}) / (1+1.7 \cdot 3.4 \cdot 0.191)]$$
$$= 1.142$$

Natural Frequency of Vessel (Operating)	0.686 Hz
Natural Frequency of Vessel (Empty)	0.686 Hz
Natural Frequency of Vessel (Test)	0.625 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.700
Structure Height to Diameter ratio	28.923
Height to top of Structure	22134.000 mm

This is classified as a flexible structure. Dynamic analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 274320.0$ mm

Effective Height $[z]$:
= Centroid Height + Vessel Base Elevation
= $1500.0 + 11000.0 = 12500.0$ mm
= 41.011 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
Because z (41.011 ft.) > 15 ft.
= $2.01(z / Zg)^{2/\alpha}$
= $2.01(41.011/900.0)^{2/9.5}$
= 1.049

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
= 1, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.049 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [0.766] \text{ kPa}$$

Force on the first element [F]:

$$= qz \cdot Gf \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 1.142 \cdot 0.7 \cdot 37.514$$

$$= 479.9 \text{ lbs. } [2134.6] \text{ N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-2	12500.0	0.000	0.000	0.000	1.049	1.000	0.766
Skirt -1	14500.0	0.000	0.000	0.000	1.082	1.000	0.766
Bottom head	15025.0	0.000	0.000	0.000	1.091	1.000	0.766
Shell-1	15950.0	0.000	0.000	0.000	1.104	1.000	0.766
Body flange -1	16947.6	0.000	0.000	0.000	1.119	1.000	0.766
Body flange -2	17133.9	0.000	0.000	0.000	1.121	1.000	0.766
Shell-2	18727.1	0.000	0.000	0.000	1.142	1.000	0.766
Shell-3	20917.1	0.000	0.000	0.000	1.169	1.000	0.766
Body flange-3	21704.7	0.000	0.000	0.000	1.178	1.000	0.766
Body flange-4	21900.0	0.000	0.000	0.000	1.181	1.000	0.766
Shell-4	23502.2	0.000	0.000	0.000	1.198	1.000	0.766
Shell-5	25652.2	0.000	0.000	0.000	1.220	1.000	0.766
Body flange-5	26413.8	0.000	0.000	0.000	1.228	1.000	0.766
Body flange-6	26628.1	0.000	0.000	0.000	1.230	1.000	0.766
Shell-6	28235.3	0.000	0.000	0.000	1.245	1.000	0.766
Shell-7	30585.3	0.000	0.000	0.000	1.267	1.000	0.766
Body flange-7	31546.9	0.000	0.000	0.000	1.275	1.000	0.766
Body flange-8	31760.2	0.000	0.000	0.000	1.277	1.000	0.766
Shell-8	32350.5	0.000	0.000	0.000	1.282	1.000	0.766
Body flange-9	32943.9	0.000	0.000	0.000	1.287	1.000	0.766
Body flange-10	33092.0	0.000	0.000	0.000	1.288	1.000	0.766

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm
- Lc - Total length of conical section(s) of vessel, mm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, N
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
- Z - Maximum amplitude of vibration at top of vessel, mm
- D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.000003 (High); 0.000003 < 0.000004 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5



:: 0.0/22134.0 = 0.0

:: [D / L²] • 10⁴ < 8 (English Units, ft.)
 :: [2.52/72.62²] • 10⁴ = 4.772

Compute the vibration possibility. If Vp > 0.000004 no chance. [Vp]:
 = W / (L • Dr²)
 = 134207/(22134.0 • 761.352²)
 = 0.10460E-04

Since Vp is > 0.000004 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Tailing lug	0.00	11000.00	0.77	0.00
Tracing coil	0.00	15000.00	0.77	0.00
H3 Davit	0.00	18877.10	0.77	0.00
N5Collector	0.00	17352.10	0.77	0.00
N5Collector	0.00	17327.10	0.77	0.00
Bed3 Sup	0.00	17777.10	0.77	0.00
Bed3 sup grid	0.00	17827.10	0.77	0.00
Bed3 limiter	0.00	21587.10	0.77	0.00
Bed3 limiter	0.00	21597.10	0.77	0.00
Dist sup	0.00	21827.10	0.77	0.00
Distributor	0.00	21797.90	0.77	0.00
H2 Davit	0.00	23652.20	0.77	0.00
Bed2 sup	0.00	22452.20	0.77	0.00
Bed sup grid	0.00	22502.20	0.77	0.00
Clip weight	0.00	26052.20	0.77	0.00
Bed2 limiter	0.00	26232.20	0.77	0.00
Bed 2 limiter	0.00	26252.20	0.77	0.00
Misc wt 7%	0.00	26000.00	0.77	0.00
Dist2 sup	0.00	26532.20	0.77	0.00
Dist wt	0.00	26521.00	0.77	0.00
H1 Davit	0.00	28985.30	0.77	0.00
N1 pipe sup	0.00	27435.30	0.77	0.00
N1 pipe	0.00	27450.30	0.77	0.00
Bed1 sup	0.00	27535.30	0.77	0.00
Bed 1 sup gri	0.00	27585.30	0.77	0.00
Bed1 limiter	0.00	31355.30	0.77	0.00
Bed1 limiter	0.00	31385.30	0.77	0.00
Trunnion 1	0.00	31235.30	0.77	0.00
Trunnion 2	0.00	31235.30	0.77	0.00
N4 pipe sup	0.00	31916.40	0.77	0.00
N4 pipe	0.00	31866.40	0.77	0.00
Demister sup	0.00	32266.40	0.77	0.00
Demister	0.00	32316.40	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 0.6856 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	12500	1161.71	3485129	0.76606	2348.01
20	30	14500	1304.91	1304910	0.76606	879.146
30	40	15025	1442.03	72101.3	0.76606	48.5762
40	50	15950	1434.87	2582758	0.76606	1740.06
50	60	16947.6	1292.4	240772	0.76606	162.213
60	70	17133.9	1292.4	240772	0.76606	162.213
70	80	18727.1	1434.87	4304597	0.76606	2900.1
80	90	20917.1	1434.87	1980114	0.76606	1334.05
90	100	21704.7	1292.4	240772	0.76606	162.213
100	110	21900	1283.95	262308	0.76606	176.722



110	120	23502.2	1420.55	4261637	0.76606	2871.16
120	130	25652.2	1420.55	1846709	0.76606	1244.17
130	140	26413.8	1272.95	272790	0.76606	183.784
140	150	26628.1	1272.95	272790	0.76606	183.784
150	160	28235.3	1420.55	4261637	0.76606	2871.16
160	170	30585.3	1420.55	2414927	0.76606	1626.99
170	180	31546.9	1272.95	272790	0.76606	183.784
180	190	31760.2	1275.06	270693	0.76606	182.372
190	200	32350.5	1420.55	1375287	0.76606	926.56
200	210	32943.9	1275.06	270693	0.76606	182.372
210	220	33092	1874.93	157494	0.76606	106.107

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(72.61810.75)$
= 0.498 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/0.686)
= 1.459

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.498$, 1.459) per 12.8.2
= 0.846

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= $Sds / (R / I)$
= $0.157 / (2.0 / 1.25)$
= 0.098

Maximum value of C_s per equation 12.8-3 [Cs]:
= $Sd1 / (T(R / I_e))$
= $0.104 / (0.846(2.0 / 1.25))$
= 0.077

Minimum value of C_s per equation 15.4-1 [Cs]:
= $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
= $\max((0.044 \cdot 0.1568 \cdot 1.25), 0.030)$
= 0.030

Total Base Shear [V]:
= $C_s \cdot W$ (Equation 12.8-1):
= $0.0769 \cdot 143957.5$
= 11063.311 N

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.157 \cdot 143958$



= 4514.5 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.4793

The Natural Frequency for the Vessel (Ope...) is 0.6856 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	1500	16006.6	52.5904	51.4677	-3732.3
20	30	3500	3320.25	38.2053	37.3829	-4234.27
30	40	4025	4892.75	69.2302	62.694	-4338.39
40	50	4950	10231.6	196.601	113.289	-4491.83
50	60	5947.65	3432.73	86.5447	84.672	-4812.69
60	70	6133.95	3432.73	90.5847	88.6241	-4920.34
70	80	7727.1	13557.4	503.42	456.877	-5027.99
80	90	9917.1	5225.62	280.672	269.417	-5453.15
90	100	10704.7	3685.12	221.622	216.799	-5617.03
100	110	10900	4241.98	262.027	238.547	-5732.59
110	120	12502.2	11201.1	847.506	778.231	-5865.62
120	130	14652.2	16848	1612.04	1567.68	-6216.89
130	140	15413.8	4143.3	427.299	417.967	-6745.24
140	150	15628.1	4535.54	477.403	436.687	-6875.18
150	160	17235.3	11276.9	1371.93	1248.58	-7017.41
160	170	19585.3	7265.59	1067.9	1030.43	-7371.05
170	180	20546.9	4143.3	653.734	639.418	-7598.9
180	190	20760.2	4084.56	654.388	640.056	-7728.84
190	200	21350.5	3116.8	520.485	461.033	-7856.93
200	210	21943.8	4084.56	710.326	694.76	-7954.67
210	220	22092	5231.04	918.802	898.667	-8082.76



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	20032.2	20032.2	220874480	220874480
20	30	3500	20032.2	20032.2	160753584	160753584
30	40	4025	20032.2	20032.2	140713280	140713280
40	50	4950	7775.19	7775.19	142188192	142188192
50	60	5947.65	7775.19	7775.19	128152168	128152168
60	70	6133.95	7775.19	7775.19	126703072	126703072
70	80	7727.1	7775.19	7775.19	125253992	125253992
80	90	9917.1	7775.19	7775.19	101918968	101918968
90	100	10704.7	7775.19	7775.19	91149840	91149840
100	110	10900	7775.19	7775.19	89700760	89700760
110	120	12502.2	7775.19	7775.19	88111656	88111656
120	130	14652.2	7775.19	7775.19	64776624	64776624
130	140	15413.8	7775.19	7775.19	54629772	54629772
140	150	15628.1	7775.19	7775.19	52962888	52962888
150	160	17235.3	7775.19	7775.19	51296000	51296000
160	170	19585.3	7775.19	7775.19	27960960	27960960
170	180	20546.9	7775.19	7775.19	14702776	14702776
180	190	20760.2	7775.19	7775.19	13035889	13035889
190	200	21350.5	7775.19	7775.19	11384556	11384556
200	210	21943.8	7775.19	7775.19	3829334	3829334
210	220	22092	7775.19	7775.19	2178002	2178002

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;

Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1500	20475.5	11063.3	231635040	181349360
20	30	3500	18127.5	11010.7	173707008	148224896
30	40	4025	17248.4	10972.5	156011872	137228816
40	50	4950	17199.8	10903.3	155150336	136681712
50	60	5947.65	15459.7	10706.7	125675240	117176656
60	70	6133.95	15297.5	10620.1	122809072	115189288
70	80	7727.1	15135.3	10529.6	119973144	113218408
80	90	9917.1	12235.2	10026.1	78900712	82372376
90	100	10704.7	10901.2	9745.46	62881060	68680568
100	110	10900	10739	9523.84	60864496	66884932
110	120	12502.2	10562.2	9261.81	58687712	64965216
120	130	14652.2	7691.08	8414.31	31296648	38440276
130	140	15413.8	6446.91	6802.27	22074216	28514874
140	150	15628.1	6263.12	6374.97	20711792	27102370
150	160	17235.3	6079.34	5897.56	19388768	25786846
160	170	19585.3	3208.18	4525.64	5451829	10145697
170	180	20546.9	1581.2	3457.73	1372087	3341516
180	190	20760.2	1397.41	2804	1052802	2670303
190	200	21350.5	1215.04	2149.61	775380	2144267
200	210	21943.8	288.479	1629.13	46360.3	309176
210	220	22092	106.107	918.802	4458.28	38605.1

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	20032.2	0.64492	2.45708	0.0015563
20	30	20032.2	3.28874	4.22285	0.0019661
30	40	20032.2	4.27215	4.32175	0.00199
40	50	7775.19	6.33395	8.77373	0.0029402
50	60	7775.19	9.04787	9.32256	0.0029518
60	70	7775.19	9.59779	9.87355	0.0029633
70	80	7775.19	14.8518	20.8282	0.0042942
80	90	7775.19	23.8843	27.1199	0.0048145
90	100	7775.19	27.5685	28.0176	0.0048227
100	110	7775.19	28.5104	29.0035	0.0048293
110	120	7775.19	36.7924	45.5707	0.0061484
120	130	7775.19	49.6437	53.8619	0.0065947
130	140	7775.19	54.5686	55.2755	0.0065983
140	150	7775.19	55.9826	56.6899	0.0066018
150	160	7775.19	66.8994	77.6218	0.0072857
160	170	7775.19	83.8681	90.203	0.0074941
170	180	7775.19	91.006	91.809	0.007495
180	190	7775.19	92.6047	93.4003	0.0074959
190	200	7775.19	97.0357	100.681	0.0075382
200	210	7775.19	101.481	102.282	0.0075384
210	220	7775.19	102.598	102.915	0.0075384

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 110.670 Actual deflection : 102.915 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	20475.5	0.73013	2.79132	0.001776
20	30	18127.5	3.74184	4.81303	0.0022588
30	40	17248.4	4.86967	4.92667	0.0022869
40	50	17199.8	7.2405	10.0318	0.0033483
50	60	15459.7	10.3439	10.6567	0.0033605
60	70	15297.5	10.97	11.2839	0.0033724
70	80	15135.3	16.8756	23.4003	0.0046188
80	90	12235.2	26.6634	30.0682	0.0050294
90	100	10901.2	30.5368	31.0058	0.0050355
100	110	10739	31.5203	32.035	0.0050403
110	120	10562.2	39.969	48.518	0.0058635
120	130	7691.08	52.3681	56.2875	0.0060773
130	140	6446.91	56.9387	57.5901	0.0060789
140	150	6263.12	58.2414	58.8929	0.0060803
150	160	6079.34	68.1286	77.525	0.0062979
160	170	3208.18	82.8885	88.266	0.0063314
170	180	1581.2	88.9444	89.6228	0.0063315
180	190	1397.41	90.2949	90.967	0.0063315
190	200	1215.04	94.0323	97.0981	0.0063336
200	210	288.479	97.7704	98.4427	0.0063336
210	220	106.107	98.7087	98.9747	0.0063336

Critical Wind Velocity for Tower Vibration:



From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	14.2977	89.3607
20	30	16.0601	100.376
30	40	17.7477	110.923
40	50	17.6596	110.372
50	60	15.9062	99.4137
60	70	15.9062	99.4137
70	80	17.6596	110.372
80	90	17.6596	110.372
90	100	15.9062	99.4137
100	110	15.8021	98.7633
110	120	17.4833	109.271
120	130	17.4833	109.271
130	140	15.6668	97.9172
140	150	15.6668	97.9172
150	160	17.4833	109.271
160	170	17.4833	109.271
170	180	15.6668	97.9172
180	190	15.6928	98.0801
190	200	17.4833	109.271
200	210	15.6928	98.0801
210	220	23.0757	144.223

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 110.670 Actual deflection : 98.975 mm

Total Deflection in the Operating Condition + Applied Forces:
Allowable deflection : 110.670 Actual deflection : 201.890 mm

**** Warning: Allowable Deflection Criteria Exceeded ****

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	39521.2	33753.2	6029285	5197076
20	30	39521.2	33753.2	6029285	5197076
30	40	35490	29716.3	5417588	4578529
40	50	31446.1	25672.5	4798670	3954380
50	60	31446.1	25672.5	4798670	3954380
60	70	31446.1	25672.5	4798670	3954380
70	80	31446.1	25672.5	4798670	3954380
80	90	31446.1	25672.5	4798670	3954380
90	100	31446.1	25672.5	4798670	3954380
100	110	23433.8	17660.1	3573978	2719028
110	120	23433.8	17660.1	3573978	2719028
120	130	23433.8	17660.1	3573978	2719028
130	140	23433.8	17660.1	3573978	2719028
140	150	23433.8	17660.1	3573978	2719028
150	160	23433.8	17660.1	3573978	2719028
160	170	23433.8	17660.1	3573978	2719028
170	180	23433.8	17660.1	3573978	2719028
180	190	23433.8	17660.1	3573978	2719028
190	200	23433.8	17660.1	3573978	2719028
200	210	23433.8	17660.1	3573978	2719028
210	220	23433.8	17660.1	3573978	2719028



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	115.247	196.556	-113.714	-146.514
20	30	82.3191	140.397	-113.714	-146.514
30	40	164.638	280.795	-111.677	-145.201
40	50	164.638	280.795	-109.343	-143.451
50	60	159.841	269.995	-109.343	-143.451
60	70	159.841	269.995	-109.343	-143.451
70	80	164.638	280.795	-109.343	-143.451
80	90	164.638	280.795	-109.343	-143.451
90	100	159.841	269.995	-109.343	-143.451
100	110	159.841	269.995	-103.244	-137.664
110	120	164.638	280.795	-103.244	-137.664
120	130	164.638	280.795	-103.244	-137.664
130	140	159.841	269.995	-103.244	-137.664
140	150	159.841	269.995	-103.244	-137.664
150	160	164.638	280.795	-103.244	-137.664
160	170	164.638	280.795	-103.244	-137.664
170	180	159.841	269.995	-103.244	-137.664
180	190	159.841	269.995	-103.244	-137.664
190	200	164.638	280.795	-103.244	-137.664
200	210	159.841	269.995	-103.244	-137.664
210	220	159.841	269.995	-103.244	-137.664



Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	22.2809	-1.13928	29.1339
40	50	25.7769	-1.30245	33.7051
50	60	...	...	...
60	70	...	...	...
70	80	25.7769	-1.30245	33.7051
80	90	25.7769	-1.30245	33.7051
90	100	...	...	...
100	110	...	...	...
110	120	37.4302	-1.84644	48.9426
120	130	37.4302	-1.84644	48.9426
130	140	...	...	...
140	150	...	...	...
150	160	37.4302	-1.84644	48.9426
160	170	37.4302	-1.84644	48.9426
170	180	...	...	...
180	190	...	...	...
190	200	37.4302	-1.84644	48.9426
200	210	...	...	...
210	220	...	...	...

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-4.12996	-4.26536	-5.33855	0.34226	0.34226
20	30	-3.65569	-3.7911	-4.86429	0.34226	0.34226
30	40	-4.04057	-4.04057	-4.04057	0.3885	0.3885
40	50	-4.50062	-4.50062	-4.50062	0.44982	0.44982
50	60	-4.26588	-4.26588	-4.26588	0.42625	0.42625
60	70	-4.13215	-4.13215	-4.13215	0.42625	0.42625
70	80	-3.99843	-3.99843	-3.99843	0.42625	0.42625
80	90	-3.47029	-3.47029	-3.47029	0.26195	0.26195
90	100	-3.26673	-3.26673	-3.26673	0.26195	0.26195
100	110	-4.54014	-4.54014	-4.54014	0.38096	0.38096
110	120	-4.29992	-4.29992	-4.29992	0.38096	0.38096
120	130	-3.6656	-3.6656	-3.6656	0.17718	0.17718
130	140	-2.71151	-2.71151	-2.71151	0.17718	0.17718
140	150	-2.47688	-2.47688	-2.47688	0.17718	0.17718
150	160	-2.22003	-2.22003	-2.22003	0.17718	0.17718
160	170	-1.58143	-1.58143	-1.58143	0.015385	0.015385
170	180	-1.16998	-1.16998	-1.16998	0.015385	0.015385
180	190	-0.93535	-0.93535	-0.93535	0.015385	0.015385
190	200	-0.70404	-0.70404	-0.70404	0.015385	0.015385
200	210	-0.52754	-0.52754	-0.52754	...	...
210	220	-0.29623	-0.29623	-0.29623	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

		Wght. Str.	Bend. Str.	Bend. Str.	Bend. Str.	Bend. Str.
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From	To	Hyd. Mom. MPa	Oper. Wind MPa	Oper. Equ. MPa	Hyd. Wind MPa	Hyd. Equ. MPa
10	20	0.34226	44.556	34.8833	22.278	...
20	30	0.34226	33.4133	28.5117	16.7066	...
30	40	0.3885	34.0638	29.9626	17.0319	...
40	50	0.44982	39.2225	34.5536	19.6112	...
50	60	0.42625	31.7711	29.6226	15.8855	...
60	70	0.42625	31.0465	29.1202	15.5233	...
70	80	0.42625	30.3296	28.622	15.1648	...
80	90	0.26195	19.9463	20.824	9.97317	...
90	100	0.26195	15.8965	17.3627	7.94826	...
100	110	0.38096	22.3775	24.591	11.1887	...
110	120	0.38096	21.5772	23.8852	10.7886	...
120	130	0.17718	11.5065	14.133	5.75327	...
130	140	0.17718	8.11582	10.4838	4.05791	...
140	150	0.17718	7.61491	9.96448	3.80746	...
150	160	0.17718	7.12849	9.48081	3.56424	...
160	170	0.015385	2.00442	3.73017	1.00221	...
170	180	0.015385	0.50446	1.22854	0.25223	...
180	190	0.015385	0.38707	0.98177	0.19354	...
190	200	0.015385	0.28508	0.78836	0.14254	...
200	210	...	0.017045	0.11367	0.0085224	...
210	220	...	0.0016391	0.014194	0.00081957	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	33.142
20	30	...	...	...	27.1338
30	40	...	...	...	28.5359
40	50	...	...	...	32.9095
50	60	...	...	...	28.2432
60	70	...	...	...	27.7661
70	80	...	...	...	27.2931
80	90	...	...	...	19.8829
90	100	...	...	...	16.5904
100	110	...	...	...	23.5003
110	120	...	...	...	22.8291
120	130	...	...	...	13.5334
130	140	...	...	...	10.0554
140	150	...	...	...	9.56208
150	160	...	...	...	9.10245
160	170	...	...	...	3.59783
170	180	...	...	...	1.18553
180	190	...	...	...	0.94794
190	200	...	...	...	0.76253
200	210	...	...	...	0.11118
210	220	...	...	...	0.013883

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	-0.24435	-0.11059
20	30	-0.24435	-0.12546
30	40	-0.27754	-0.14601
40	50	-0.32126	-0.17498
50	60	-0.32126	-0.18748
60	70	-0.32126	-0.19167
70	80	-0.32126	-0.19587
80	90	-0.32126	-0.21243
90	100	-0.32126	-0.21881

100	110	-0.46701	-0.32463
110	120	-0.46701	-0.33217
120	130	-0.46701	-0.35206
130	140	-0.46701	-0.38198
140	150	-0.46701	-0.38934
150	160	-0.46701	-0.39739
160	170	-0.46701	-0.41742
170	180	-0.46701	-0.43032
180	190	-0.46701	-0.43768
190	200	-0.46701	-0.44493
200	210	-0.46701	-0.45047
210	220	-0.46701	-0.45772

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	42.4861	42.4861	36.6219	36.6219
20	30	30.9216	30.9216	26.6536	26.6536
30	40	30.7234	30.7234	25.9651	25.9651
40	50	35.9456	35.9456	29.6213	29.6213
50	60	32.3973	32.3973	26.6972	26.6972
60	70	32.0309	32.0309	26.3953	26.3953
70	80	31.6646	31.6646	26.0935	26.0935
80	90	25.7654	25.7654	21.2322	21.2322
90	100	23.043	23.043	18.9887	18.9887
100	110	32.9794	32.9794	25.0903	25.0903
110	120	32.3952	32.3952	24.6458	24.6458
120	130	23.8158	23.8158	18.1187	18.1187
130	140	20.0852	20.0852	15.2805	15.2805
140	150	19.4724	19.4724	14.8143	14.8143
150	160	18.8595	18.8595	14.348	14.348
160	170	10.2801	10.2801	7.82098	7.82098
170	180	5.40563	5.40563	4.11252	4.11252
180	190	4.79278	4.79278	3.64628	3.64628
190	200	4.18565	4.18565	3.18438	3.18438
200	210	1.4079	1.4079	1.07111	1.07111
210	220	0.80077	0.80077	0.60921	0.60921

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...
160	170	...
170	180	...
180	190	...
190	200	...
200	210	...
210	220	...





Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	83.01	115.25	-91.76	113.71	0.7203	0.8069
20	60.78	82.32	-68.58	113.71	0.7383	0.6031
30	60.86	164.64	-69.49	111.68	0.3696	0.6223
40	70.80	164.64	-80.44	109.34	0.4300	0.7357
70	58.10	164.64	-66.74	109.34	0.3529	0.6104
80	42.18	164.64	-49.77	109.34	0.2562	0.4551
110	49.59	164.64	-59.12	103.24	0.3012	0.5726
120	31.37	164.64	-39.63	103.24	0.1905	0.3839
150	23.48	164.64	-28.85	103.24	0.1426	0.2795
160	10.25	164.64	-14.35	103.24	0.0623	0.1390
190	3.32	164.64	-5.66	103.24	0.0201	0.0548

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	71.73	115.25	-80.21	113.71	0.6224	0.7054
20	54.62	82.32	-62.18	113.71	0.6635	0.5468
30	55.46	164.64	-63.83	111.68	0.3369	0.5716
40	64.63	164.64	-73.98	109.34	0.3926	0.6766
70	55.19	164.64	-63.58	109.34	0.3352	0.5815
80	42.23	164.64	-49.59	109.34	0.2565	0.4536
110	50.97	164.64	-60.24	103.24	0.3096	0.5834
120	33.51	164.64	-41.54	103.24	0.2035	0.4024



150	25.52	164.64	-30.76	103.24	0.1550	0.2979
160	11.89	164.64	-15.89	103.24	0.0722	0.1539
190	3.81	164.64	-6.11	103.24	0.0232	0.0592

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	82.87	115.25	-91.89	113.71	0.7191	0.8081
20	60.64	82.32	-68.71	113.71	0.7367	0.6043
30	60.86	164.64	-69.49	111.68	0.3696	0.6223
40	70.80	164.64	-80.44	109.34	0.4300	0.7357
70	58.10	164.64	-66.74	109.34	0.3529	0.6104
80	42.18	164.64	-49.77	109.34	0.2562	0.4551
110	49.59	164.64	-59.12	103.24	0.3012	0.5726
120	31.37	164.64	-39.63	103.24	0.1905	0.3839
150	23.48	164.64	-28.85	103.24	0.1426	0.2795
160	10.25	164.64	-14.35	103.24	0.0623	0.1390
190	3.32	164.64	-5.66	103.24	0.0201	0.0548

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	73.34	115.25	-82.09	113.71	0.6363	0.7219
20	55.86	82.32	-63.69	113.71	0.6786	0.5601
30	56.89	164.64	-65.26	111.68	0.3455	0.5844
40	66.27	164.64	-75.62	109.34	0.4025	0.6916
70	56.52	164.64	-64.91	109.34	0.3433	0.5936
80	43.17	164.64	-50.53	109.34	0.2622	0.4622
110	52.03	164.64	-61.29	103.24	0.3160	0.5937
120	34.11	164.64	-42.14	103.24	0.2072	0.4082
150	25.90	164.64	-31.13	103.24	0.1573	0.3016
160	12.03	164.64	-16.02	103.24	0.0731	0.1552
190	3.84	164.64	-6.14	103.24	0.0233	0.0595

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	17.28	196.56	-27.96	146.51	0.0879	0.1908
20	12.18	140.40	-21.91	146.51	0.0868	0.1496
30	13.38	280.79	-21.46	145.20	0.0476	0.1478
40	15.56	280.79	-24.56	143.45	0.0554	0.1712
70	11.59	280.79	-19.59	143.45	0.0413	0.1366
80	6.76	280.79	-13.71	143.45	0.0241	0.0955
110	6.87	280.79	-15.47	137.66	0.0245	0.1124
120	2.26	280.79	-9.60	137.66	0.0081	0.0697
150	1.52	280.79	-5.96	137.66	0.0054	0.0433
160		280.79	-2.60	137.66		0.0189
190		280.79	-0.86	137.66		0.0063

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-5.68	146.51		0.0388
20		140.40	-5.21	146.51		0.0355
30		280.79	-4.43	145.20		0.0305
40		280.79	-4.95	143.45		0.0345
70		280.79	-4.42	143.45		0.0308
80		280.79	-3.73	143.45		0.0260
110		280.79	-4.68	137.66		0.0340
120		280.79	-3.84	137.66		0.0279
150		280.79	-2.40	137.66		0.0174
160		280.79	-1.60	137.66		0.0116
190		280.79	-0.72	137.66		0.0052

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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Node	Stress	Stress	Stress	Stress	Ratio	Ratio
10	82.87	115.25	-91.89	113.71	0.7191	0.8081
20	60.64	82.32	-68.71	113.71	0.7367	0.6043
30	83.14	164.64	-47.21	111.68	0.5050	0.4228
40	96.57	164.64	-54.66	109.34	0.5866	0.4999
70	83.88	164.64	-40.96	109.34	0.5095	0.3746
80	67.96	164.64	-23.99	109.34	0.4128	0.2194
110	87.02	164.64	-21.69	103.24	0.5285	0.2101
120	68.80	164.64	-2.20	103.24	0.4179	0.0213
150	60.91	164.64		103.24	0.3700	
160	47.68	164.64		103.24	0.2896	
190	40.75	164.64		103.24	0.2475	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	73.34	115.25	-82.09	113.71	0.6363	0.7219
20	55.86	82.32	-63.69	113.71	0.6786	0.5601
30	79.17	164.64	-42.98	111.68	0.4809	0.3849
40	92.05	164.64	-49.85	109.34	0.5591	0.4559
70	82.30	164.64	-39.13	109.34	0.4999	0.3579
80	68.95	164.64	-24.76	109.34	0.4188	0.2264
110	89.46	164.64	-23.86	103.24	0.5434	0.2311
120	71.54	164.64	-4.71	103.24	0.4345	0.0457
150	63.33	164.64		103.24	0.3847	
160	49.46	164.64		103.24	0.3004	
190	41.27	164.64		103.24	0.2507	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	82.87	115.25	-91.89	132.27	0.7191	0.6947
20	60.64	82.32	-68.71	132.27	0.7367	0.5195
30	59.72	164.64	-70.63	130.61	0.3627	0.5408
40	69.49	164.64	-81.74	128.58	0.4221	0.6357
70	56.80	164.64	-68.04	128.58	0.3450	0.5292
80	40.88	164.64	-51.07	128.58	0.2483	0.3972
110	47.74	164.64	-60.97	122.62	0.2900	0.4972
120	29.52	164.64	-41.48	122.62	0.1793	0.3383
150	21.63	164.64	-30.70	122.62	0.1314	0.2504
160	8.41	164.64	-16.19	122.62	0.0511	0.1321
190	1.47	164.64	-7.50	122.62	0.0089	0.0612

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	73.34	115.25	-82.09	132.27	0.6363	0.6206
20	55.86	82.32	-63.69	132.27	0.6786	0.4815
30	55.75	164.64	-66.40	130.61	0.3386	0.5084
40	64.97	164.64	-76.93	128.58	0.3946	0.5983
70	55.22	164.64	-66.21	128.58	0.3354	0.5149
80	41.87	164.64	-51.84	128.58	0.2543	0.4031
110	50.18	164.64	-63.14	122.62	0.3048	0.5149
120	32.26	164.64	-43.99	122.62	0.1960	0.3588
150	24.05	164.64	-32.98	122.62	0.1461	0.2690
160	10.18	164.64	-17.87	122.62	0.0618	0.1457
190	1.99	164.64	-7.98	122.62	0.0121	0.0651

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	17.28	196.56	-27.96	146.51	0.0879	0.1908
20	12.18	140.40	-21.91	146.51	0.0868	0.1496
30	42.51	280.79		145.20	0.1514	
40	49.27	280.79		143.45	0.1755	



70	45.30	280.79	143.45	0.1613
80	40.47	280.79	143.45	0.1441
110	55.81	280.79	137.66	0.1988
120	51.21	280.79	137.66	0.1824
150	50.46	280.79	137.66	0.1797
160	48.38	280.79	137.66	0.1723
190	48.40	280.79	137.66	0.1724

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-5.68	146.51		0.0388
20		140.40	-5.21	146.51		0.0355
30	25.48	280.79		145.20	0.0907	
40	29.65	280.79		143.45	0.1056	
70	30.13	280.79		143.45	0.1073	
80	30.50	280.79		143.45	0.1086	
110	45.02	280.79		137.66	0.1603	
120	45.45	280.79		137.66	0.1619	
150	46.90	280.79		137.66	0.1670	
160	47.38	280.79		137.66	0.1687	
190	48.25	280.79		137.66	0.1718	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.25	-4.47	113.71		0.0393
20		82.32	-4.00	113.71		0.0352
30	18.63	164.64		111.68	0.1132	
40	21.73	164.64		109.34	0.1320	
70	22.20	164.64		109.34	0.1349	
80	22.57	164.64		109.34	0.1371	
110	33.51	164.64		103.24	0.2035	
120	33.94	164.64		103.24	0.2062	
150	35.39	164.64		103.24	0.2149	
160	35.86	164.64		103.24	0.2178	
190	36.74	164.64		103.24	0.2232	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.25	-3.64	113.71		0.0320
20		82.32	-3.24	113.71		0.0285
30	18.90	164.64		111.68	0.1148	
40	22.10	164.64		109.34	0.1342	
70	22.51	164.64		109.34	0.1367	
80	22.94	164.64		109.34	0.1394	
110	34.19	164.64		103.24	0.2077	
120	34.67	164.64		103.24	0.2106	
150	35.76	164.64		103.24	0.2172	
160	36.24	164.64		103.24	0.2201	
190	36.90	164.64		103.24	0.2241	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.25	-4.61	113.71		0.0405
20		82.32	-4.13	113.71		0.0363
30	18.63	164.64		111.68	0.1132	
40	21.73	164.64		109.34	0.1320	
70	22.20	164.64		109.34	0.1349	
80	22.57	164.64		109.34	0.1371	
110	33.51	164.64		103.24	0.2035	
120	33.94	164.64		103.24	0.2062	
150	35.39	164.64		103.24	0.2149	
160	35.86	164.64		103.24	0.2178	



190 36.74 164.64 103.24 0.2232

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.25	-4.47	113.71		0.0393
20		82.32	-4.00	113.71		0.0352
30	18.63	164.64		111.68	0.1132	
40	21.73	164.64		109.34	0.1320	
70	22.20	164.64		109.34	0.1349	
80	22.57	164.64		109.34	0.1371	
110	33.51	164.64		103.24	0.2035	
120	33.94	164.64		103.24	0.2062	
150	35.39	164.64		103.24	0.2149	
160	35.86	164.64		103.24	0.2178	
190	36.74	164.64		103.24	0.2232	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.25	-4.61	113.71		0.0405
20		82.32	-4.13	113.71		0.0363
30		164.64	-4.43	111.68		0.0397
40		164.64	-4.95	109.34		0.0453
70		164.64	-4.42	109.34		0.0405
80		164.64	-3.73	109.34		0.0341
110		164.64	-4.68	103.24		0.0453
120		164.64	-3.84	103.24		0.0372
150		164.64	-2.40	103.24		0.0232
160		164.64	-1.60	103.24		0.0155
190		164.64	-0.72	103.24		0.0070

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	38.45	96.04	-47.20	94.76	0.4004	0.4981
20	27.35	68.60	-35.18	94.76	0.3987	0.3713
30	49.21	137.20	-13.02	93.06	0.3587	0.1399
40	57.50	137.20	-15.29	91.12	0.4191	0.1678
70	53.67	137.20	-10.51	91.12	0.3912	0.1153
80	48.12	137.20	-3.93	91.12	0.3507	0.0432
110	65.57	137.20		86.04	0.4780	
120	57.41	137.20		86.04	0.4184	
150	53.85	137.20		86.04	0.3925	
160	45.73	137.20		86.04	0.3333	
190	40.48	137.20		86.04	0.2951	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	38.45	96.04	-47.20	110.23	0.4004	0.4283
20	27.35	68.60	-35.18	110.23	0.3987	0.3192
30	25.79	137.20	-36.44	108.84	0.1879	0.3348
40	30.42	137.20	-42.37	107.15	0.2217	0.3955
70	26.59	137.20	-37.59	107.15	0.1938	0.3508
80	21.04	137.20	-31.01	107.15	0.1534	0.2894
110	26.30	137.20	-39.25	102.18	0.1917	0.3842
120	18.13	137.20	-29.86	102.18	0.1321	0.2922
150	14.57	137.20	-23.50	102.18	0.1062	0.2300
160	6.45	137.20	-14.14	102.18	0.0470	0.1384
190	1.21	137.20	-7.20	102.18	0.0088	0.0704

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10		115.25	-4.61	113.71	0.0405
20		82.32	-4.13	113.71	0.0363
30	18.63	164.64		111.68	0.1132
40	21.73	164.64		109.34	0.1320
70	22.20	164.64		109.34	0.1349
80	22.57	164.64		109.34	0.1371
110	33.51	164.64		103.24	0.2035
120	33.94	164.64		103.24	0.2062
150	35.39	164.64		103.24	0.2149
160	35.86	164.64		103.24	0.2178
190	36.74	164.64		103.24	0.2232

Absolute Maximum of the all of the Stress Ratio's 0.8081

Governing Element: Skirt-2

Governing Load Case 3 : NP+OW+WI+FW+BW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-2	...	-4.5	0.3	87.0	82.9	+ Tens
Skirt-2	...	-4.5	-0.3	-87.0	-91.9	- Comp
Skirt -1	...	-4.0	0.3	64.3	60.6	+ Tens
Skirt -1	...	-4.0	-0.3	-64.3	-68.7	- Comp
Bottom head	...	-4.3	0.4	64.8	60.9	+ Tens
Bottom head	...	-4.3	-0.4	-64.8	-69.5	- Comp
Shell-1	...	-4.8	0.4	75.2	70.8	+ Tens
Shell-1	...	-4.8	-0.4	-75.2	-80.4	- Comp
Shell-2	...	-4.3	0.4	62.0	58.1	+ Tens
Shell-2	...	-4.3	-0.4	-62.0	-66.7	- Comp
Shell-3	...	-3.8	0.3	45.7	42.2	+ Tens
Shell-3	...	-3.8	-0.3	-45.7	-49.8	- Comp
Shell-4	...	-4.8	0.4	54.0	49.6	+ Tens
Shell-4	...	-4.8	-0.4	-54.0	-59.1	- Comp
Shell-5	...	-4.1	0.2	35.3	31.4	+ Tens
Shell-5	...	-4.1	-0.2	-35.3	-39.6	- Comp
Shell-6	...	-2.7	0.2	26.0	23.5	+ Tens
Shell-6	...	-2.7	-0.2	-26.0	-28.9	- Comp
Shell-7	...	-2.0	...	12.3	10.3	+ Tens
Shell-7	...	-2.0	...	-12.3	-14.3	- Comp
Shell-8	...	-1.2	...	4.5	3.3	+ Tens
Shell-8	...	-1.2	...	-4.5	-5.7	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

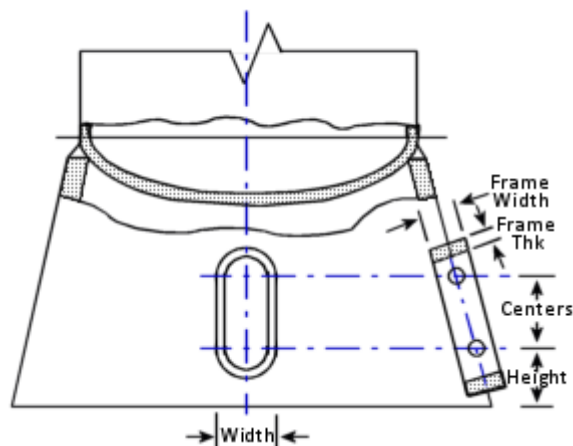
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt-2

Total vessel (empty) weight	Fg	122338.539 N
Weight of vessel contents (liquid)	Ff	4569.903 N
Vessel Gross Weight	Fg + Ff	126908.438 N
Skirt Mean Diameter	Dz	632.000 mm
Skirt Analysis Thickness (Corroded)	ez	17.000 mm
Skirt Height	H	3000.000 mm
Additional Axial Force at section 1-1	F1	-8246.810 N
Moment at section 1-1	M1	334460608.000 N-mm
Applied Moment - Bottom of Skirt	Mb	452509536.000 N-mm
Weight at Bottom of Skirt	W	143957.516 N



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	150.0000	270.0	500.00	0.00	180.000	16.000

Section properties at least metal area [A]:

Number of Openings At Critical Height = 1

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	256.2703	283.7297

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
-76.270	256.270



Height of Weakest Section	H	500.100	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	0.000	mm
Centroid Position in Y direction	h	-26.053	mm
Principle Moment of Inertia	Ixx	1966261376.000	mm ⁴
Principle Moment of Inertia	Iyy	1706346496.000	mm ⁴
Distance to Extreme Fiber	ybar	350.553	mm
Distance to Extreme Fiber	xbar	324.500	mm
Minimum Section Modulus	W4	5258391.500	mm ³
Metal Area at the Cutting Plane	A4	36938.707	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(150.0/632.0)$$

$$= 0.2396 \text{ rad } (13.7297 \text{ deg})$$

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 632.0/2$$

$$= 316.0 \text{ mm}$$

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.24 \cdot \sqrt{(316.0/17.0)}$$

$$= 1.033$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= e_{x(\text{new})} - c_i - c_{\text{ext}}$$

$$= 20.0 - 3.0 - 0.0$$

$$= 17.0 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (M_{\text{bottom}} - M_{\text{top}}) \cdot (H - \text{CutHeight}) / H + M_{\text{top}}$$

$$= (.45251\text{E}+09 - 0.33446\text{E}+09) \cdot (3000.0 - 500.1) / 3000.0 + 0.33446\text{E}+09$$

$$= 0.43283\text{E}+09 \text{ N-mm}$$

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx} / y_{\text{bar}} = 0.19663\text{E}+10 / 350.553 = 5609037 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / x_{\text{bar}} = 0.17063\text{E}+10 / 324.5 = 5258392 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(5609037, 5258392) = 5258392 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A4))}$$

$$= \sqrt{(\min(0.19663\text{E}+10, 0.17063\text{E}+10) / 36938.707)}$$

$$= 214.928 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 214.928 / 3000.0$$

$$= 0.072$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(e_x^2 + e_y^2)}$$

$$= \sqrt{(0.^2 + -26.053^2)}$$

$$= 20.900 \text{ mm}$$

$$\text{Where } e_x = 0. : e_y = -26.053$$

Moment Increment at Centroid [$\Delta M4$]:



$$\begin{aligned}
 &= \text{abs}(F1 - Ff - Fg) \cdot e = F4 \cdot e \\
 &= \text{abs}(-8246.81 - 122338.54 - 4569.9) \cdot 26.053 \\
 &= 3522587.500 \text{ N-mm}
 \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Protrusion, Figure 16.12-5 [ht]:

$$\begin{aligned}
 &= \text{abs}(lt - ea3) / 2 \\
 &= \text{abs}(180.0 - 17.0) / 2 \\
 &= 81.500 \text{ mm}
 \end{aligned}$$

Mean Diameter of Opening, Figure 16.12-5 [d]:

$$\begin{aligned}
 &= \text{opening width} - et \\
 &= 150.0 - 16.0 \\
 &= 134.000 \text{ mm}
 \end{aligned}$$

Angle, 16.12-72 [δ]:

$$\begin{aligned}
 &= \sin^{-1}(d / D3) \\
 &= \sin^{-1}(134.0 / 632.0) \\
 &= 0.214 \text{ radians}
 \end{aligned}$$

Area in the Frame [At]:

$$\begin{aligned}
 &= 2 \cdot lt \cdot et \\
 &= 2 \cdot 180.0 \cdot 16.0 \\
 &= 5760.000 \text{ mm}^2
 \end{aligned}$$

Cross Sectional Shell Area at Opening, 16.12-73(b) [AS]:

$$\begin{aligned}
 &= (\pi - \delta) \cdot D3 \cdot ea3 \\
 &= (\pi - 0.214) \cdot 632.0 \cdot 17.0 \\
 &= 31457.850 \text{ mm}^2
 \end{aligned}$$

Combined Area at Opening, 16.12-73 [A4]:

$$\begin{aligned}
 &= AS + AT \\
 &= 31457.85 + 5760.0 \\
 &= 37217.848 \text{ mm}^2
 \end{aligned}$$

Computed Offset, 16.12-74(b) [yt]:

$$\begin{aligned}
 &= 0.5 \cdot D3 \cdot \cos(\delta) + ht - 0.5 \cdot lt \\
 &= 0.5 \cdot 632.0 \cdot \cos(0.214) + 81.5 - 0.5 \cdot 180.0 \\
 &= 300.315 \text{ mm}
 \end{aligned}$$

Distance to Neutral Axis, 16.12-74 [yG]:

$$\begin{aligned}
 &= (0.5 \cdot D3 \cdot ea3 \cdot d - 2 \cdot lt \cdot eat \cdot yt) / A4 \\
 &= (0.5 \cdot 632.0 \cdot 17.0 \cdot 134.0 - 2 \cdot 180.0 \cdot 16.0 \cdot 300.315) / 37217.848 \\
 &= -27.137 \text{ mm}
 \end{aligned}$$

Computed Offset [ymax]:

$$\begin{aligned}
 &= \max[\text{abs}(0.5 \cdot d3 \cdot \cos(\delta) + ht + yG); \text{abs}(0.5 \cdot D3 - yG)] \\
 &= \max[\text{abs}(0.5 \cdot 632.0 \cdot \cos(0.214) + 81.5 + -27.137); \text{abs}(0.5 \cdot 632.0 - -27.137)] \\
 &= 363.179 \text{ mm}
 \end{aligned}$$

Inertia Value, 16.12-75(b) [IS]:

$$\begin{aligned}
 &= (\pi - \delta - \sin(\delta)) \cdot \cos(\delta) \cdot ea3 \cdot (D3/2)^3 \\
 &= (\pi - 0.214 - \sin(0.214)) \cdot \cos(0.214) \cdot 17.0 \cdot (632.0/2)^3 \\
 &= 1459476224.0 \text{ mm}^4
 \end{aligned}$$

Section Modulus [W4]:

$$\begin{aligned}
 &= (Is + At(yt^2 + lt^2/12) - A4 \cdot yG^2) / ymax \\
 &= (0.14595E+10 + 5760.0(300.315^2 + 180.0^2/12) - 37217.848 \cdot -27.137^2) / 363.179 \\
 &= 5416376.500 \text{ mm}^3
 \end{aligned}$$

Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned}
 &= \min(1; A4 / (\pi \cdot D3 \cdot ea3)) \\
 &= \min(1; 37217.848 / (\pi \cdot 632.0 \cdot 17.0)) \\
 &= 1.000
 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned}
 &= \min(1; 4 \cdot W4 / (\pi \cdot D^2/3 / (\pi \cdot D3 \cdot ea3))) \\
 &= \min(1; 4.0 \cdot 5416377 / (\pi \cdot 632.0^2 \cdot 17.0)) \\
 &= 1.000
 \end{aligned}$$



Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi/4 \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi/4 \cdot 632.0^2 \cdot 17.0 \cdot 113.7 \\ &= 606632640.0 \text{ N-mm} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d^3 \cdot e a^3 \cdot \sigma_{call} \\ &= \pi \cdot 632.0 \cdot 17.0 \cdot 113.7 \\ &= 3837891.2 \text{ N} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F_4 / (\psi_1 \cdot F_{c,max}) + (\text{abs}(M_4) + F_4 \cdot \text{abs}(y_q)) / (\psi_2 \cdot M_{max}) \\ &= 135155.2 / (1.0 \cdot 3837891) + (\text{abs}(0.43283\text{E}+09) + 135155.2 \cdot \text{abs}(-27.137)) / \\ &\quad (1.0 \cdot 0.60663\text{E}+09) \\ &= 0.755 \text{ must be } \leq \text{ to } 1, \text{ Passed} \end{aligned}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Skirt -1

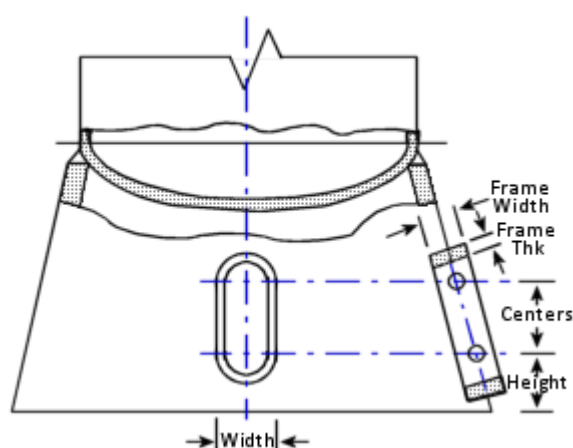
Total vessel (empty) weight	Fg	122338.539 N
Head Weight below section 2-2	DeltaFg	4528.424 N
Weight of vessel contents (liquid)	Ff	4569.903 N
Vessel Gross Weight	Fg + Ff	126908.438 N
Skirt Mean Diameter	Dz	632.000 mm
Skirt Analysis Thickness (Corroded)	ez	17.000 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	-8246.810 N
Moment at section 1-1	M1	296725184.000 N-mm
Applied Moment - Bottom of Skirt	Mb	334460608.000 N-mm
Weight at Bottom of Skirt	W	127950.930 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:	Elliptical
Head Mean Diameter at Skirt Top	Db 630.601 mm
Head Corroded Thickness	eb 15.000 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 155.4 \text{ mm} \\
 K &= D_i / (2 * h_i) = 615.601 / (2 * 155.4) = 1.9807 \\
 R &= D_i (0.44 * K + 0.02) = 615.601 * (0.44 * 1.981 + 0.02) = 548.8119 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 615.601 * ((0.5 / 1.981) - 0.08) = 106.1522 \text{ mm}
 \end{aligned}$$



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	80.0000	0.0	800.00	0.00	80.000	16.000
2	150.0000	0.0	200.00	0.00	80.000	16.000
3	80.0000	180.0	800.00	0.00	80.000	16.000

Interference check:

Access	Access	Pitch	Ligament
--------	--------	-------	----------



		mm	mm
1	2	600.0000	485.0000
2	3	1159.9595	1044.9596
1	3	992.7266	912.7266

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 600.0000 mm
 Minimum Ligament between Openings is: 485.0000 mm

Recommended minimum Ligament [IMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(632.0 \cdot 17.0)} \text{ mm}$$

$$= 207.307 \text{ mm}$$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 3

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-7.2721	7.2721
3	172.7279	187.2721

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
7.272	0.000
0.000	172.728
-172.728	-7.272

Height of Weakest Section	H	799.970	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	20.900	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	1689344768.000	mm ⁴
Principle Moment of Inertia	Iyy	2160241152.000	mm ⁴
Distance to Extreme Fiber	ybar	324.500	mm
Distance to Extreme Fiber	xbar	345.400	mm
Minimum Section Modulus	W4	5205998.000	mm ³
Metal Area at the Cutting Plane	A4	38705.957	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(150.0/632.0)$$

$$= 0.2396 \text{ rad (13.7297 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 632.0/2$$

$$= 316.0 \text{ mm}$$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
2560.0	1089.9	...	...
2560.0	2295.4	...	...
2560.0	1089.9	...	...

This calculation of ratio below is per an earlier version of EN 13445.



Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.24 \cdot \sqrt{(316.0 / 17.0)}$$

$$= 1.033$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= e_{x(new)} - c_i - c_{ext}$$

$$= 20.0 - 3.0 - 0.0$$

$$= 17.0 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (M_{bottom} - M_{top}) \cdot (H - CutHeight) / H + M_{top}$$

$$= (.33446E+09 - 0.29673E+09) \cdot (1000.0 - 799.97) / 1000.0 + 0.29673E+09$$

$$= 0.30427E+09 \text{ N-mm}$$

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx} / y_{bar} = 0.16893E+10 / 324.5 = 5205998 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / x_{bar} = 0.21602E+10 / 345.4 = 6254320 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(5205998, 6254320) = 5205998 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A_4))}$$

$$= \sqrt{(\min(0.16893E+10, 0.21602E+10) / 38705.957)}$$

$$= 208.915 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 208.915 / 1000.0$$

$$= 0.209$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(e_x^2 + e_y^2)}$$

$$= \sqrt{(20.9^2 + 0.^2)}$$

$$= 26.053 \text{ mm}$$

Where $e_x = 20.9$: $e_y = 0$.

Moment Increment at Centroid [ΔM_4]:

$$= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e$$

$$= \text{abs}(-8246.81 - 122338.54 - 4569.9) \cdot 20.9$$

$$= 2825909.250 \text{ N-mm}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= -8246.81 - 122338.539 - 4569.903 + 4 \cdot 0.29673E+09 / 632.0$$

$$= 1742091 \text{ N}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= -8246.81 - 122338.539 - 4569.903 - 4 \cdot 0.29673E+09 / 632.0$$

$$= -2012401.4 \text{ N}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (1742091 + 4528.424 + 4569.903) / (\pi \cdot 630.601 \cdot 15.0) + 2.215 \cdot 630.601 / (4 \cdot 15.0)$$



$$= 82.21 \text{ MPa}$$

Stress [σ_{1qm}]:

$$= (Fzq + \Delta Fq + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-0.2E+07 + 4528.424 + 4569.903) / (\pi \cdot 630.601 \cdot 15.0) + 2.215 \cdot 630.601 / (4 \cdot 15.0)$$

$$= -44.14 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (Ff + \Delta Fq) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (4569.903 + 4528.424) / (\pi \cdot 630.601 \cdot 15.0) + 2.215 \cdot 630.601 / (4 \cdot 15.0)$$

$$= 23.58 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= Fzp / (\pi \cdot Dz \cdot ez)$$

$$= 1742091 / (\pi \cdot 632.0 \cdot 17.0)$$

$$= 51.62 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= Fzq / (\pi \cdot Dz \cdot ez)$$

$$= -2012401.375 / (\pi \cdot 632.0 \cdot 17.0)$$

$$= -59.63 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$$

$$= 1 - 0.5(630.6 + 15.0 - 632.0 + 17.0) / (106.15 + 15.0)$$

$$= 0.8737$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.874) = 29.108 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma)}$$

$$= 0.5 \cdot \sqrt{15.0^2 + 17.0^2 + 2 \cdot 15.0 \cdot 17.0 \cdot 0.874}$$

$$= 15.489 \text{ mm}$$

Bending moment at verification points p and q [M_p & M_q]:

$$M_p = a \cdot Fzp = 15.48 \cdot 1742090.9 = 26.9935 \times 10^6 \text{ N-mm}$$

$$M_q = a \cdot Fzq = 15.48 \cdot -2012401.4 = -31.1820 \times 10^6 \text{ N-mm}$$

Correction factor [C]:

$$= 0.63 - 0.057(eb/ez)^2$$

$$= 0.63 - 0.057(15.0/17.0)^2$$

$$= 0.586$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot Db \cdot eb^2)$$

$$= 0.586 \cdot 6 \cdot 26993550 / (\pi \cdot 630.601 \cdot 15.0^2)$$

$$= 212.72 \text{ MPa}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot Db \cdot eb^2)$$

$$= 0.586 \cdot 6 \cdot -0.3E+08 / (\pi \cdot 630.601 \cdot 15.0^2)$$

$$= -245.72 \text{ MPa}$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot Dz \cdot ez^2)$$

$$= 0.586 \cdot 6 \cdot 26993550 / (\pi \cdot 632.0 \cdot 17.0^2)$$

$$= 165.24 \text{ MPa}$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot Dz \cdot ez^2)$$

$$= 0.586 \cdot 6 \cdot -0.3E+08 / (\pi \cdot 632.0 \cdot 17.0^2)$$

$$= -190.88 \text{ MPa}$$



Geometry variables, 16.12-32 to 36:

$$eb/Db = 0.591/24.827 = 0.0238$$

$$y = 125 * eb/Db = 125 * 15.0/630.601 = 2.9734$$

As this an elliptical head, it will be treated a a Korbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 * y = 4.2 - 0.2 * 2.973 = 3.6053$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) * Db ((y / yA) * \alpha - 1) / (4 * eb)$$

$$= (2.215 + 0.0) * 630.6 ((29.11/40.0) * 3.605 - 1) / (4 * 15.0)$$

$$= 37.79 \text{ MPa}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 82.213 - 212.717 + 37.792$$

$$= -92.71 \text{ MPa}$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 82.213 + 212.717 - 37.792$$

$$= 257.14 \text{ MPa}$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= -44.142 - -245.724 + 37.792$$

$$= 239.37 \text{ MPa}$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= -44.142 + -245.724 - 37.792$$

$$= -327.66 \text{ MPa}$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 23.584 + 212.717 + 37.792$$

$$= 274.09 \text{ MPa}$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 23.584 - 212.717 - 37.792$$

$$= -226.93 \text{ MPa}$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 23.584 + -245.724 + 37.792$$

$$= -184.35 \text{ MPa}$$

Stress [σ_{2qotot}]:

$$= \sigma_{2qm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 23.584 - -245.724 - 37.792$$

$$= 231.51 \text{ MPa}$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= 51.617 - 165.244$$

$$= -113.63 \text{ MPa}$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= 51.617 + 165.244$$

$$= 216.86 \text{ MPa}$$

Stress [σ_{3qitot}]:



$$\begin{aligned}
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -59.626 - -190.884 \\
 &= 131.26 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qotot}] &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -59.626 + -190.884 \\
 &= -250.51 \text{ MPa}
 \end{aligned}$$

Allowable stresses:

$$\begin{aligned}
 \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 137.198(3 - (1/1.5)) \cdot (82.213/137.198)^2 \\
 &= 378.75 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 137.198(3 - (1/1.5)) \cdot (-44.142/137.198)^2 \\
 &= 402.13 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 137.198(3 - (1/1.5)) \cdot (23.584/137.198)^2 \\
 &= 408.89 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\
 &= 137.198(3 - (1/1.5)) \cdot (23.584/137.198)^2 \\
 &= 408.89 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\
 &= 137.198(3 - (1/1.5)) \cdot (51.617/137.198)^2 \\
 &= 398.65 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\
 &= 137.198(3 - (1/1.5)) \cdot (-59.626/137.198)^2 \\
 &= 394.32 \text{ MPa}
 \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

$$\begin{aligned}
 \text{Weakening Factor, 16.12-70(b) [psi1]:} \\
 &= \min(1; A4 / (\pi \cdot D3 \cdot ea3)) \\
 &= \min(1; 38705.957 / (\pi \cdot 632.0 \cdot 17.0)) \\
 &= 1.000
 \end{aligned}$$

$$\begin{aligned}
 \text{Weakening Factor, 16.12-70(c) [psi2]:} \\
 &= \min(1; 4 \cdot W4 / (\pi \cdot D^2 / 3 / (\pi \cdot D3 \cdot ea3))) \\
 &= \min(1; 4.0 \cdot 6254320 / (\pi \cdot 632.0^2 \cdot 17.0)) \\
 &= 1.000
 \end{aligned}$$

$$\begin{aligned}
 \text{Maximum Bending Moment [Mmax]:} \\
 &= \pi / 4 \cdot d3^2 \cdot ea3 \cdot \sigma_{call} \\
 &= \pi / 4 \cdot 632.0^2 \cdot 17.0 \cdot 113.7 \\
 &= 606632640.0 \text{ N-mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Maximum Compressive Force [Fc,max]:} \\
 &= \pi \cdot d3 \cdot ea3 \cdot \sigma_{call} \\
 &= \pi \cdot 632.0 \cdot 17.0 \cdot 113.7 \\
 &= 3837891.2 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Unity Check of Weakened Section, 16.12-70 [UC1]:} \\
 &= F4 / (\psi1 \cdot Fc,max) + (abs(M4) + F4 \cdot abs(yq)) / (\psi2 \cdot Mmax) \\
 &= 135155.2 / (1.0 \cdot 3837891) + (abs(0.30427E+09) + 135155.2 \cdot abs(20.9)) / \\
 &\quad (1.0 \cdot 0.60663E+09) \\
 &= 0.541 \text{ must be } \leq \text{ to } 1, \text{ Passed}
 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is



an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	-92.71	378.75	Pass	46
Stot1po	257.14	378.75	Pass	47
Stot1qi	239.37	402.13	Pass	48
Stot1qo	-327.66	402.13	Pass	49
Stot2pi	274.09	408.89	Pass	50
Stot2po	-226.93	408.89	Pass	51
Stot2qi	-184.35	408.89	Pass	52
Stot2qo	231.51	408.89	Pass	53
Stot3pi	-113.63	398.65	Pass	54
Stot3po	216.86	398.65	Pass	55
Stot3qi	131.26	394.32	Pass	56
Stot3qo	-250.51	394.32	Pass	57

The stresses are satisfactory in the region without openings.
The stresses are satisfactory in the region with the opening(s).

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**Shop/Field Installation Options :**

Packing is installed in the Shop.

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Packing	13528.192 mm
Center of Gravity of the Liquid	4845.181 mm
Center of Gravity of the Insulation	12765.857 mm
Center of Gravity of the Nozzles	12332.549 mm
Center of Gravity of the Added Weights (Operating)	12556.648 mm
Center of Gravity of the Added Weights (Empty)	12540.107 mm
Center of Gravity of Bare Shell New and Cold	11555.315 mm
Center of Gravity of Bare Shell Corroded	11631.950 mm
Vessel CG in the Operating Condition	11815.082 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	11963.853 mm
Vessel CG in the Test Condition	12239.775 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	649.0000	mm
Skirt Thickness	STHK	20.0000	mm
Skirt Internal Corrosion Allowance	SCA	3.0000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	54.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	MPa
Basing Yield Stress		262.00	MPa
Inside Diameter of Basing	DI, D4	465.0000	mm
Outside Diameter of Basing	DOU, D5	985.0001	mm

Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	138.00	MPa
Number of Bolts	RN, nb	12	
Diameter of Bolt Circle	DC, DBC	817.0000	mm
Bolt Allowable Shear Stress		82.800	MPa

Ultimate Comp. Strength of Concrete	FPC	41.0	MPa
Allowable Comp. Strength of Concrete	FC	22.2	MPa
Modular ratio Steel/Concrete		9.833	

Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	158.0000	mm
Width of Gussets at Base Plate	BWDT	158.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	262.0	MPa
Height of Chair	h1	350.0000	mm
Distance between Gussets	RG, b6	96.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	48.0000	mm
Number of Gussets per bolt	NG	2	

Thickness of Top Plate or Ring	TTA, e5	44.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	168.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm

External Corrosion Allowance	CA	3.0000	mm
------------------------------	----	--------	----

Dead Weight of Vessel	DW	144006.2	N
Operating Weight of Vessel	ROW	152204.3	N
Earthquake Moment on Basing	EQMOM	182232256.0	N-mm
Wind Moment on Basing	WIMOM	232517936.0	N-mm

User Moment at the Base [Earthquake Case]		220874480.0	N-mm (UEQMOM)
User Moment at the Base [WindLoad Case]		220874480.0	N-mm (UWIMOM)
Percent Bolt Preload	ppl	100.0	

Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	

Factor for Increase of Allowables	Fact	1.0000	
-----------------------------------	------	--------	--

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 0.40311E+09/817.0) - 147690)/12$$

$$= 152092.1 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.45339E+09/817.0) - 144006)/12$$

$$= 172907.1 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/817.0) - 152204)/12$$

$$= -12683.7 \text{ N}$$

Governing Bolt Load Condition, Wind + Dead Weight Condition:

Area Available in 1 Bolt (corroded) Abss : 1155.1653 mm²

Area Available in all the Bolts Abss * RN : 13861.9834 mm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.299	0.149	1.506	2.445	0.438	0.781	629433.7	773439.9	75287.7
4	0.224	0.261	1.293	2.608	0.454	0.777	629728.5	773734.7	70617.8
6	0.280	0.271	1.455	2.485	0.442	0.780	629387.6	773393.7	74052.1
8	0.275	0.273	1.442	2.496	0.443	0.780	629387.8	773393.9	73750.8
10	0.272	0.271	1.432	2.503	0.444	0.780	629391.3	773397.4	73526.6
12	0.271	0.271	1.430	2.505	0.444	0.780	629392.7	773398.8	73470.9

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 629393 / (5.401 \cdot 817.0 \cdot 2.505)$$

$$= 113.89 \text{ MPa, Should be } \leq 138.0 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 12 \cdot 1155.165 / (\pi \cdot 817.0)$$

$$= 5.401 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 4.304[(2 \cdot 0.271 \cdot 817.0 + 260.0) / (2 \cdot 0.271 \cdot 817.0)]$$

$$= 6.831 \text{ MPa, Should be } \leq 22.2 \text{ MPa}$$

Values for table 10.3, l = 168.0, b = 106.945, l/b = 1.5709

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(9697.655, 23932.594) = 23932.594 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 23932.594 / 172.9)} + 3.0$$

$$= 31.818 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 168.0, b = 96.0, b/l = 0.571429

As b/l (0.571) is less than 1, inverting b/l = 1.75.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 159399.25 / (4 \cdot \pi) [1.3(\ln((2 \cdot 168.0 \cdot \sin(\pi \cdot 84.0/168.0)) / (\pi \cdot 40.0))) + 1] - [(0.7 - 0.044) \cdot 159399.25 / (4 \cdot \pi)]$$

$$= 20587.5703 \text{ N}$$



Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(20587.57) / 174.66) + 3.0}$$

$$= 29.5947 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 159399.25 / (124.11 \cdot 158.0) + 3.0$$

$$= 12.525 \text{ (not less than } 9.525 + 3.0 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 817.0 / 12 = 213.89 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76(P \cdot a / (m \cdot (h + t_{ba}) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76(159399 \cdot 84.0 / (213.89 \cdot 306.0 \cdot 206))^{2/3} \cdot 324.51^{1/3} + \text{Ca}$$

$$= 12.047 + 3.0 = 15.047 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 40507.7 / (1155.165 \cdot 12)$$

$$= 2.9 \text{ MPa. Must be less than } 82.8 \text{ MPa.}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	31.8184	mm
Actual Basing Thickness as entered by user	54.0000	mm
Required Thickness of Chair Cap	29.5947	mm
Actual Top Ring Thickness as entered by user	44.0000	mm
Required Gusset thickness, + CA	12.5250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	15.0471	mm
Given Thickness of Skirt	20.0000	mm
Required Gusset Height to meet local stress	128.5469	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 91.894 \cdot (20.0 - 3.0)$$

$$= 1562.07 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 1562 / [(0.4 \cdot 211) \cdot 2 \cdot 0.707]$$

$$= 13.078 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 159399.2 / (180.0 + 2 \cdot (252.0 + 44.0))$$

$$= 206.476 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 159399.2 \cdot 84.0 / (180.0 \cdot (296.0) + 0.6667 \cdot (296.0)^2)$$

$$= 119.88 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{W_v^2 + W_h^2}$$

$$= \sqrt{206.48^2 + 119.88^2}$$

$$= 238.754 \text{ N/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$



$$= 238.75 / [(0.4 \cdot 211) \cdot 2 \cdot 0.707]$$
$$= 2.0 \text{ mm}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$
$$= 159399.2 / (2 \cdot 158.0)$$
$$= 504.428 \text{ N/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$
$$= 159399.2 \cdot 84.0 / (2 \cdot 296.0 \cdot 158.0)$$
$$= 143.148 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$
$$= \sqrt{504.43^2 + 143.15^2}$$
$$= 524.346 \text{ N/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$
$$= 524.35 / [(0.4 \cdot 211) \cdot 2 \cdot 0.707]$$
$$= 4.39 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	13.0777	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	4.3898	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N380

From: 30

Pressure for Reinforcement Calculations	P	2.2309	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	609.60	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	18.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
-----------------	--	--------	--

Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

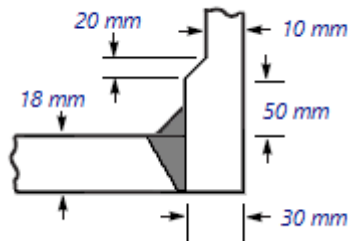
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		500.0	mm
Centerline Distance to Tangent		800.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N380

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	80.000 mm.
Actual Thickness Used in Calculation	10.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 2.23 \cdot 615.601 \cdot 0.8917 / (2 \cdot 137.2 \cdot 1.0 - 0.2 \cdot 2.23)$
 $= 4.4702 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.23 \cdot 43.0 / (133 \cdot 1.0 - 0.6 \cdot 2.23)$
 $= 0.7275 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4327 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 20.0 / 0.5773$
 $= 34.6410 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.0000 mm
Parallel to Vessel Wall, opening length	d	86.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm

Weld Strength Reduction Factor [fr1]:

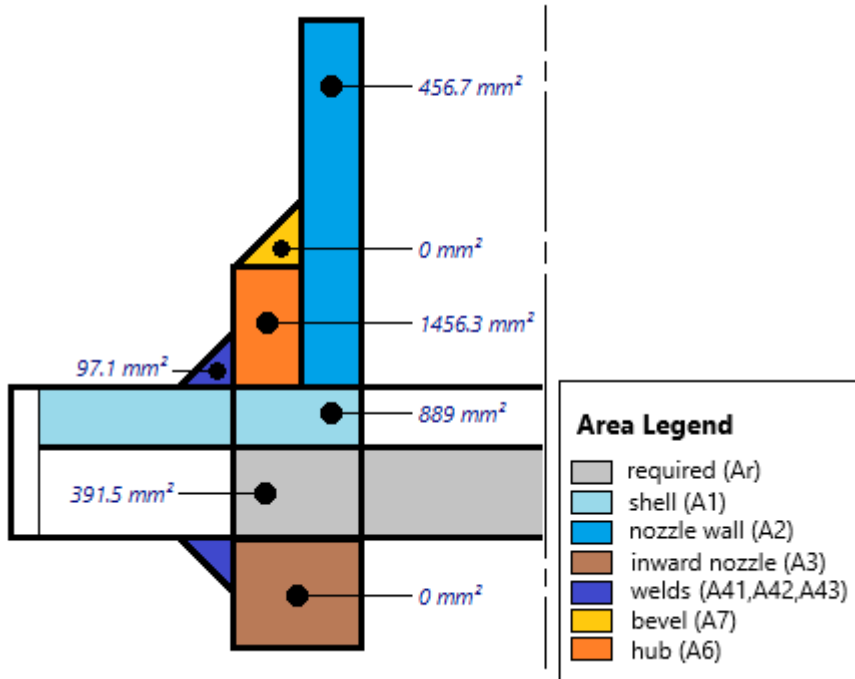
$= \min(1, Sn/Sv)$
 $= \min(1, 133.2/137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:

$= \min(1, Sn/Sv)$
 $= \min(1, 133.2/137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	391.472	74.381	NA
Area in Shell	A1	888.992	1122.979	NA
Area in Nozzle Wall	A2	456.728	478.192	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1456.285	1456.285	NA
TOTAL AREA AVAILABLE	Atot	2899.091	3154.542	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (86.0 \cdot 4.4702 \cdot 1.0 + 2 \cdot 27.0 \cdot 4.4702 \cdot 1.0 \cdot (1 - 0.97))$
 $= 391.472 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 86.0(1.0 \cdot 15.0 - 1.0 \cdot 4.47) - 2 \cdot 27.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 4.4702) \cdot (1 - 0.971)$
 $= 888.992 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 37.5)(7.0 - 0.73) \cdot 0.9709$
 $= 456.728 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$
 $= 97.086 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(37.5, 200.0, 50.0)) \cdot (30.0 - 10.0) \cdot 0.971$$

$$= 1456.285 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.7275 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.9489 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.9489 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.02, \max(7.9489, 4.5)]$$

$$= 7.9489 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.7275, 7.9489)$$

$$= 7.9489 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.9,	Allowable	: 133.2 MPa	Passed
Expansion	: 0.0,	Allowable	: 307.1 MPa	Passed
Occasional	: 6.3,	Allowable	: 177.2 MPa	Passed
Shear	: 21.7,	Allowable	: 93.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.702, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.1, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 4.444, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.296, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 2.23/10.20 = 0.219$$

Weld Size Calculations, Description: N380

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (391.4716 - 888.9921 + 2 • 27.0 • 0.9709 •
(1.0 • 15.0 - 4.4702))137)
= 7479.12 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (1913.0131 + 97.0856 - 0.0 • 0.97) • 137
= 275758.94 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1913.0131 + 0.0 + 97.0856 + (786.3938)) • 137
= 383641.72 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1913.0131 + 0.0 + 97.0856 + 0.0 + (786.3938)) • 137
= 383641.72 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 140.0 • 10.0 • 0.49 • 133
= 143520. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 56.5) • (30.0 - 3.0) • 0.7 • 133
= 446815. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 140.0 • (10.0 - 0.0) • 0.74 • 137
= 223250. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (143520 + 446815) = 590334 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (143520 + 0 + 223250 + 0) = 366770 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (143520 + 223250 + 0) = 366770 N

Summary of Failure Path Calculations:

Path 1-1 = 590334 N	, must exceed W = 7479 N	or W1 = 275758 N
Path 2-2 = 366769 N	, must exceed W = 7479 N	or W2 = 383641 N
Path 3-3 = 366769 N	, must exceed W = 7479 N	or W3 = 383641 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 7 MPa



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 4.3683 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.3683 mm

Input Echo, WRC107/537 Item 1, Description: N380 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	1097.282	mm
Vessel Thickness	Tv	18.000	mm
Design Temperature	T1	270.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	137.20	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	80.000	mm
Nozzle Thickness	Tn	30.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	133.20	MPa
Design Internal Pressure	Dp	2.231	MPa
Include Pressure Thrust		No	

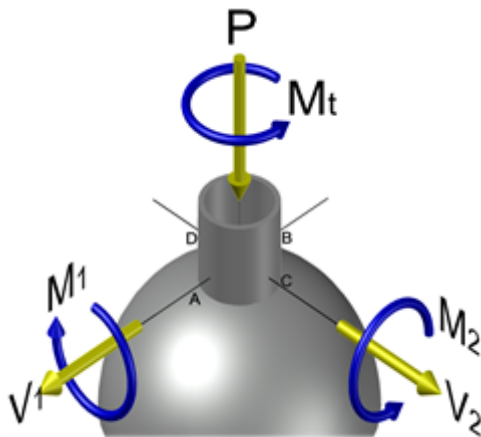
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-5997.7	N
Longitudinal Shear (SUS)	(V1) V1	3998.5	N
Circumferential Shear (SUS)	(Vc) V2	3998.5	N
Circumferential Moment (SUS)	(Mc) M1	896046.4	N-mm
Longitudinal Moment (SUS)	(Ml) M2	896046.4	N-mm
Torsional Moment (SUS)	Mt	1280070.8	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 140.0 + 2 \cdot 1.65 \cdot \sqrt{(559.141 (18.0 - 3.0))}$$

$$= 442.218 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-5997.7	N
Circumferential Shear	(VC) V2	3998.5	N
Longitudinal Shear	(VL) V1	3998.5	N
Circumferential Moment	(MC) M1	896046.4	N-mm
Longitudinal Moment	(ML) M2	896046.4	N-mm
Torsional Moment	MT	1280070.8	N-mm

Dimensionless Param: U = 0.76 TAU = 5.00 (2.09) RHO = 0.56

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.05237	(A,B,C,D)
$M(x) / P$	SP 1	0.05837	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.07051	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.12193	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.07051	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.12193	(A,B,C,D)
$N(y) * T / P$	SP 1	0.08892	(A,B,C,D)
$M(y) / P$	SP 1	0.01799	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.08602	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.04361	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.08602	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.04361	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Rad. Bend.	P	9.3	-9.3	9.3	-9.3	9.3	-9.3	9.3	-9.3
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-3.1	-3.1	3.1	3.1
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-31.8	31.8	31.8	-31.8
Rad. Memb.	ML	-3.1	-3.1	3.1	3.1	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-31.8	31.8	31.8	-31.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-24.1	20.8	45.6	-36.7	-24.1	20.8	45.6	-36.7



Tang. Memb. P	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Tang. Bend. P	2.9	-2.9	2.9	-2.9	2.9	-2.9	2.9	-2.9
Tang. Memb. MC	0.0	0.0	0.0	0.0	-3.7	-3.7	3.7	3.7
Tang. Bend. MC	0.0	0.0	0.0	0.0	-11.4	11.4	11.4	-11.4
Tang. Memb. ML	-3.7	-3.7	3.7	3.7	0.0	0.0	0.0	0.0
Tang. Bend. ML	-11.4	11.4	11.4	-11.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-9.9	7.1	20.4	-8.1	-9.9	7.1	20.4	-8.1
Shear VC	1.2	1.2	-1.2	-1.2	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear MT	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8
Tot. Shear	4.0	4.0	1.6	1.6	1.6	1.6	4.0	4.0
Str. Int.	25.2	21.9	45.7	36.8	24.3	21.0	46.2	37.2

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		41.0	41.0	41.0	41.0	41.0	41.0	41.0	41.0
Rad. Pl (SUS)		-1.7	-1.7	4.5	4.5	-1.7	-1.7	4.5	4.5
Rad. Q (SUS)		-22.5	22.5	41.1	-41.1	-22.5	22.5	41.1	-41.1
Long. Pm (SUS)		41.0	41.0	41.0	41.0	41.0	41.0	41.0	41.0
Long. Pl (SUS)		-1.4	-1.4	6.1	6.1	-1.4	-1.4	6.1	6.1
Long. Q (SUS)		-8.5	8.5	14.3	-14.3	-8.5	8.5	14.3	-14.3
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.2	1.2	-1.2	-1.2	-1.2	-1.2	1.2	1.2
Shear Q (SUS)		2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8
Pm (SUS)		41.0	41.0	41.0	41.0	41.0	41.0	41.0	41.0
Pm+Pl (SUS)		40.7	40.7	47.8	47.8	40.7	40.7	47.8	47.8
Pm+Pl+Q (Total)		32.2	62.9	86.7	33.0	31.3	62.0	87.3	33.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.04	137.20	Passed
Pm+Pl (SUS)	47.79	205.80	Passed
Pm+Pl+Q (TOTAL)	87.26	411.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N6100

From: 40

Pressure for Reinforcement Calculations	P	2.2160	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	1900.8000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1650.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		100.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
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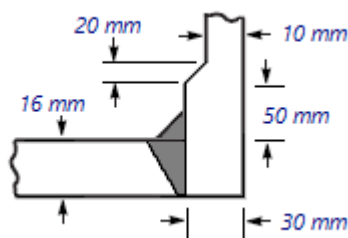
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: N6100

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 100.000 mm.
Actual Thickness Used in Calculation 10.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.22 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.22)$
= 5.0203 mm

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.7357 and the shell thk. is 16.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $2.22 \cdot 50.0 / (133 \cdot 1.0 + 0.4 \cdot 2.22)$
= 0.8263 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5139 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 20.0/0.5773
= 34.6410 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

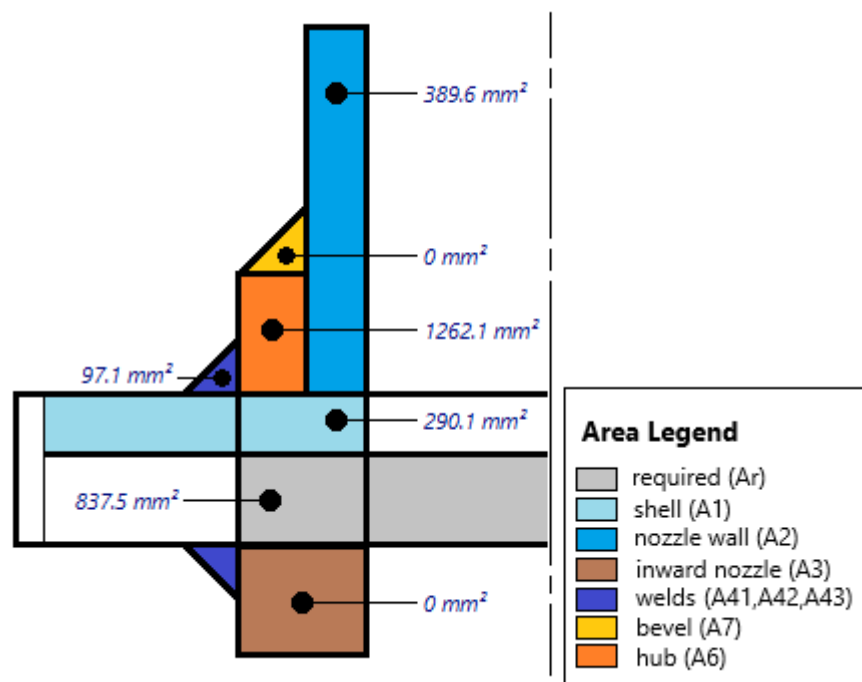
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.0000	mm
Parallel to Vessel Wall, opening length	d	86.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, $133.2/137.2$)
= 0.971

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, $133.2/137.2$)
= 0.971

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.971, 1.0)
= 0.971



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	837.525	143.674	NA
Area in Shell	A1	290.117	820.521	NA
Area in Nozzle Wall	A2	389.593	409.311	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1262.114	1262.114	NA
TOTAL AREA AVAILABLE	Atot	2038.909	2589.031	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (86.0 \cdot 9.5637 \cdot 1.0 + 2 \cdot 27.0 \cdot 9.5637 \cdot 1.0 \cdot (1 - 0.97))$
 $= 837.525 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 86.0(1.0 \cdot 13.0 - 1.0 \cdot 9.564) - 2 \cdot 27.0$
 $(1.0 \cdot 13.0 - 1.0 \cdot 9.5637) \cdot (1 - 0.971)$
 $= 290.117 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(7.0 - 0.83) \cdot 0.9709$
 $= 389.593 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$
 $= 97.086 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(32.5, 300.0, 50.0)) \cdot (30.0 - 10.0) \cdot 0.971 \\ &= 1262.114 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8263 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.0203 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.0203 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[8.02, \max(8.0203, 4.5)] \\ &= 8.0200 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(3.8263, 8.02) \\ &= 8.0200 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 47.1,	Allowable	: 133.2 MPa	Passed
Expansion	: 0.0,	Allowable	: 291.9 MPa	Passed
Occasional	: 6.3,	Allowable	: 177.2 MPa	Passed
Shear	: 32.1,	Allowable	: 93.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.798, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.114, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 4.991, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.384, Temp. Reduction = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 2.22/10.20 = 0.217$$

Weld Size Calculations, Description: N6100

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (837.5254 - 290.1174 + 2 \cdot 27.0 \cdot 0.9709 \cdot (1.0 \cdot 13.0 - 9.5637)) \cdot 137)$
 $= 99811.95 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1651.7061 + 97.0856 - 0.0 \cdot 0.97) \cdot 137$
 $= 239911.09 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1651.7061 + 0.0 + 97.0856 + (681.5412)) \cdot 137$
 $= 333409.53 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1651.7061 + 0.0 + 97.0856 + 0.0 + (681.5412)) \cdot 137$
 $= 333409.53 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 100.0 \cdot 10.0 \cdot 0.49 \cdot 133$
 $= 102514. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 46.5) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 133$
 $= 367732. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 100.0 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137$
 $= 159464. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (102514 + 367732) = 470246 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (102514 + 0 + 159464 + 0) = 261978 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (102514 + 159464 + 0) = 261978 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 470246 N	, must exceed W = 99811 N	or W1 = 239911 N
Path 2-2 = 261978 N	, must exceed W = 99811 N	or W2 = 333409 N
Path 3-3 = 261978 N	, must exceed W = 99811 N	or W3 = 333409 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa



Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 4.1290 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 320.1290 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 86.00 mm

Input Echo, WRC297 Item 1, Description: N6100 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	609.601	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	137.20 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	140.000	mm
Nozzle Thickness	30.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	133.20 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2.22	MPa
Radial Load	P	-7497.10	N
Circumferential Shear	Vc	4998.08	N
Longitudinal Shear	Vl	4998.08	N
Circumferential Moment	Mc	0.140E+07	N-mm
Longitudinal Moment	Ml	0.140E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.667) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 140.0 + 2 \cdot 1.65 \cdot \sqrt{(314.301(16.0 - 3.0))}$$

$$= 350.940 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	628.601 mm
Nozzle Outside Diameter	(d) =	140.000 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	27.000 mm
T / t	=	0.481



$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.549$$

Nr/P	=	0.080
Mr/P	=	0.169
M0/P	=	0.049
N0/P	=	0.109
MrD/Mc	=	0.295
NrDL/Mc	=	0.097
M0d/Mc	=	0.092
N0DL/Mc	=	0.060
MrD/Ml	=	0.140
NrDL/Ml	=	0.077
M0D/Ml	=	0.045
N0DL/Ml	=	0.129

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	51	53	51	53
Outplane Stress Summary	45	53	93	36

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-49	49	49	-49
Inplane Pressure Stress	25	25	25	25
Inplane Stress Summary	19	29	128	-61

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	3	3

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	47	55	128	98

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-32	32	32	-32
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	25	25	25	25

Outplane Stress Summary	7	46	79	-11
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-104	104	104	-104
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	53	51	53

Inplane Stress Summary	-10	111	210	-87
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	6	6
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	19	111	210	87

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5

Hoop Stress Summary	0	3	15	18
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-6	-6	6	6
Axial Bending (ML)	-8	8	8	-8
Axial Pressure Stress	1	1	1	1



Axial Stress Summary -4 -3 24 -7

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress) Au Al Bu Bl
 Outside Inside Outside Inside

Shear due to (Vc) 0 0 0 0
 Shear due to (Vl) 0 0 0 0
 Shear due to Torsion 2 2 2 2

Shear Stress Summary 3 3 1 1

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities Au Al Bu Bl
 Outside Inside Outside Inside

Two * Max Shear Stress 7 9 25 26

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses in the Cu Cl Du Dl
 hoop direction) Left Left Right Right
 Outside Inside Outside Inside

Hoop Membrane (P) 4 4 4 4
 Hoop Bending (P) 0 0 0 0
 Hoop Membrane (Mc) -3 -3 3 3
 Hoop Bending (Mc) 0 0 0 0
 Hoop Membrane (ML) 0 0 0 0
 Hoop Bending (ML) 0 0 0 0
 Hoop Pressure Stress 3 5 3 5

Hoop Stress Summary 4 7 11 14

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to Cu Cl Du Dl
 pipe cross-section) Left Left Right Right
 Outside Inside Outside Inside

Axial Membrane (P) 0 0 0 0
 Axial Bending (P) 7 -7 7 -7
 Axial Membrane (Mc) -6 -6 6 6
 Axial Bending (Mc) -20 20 20 -20
 Axial Membrane (ML) 0 0 0 0
 Axial Bending (ML) 0 0 0 0
 Axial Pressure Stress 1 1 1 1

Axial Stress Summary -15 8 36 -19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress) Cu Cl Du Dl
 Outside Inside Outside Inside

Shear due to (Vc) 0 0 0 0
 Shear due to (Vl) 0 0 0 0
 Shear due to Torsion 2 2 2 2

Shear Stress Summary 1 1 3 3

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stress Intensities Cu Cl Du Dl
 Outside Inside Outside Inside

Two * Max Shear Stress 20 9 37 34

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	51	53	51	53	51	53	51	53
Circ. Pl (SUS)	-3	-3	11	11	-2	-2	8	8
Circ. Q (SUS)	-3	3	29	-29	-59	59	149	-149
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. Pl (SUS)	-1	-1	7	7	1	1	7	7
Long. Q (SUS)	-4	4	94	-94	-19	19	45	-45
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	51.0	53.0	51.0	53.0	51.0	53.0	51.0	53.0
Pm+Pl (SUS)	48.0	50.0	62.0	64.0	49.0	51.0	59.0	61.0
Pm+Pl+Q (Total)	46.0	54.0	126.3	97.2	18.0	110.1	208.2	88.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	137.20	Passed
Pm+Pl (SUS)	64.03	205.80	Passed
Pm+Pl+Q (TOTAL)	208.19	411.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	5	3	5	3	5	3	5
Circ. Pl (SUS)	-3	-3	11	11	1	1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-6	-6	6	6	-6	-6	6	6
Long. Q (SUS)	-1	1	15	-15	-13	13	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	3.0	5.0	3.0	5.0	3.0	5.0	3.0	5.0
Pm+Pl (SUS)	5.0	7.0	14.0	16.0	9.0	11.0	10.0	12.0
Pm+Pl+Q (Total)	7.2	7.2	22.5	24.3	22.4	9.2	34.2	32.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	133.20	Passed
Pm+Pl (SUS)	16.00	199.80	Passed
Pm+Pl+Q (TOTAL)	34.17	399.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.





Input, Nozzle Desc: K150

From: 40

Pressure for Reinforcement Calculations	P	2.2160	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	1900.8000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1650.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

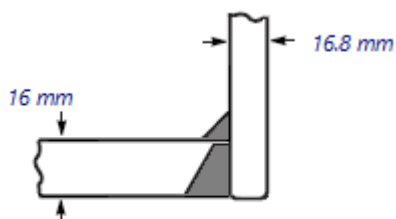
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 16.800 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.22 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.22)$
= 5.0203 mm

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.7357 and the shell thk. is 16.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.22 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 2.22)$
= 0.4773 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $133.2 / 137.2$)
= 0.971

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $133.2 / 137.2$)
= 0.971

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.971, 1.0)
= 0.971

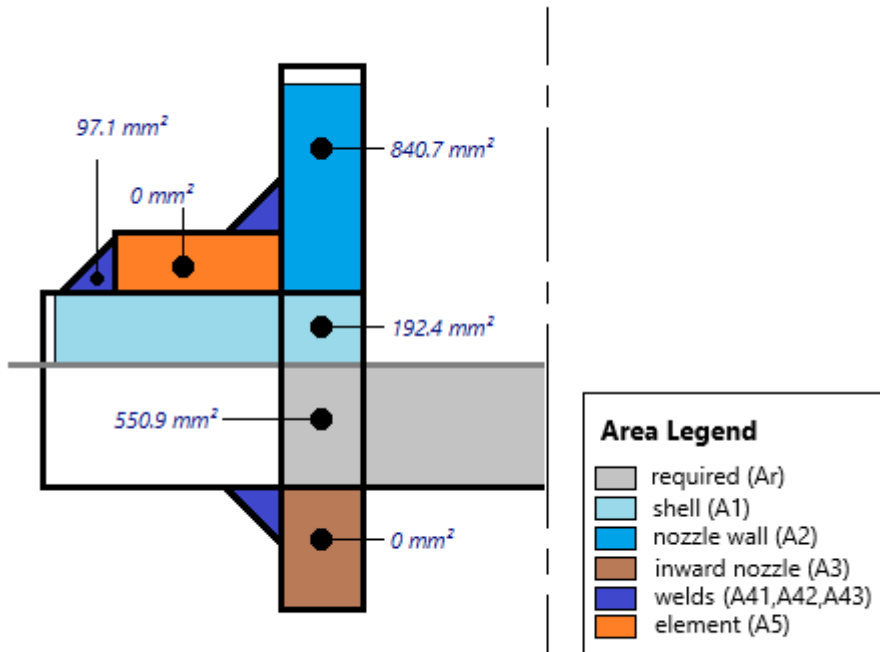


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	550.908	94.506	NA
Area in Shell	A1	192.420	544.210	NA
Area in Nozzle Wall	A2	840.741	846.093	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1130.247	1487.388	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 9.5637 \cdot 1.0 + 2 \cdot 13.8 \cdot 9.5637 \cdot 1.0 \cdot (1 - 0.97))$
 $= 550.908 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 56.8(1.0 \cdot 13.0 - 1.0 \cdot 9.564) - 2 \cdot 13.8$
 $(1.0 \cdot 13.0 - 1.0 \cdot 9.5637) \cdot (1 - 0.971)$
 $= 192.420 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(13.8 - 0.48)0.9709$
 $= 840.741 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$
 $= 97.086 \text{ mm}^2$



UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.4773$ mm
Wall Thickness per UG16(b), $tr_{16b} = 4.5000$ mm
Wall Thickness, shell/head, internal pressure $trb_1 = 8.0203$ mm
Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.0203$ mm
Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000$ mm
Wall Thickness per table UG-45 $tb_3 = 7.8000$ mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[7.8, \max(8.0203, 4.5)]$
 $= 7.8000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(3.4773, 7.8)$
 $= 7.8000$ mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 16.0$, $t_r = 4.991$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio $= t_r * E^* / (t_g - c) = 0.384$, Temp. Reduction = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $2.22/5.10 = 0.434$

Weld Size Calculations, Description: K150

Intermediate Calc. for nozzle/shell Welds $T_{min} = 13.0000$ mm

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A-A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (550.9084 - 192.4201 + 2 \cdot 13.8 \cdot 0.9709 \cdot (1.0 \cdot 13.0 - 9.5637)) \cdot 137)$
 $= 61811.87$ N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot Sv$
 $= (840.7408 + 0.0 + 97.0856 - 0.0 \cdot 0.97) \cdot 137$
 $= 128657.39$ N

Weld Load [W2]:

$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv$
 $= (840.7408 + 0.0 + 97.0856 + (348.3433)) \cdot 137$
 $= 176445.48$ N



Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S \\ &= (840.7408 + 0.0 + 97.0856 + 0.0 + (348.3433)) \cdot 137 \\ &= 176445.48 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 133 \\ &= 86522. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 35.3) \cdot (16.8 - 3.0) \cdot 0.7 \cdot 133 \\ &= 142682. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 84.4 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137 \\ &= 134588. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (86522 + 142682) = 229204 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (86522 + 0 + 134588 + 0) = 221110 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (86522 + 134588 + 0) = 221110 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 229203 N	, must exceed W = 61811 N	or W1 = 128657 N
Path 2-2 = 221109 N	, must exceed W = 61811 N	or W2 = 176445 N
Path 3-3 = 221109 N	, must exceed W = 61811 N	or W3 = 176445 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.9355 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.9354 mm

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Input, Nozzle Desc: K250

From: 40

Pressure for Reinforcement Calculations	P	2.2280	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	1900.8000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		200.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

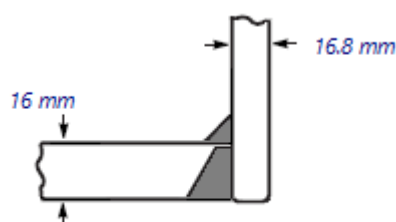
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K250

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 16.800 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.23 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.23)$
= 5.0476 mm

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.7357 and the shell thk. is 16.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.23 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 2.23)$
= 0.4798 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $133.2 / 137.2$)
= 0.971

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $133.2 / 137.2$)
= 0.971

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.971, 1.0)
= 0.971

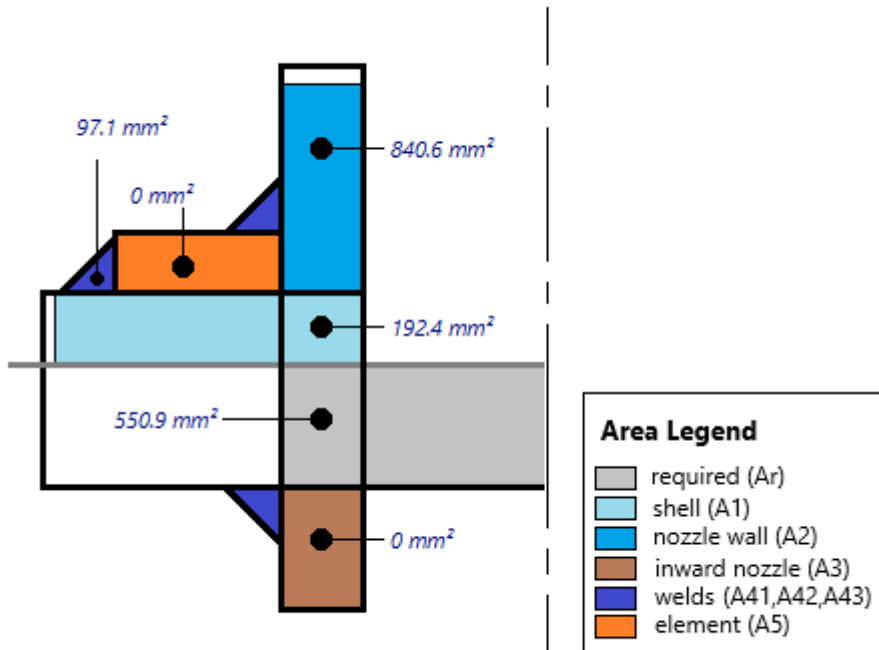


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	550.908	94.506	NA
Area in Shell	A1	192.420	544.210	NA
Area in Nozzle Wall	A2	840.577	846.093	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1130.083	1487.388	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 9.5637 \cdot 1.0 + 2 \cdot 13.8 \cdot 9.5637 \cdot 1.0 \cdot (1 - 0.97))$
 $= 550.908 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 56.8(1.0 \cdot 13.0 - 1.0 \cdot 9.564) - 2 \cdot 13.8$
 $(1.0 \cdot 13.0 - 1.0 \cdot 9.5637) \cdot (1 - 0.971)$
 $= 192.420 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(13.8 - 0.48)0.9709$
 $= 840.577 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$
 $= 97.086 \text{ mm}^2$



UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.4798$ mm
Wall Thickness per UG16(b), $tr_{16b} = 4.5000$ mm
Wall Thickness, shell/head, internal pressure $trb_1 = 8.0476$ mm
Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.0476$ mm
Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000$ mm
Wall Thickness per table UG-45 $tb_3 = 7.8000$ mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[7.8, \max(8.0476, 4.5)]$
 $= 7.8000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(3.4798, 7.8)$
 $= 7.8000$ mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 16.0$, $t_r = 5.018$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio $= t_r * E^* / (t_g - c) = 0.386$, Temp. Reduction = 54 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $2.23/5.10 = 0.437$

Weld Size Calculations, Description: K250

Intermediate Calc. for nozzle/shell Welds $T_{min} = 13.0000$ mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A-A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (550.9084 - 192.4201 + 2 \cdot 13.8 \cdot 0.9709 \cdot (1.0 \cdot 13.0 - 9.5637)) \cdot 137)$
 $= 61811.87$ N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot Sv$
 $= (840.577 + 0.0 + 97.0856 - 0.0 \cdot 0.97) \cdot 137$
 $= 128634.91$ N

Weld Load [W2]:

$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv$
 $= (840.577 + 0.0 + 97.0856 + (348.3433)) \cdot 137$
 $= 176423.00$ N



Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S \\ &= (840.577 + 0.0 + 97.0856 + 0.0 + (348.3433)) \cdot 137 \\ &= 176423.00 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 133 \\ &= 86522. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 35.3) \cdot (16.8 - 3.0) \cdot 0.7 \cdot 133 \\ &= 142682. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 84.4 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137 \\ &= 134588. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (86522 + 142682) = 229204 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (86522 + 0 + 134588 + 0) = 221110 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (86522 + 134588 + 0) = 221110 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 229203 N	, must exceed W = 61811 N	or W1 = 128634 N
Path 2-2 = 221109 N	, must exceed W = 61811 N	or W2 = 176423 N
Path 3-3 = 221109 N	, must exceed W = 61811 N	or W3 = 176423 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.9355 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.9354 mm

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Input, Nozzle Desc: K650

From: 40

Pressure for Reinforcement Calculations	P	2.2280	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	1900.8000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		200.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

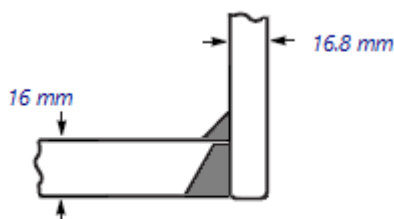
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K650

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 16.800 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.23 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.23)$
= 5.0476 mm

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.7357 and the shell thk. is 16.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.23 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 2.23)$
= 0.4798 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $133.2 / 137.2$)
= 0.971

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $133.2 / 137.2$)
= 0.971

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.971, 1.0)
= 0.971

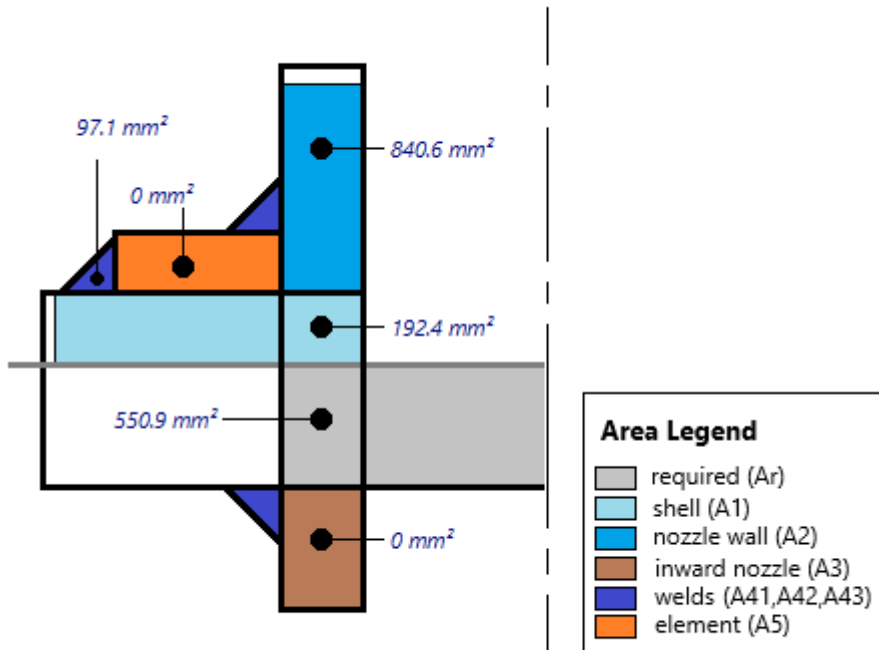


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	550.908	94.506	NA
Area in Shell	A1	192.420	544.210	NA
Area in Nozzle Wall	A2	840.577	846.093	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1130.083	1487.388	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 9.5637 \cdot 1.0 + 2 \cdot 13.8 \cdot 9.5637 \cdot 1.0 \cdot (1 - 0.97))$
 $= 550.908 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 56.8(1.0 \cdot 13.0 - 1.0 \cdot 9.564) - 2 \cdot 13.8$
 $(1.0 \cdot 13.0 - 1.0 \cdot 9.5637) \cdot (1 - 0.971)$
 $= 192.420 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(13.8 - 0.48)0.9709$
 $= 840.577 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$
 $= 97.086 \text{ mm}^2$



UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.4798$ mm
Wall Thickness per UG16(b), $tr_{16b} = 4.5000$ mm
Wall Thickness, shell/head, internal pressure $trb_1 = 8.0476$ mm
Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.0476$ mm
Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000$ mm
Wall Thickness per table UG-45 $tb_3 = 7.8000$ mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[7.8, \max(8.0476, 4.5)]$
 $= 7.8000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(3.4798, 7.8)$
 $= 7.8000$ mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 16.0$, $t_r = 5.018$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio $= t_r * E^* / (t_g - c) = 0.386$, Temp. Reduction = 54 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $2.23/5.10 = 0.437$

Weld Size Calculations, Description: K650

Intermediate Calc. for nozzle/shell Welds $T_{min} = 13.0000$ mm

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A-A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (550.9084 - 192.4201 + 2 \cdot 13.8 \cdot 0.9709 \cdot (1.0 \cdot 13.0 - 9.5637)) \cdot 137)$
 $= 61811.87$ N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot Sv$
 $= (840.577 + 0.0 + 97.0856 - 0.0 \cdot 0.97) \cdot 137$
 $= 128634.91$ N

Weld Load [W2]:

$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv$
 $= (840.577 + 0.0 + 97.0856 + (348.3433)) \cdot 137$
 $= 176423.00$ N



Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S \\ &= (840.577 + 0.0 + 97.0856 + 0.0 + (348.3433)) \cdot 137 \\ &= 176423.00 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 133 \\ &= 86522. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 35.3) \cdot (16.8 - 3.0) \cdot 0.7 \cdot 133 \\ &= 142682. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{g_{nvi}} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 84.4 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137 \\ &= 134588. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (86522 + 142682) = 229204 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + T_{pgw} + T_{ngw} + Sinw) \\ &= (86522 + 0 + 134588 + 0) = 221110 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + T_{ngw} + Sinw) \\ &= (86522 + 134588 + 0) = 221110 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 229203 N	, must exceed W = 61811 N	or W1 = 128634 N
Path 2-2 = 221109 N	, must exceed W = 61811 N	or W2 = 176423 N
Path 3-3 = 221109 N	, must exceed W = 61811 N	or W3 = 176423 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.9355 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.9354 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 56.80 mm

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Input, Nozzle Desc: N5100

From: 70

Pressure for Reinforcement Calculations	P	2.2148	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	4380.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 2352.10 mm

User Entered Minimum Design Metal Temperature -12.00 °C

Type of Element Connected to the Parent : Nozzle

Material	SA-105	+IT
Material UNS Number	K03504	
Material Specification/Type	Forgings	
Allowable Stress at Temperature	Sn	133.20 MPa
Allowable Stress At Ambient	Sna	138.00 MPa

Diameter Basis (for tr calc only)	Outside	
Layout Angle	0.00	deg
Diameter	100.0000	mm

Size and Thickness Basis		Actual
Actual Thickness	tn	10.0000 mm

Flange Material	SA-105
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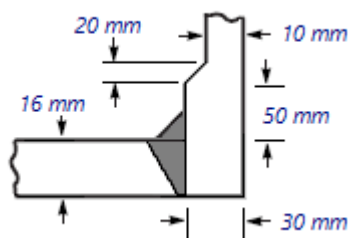
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class	600
Flange Grade	GR 1.1

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: N5100

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 100.000 mm.
Actual Thickness Used in Calculation 10.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.21 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.21)$
= 5.0174 mm

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6104 and the shell thk. is 16.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $2.21 \cdot 50.0 / (133 \cdot 1.0 + 0.4 \cdot 2.21)$
= 0.8259 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5139 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 20.0/0.5773
= 34.6410 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

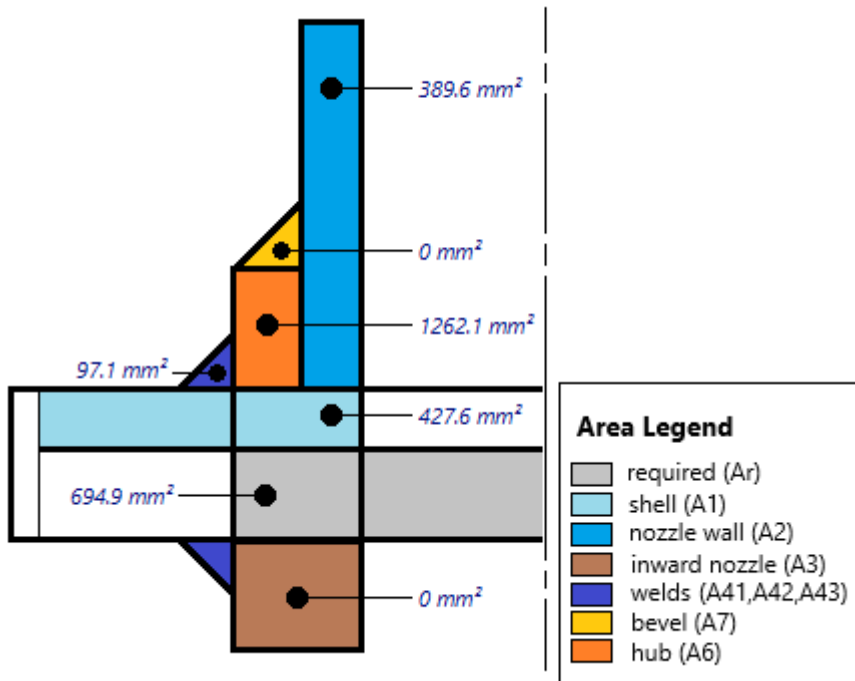
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.0000	mm
Parallel to Vessel Wall, opening length	d	86.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, $133.2/137.2$)
= 0.971

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, $133.2/137.2$)
= 0.971

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.971, 1.0)
= 0.971



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	694.887	206.578	NA
Area in Shell	A1	427.630	699.235	NA
Area in Nozzle Wall	A2	389.621	409.311	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1262.114	1262.114	NA
TOTAL AREA AVAILABLE	Atot	2176.450	2467.745	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (86.0 \cdot 7.9349 \cdot 1.0 + 2 \cdot 27.0 \cdot 7.9349 \cdot 1.0 \cdot (1 - 0.97))$$

$$= 694.887 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 86.0(1.0 \cdot 13.0 - 1.0 \cdot 7.935) - 2 \cdot 27.0$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 7.9349) \cdot (1 - 0.971)$$

$$= 427.630 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 32.5)(7.0 - 0.83)0.9709$$

$$= 389.621 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(32.5, 300.0, 50.0)) \cdot (30.0 - 10.0) \cdot 0.971$$

$$= 1262.114 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8259 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.0174 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.0174 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.02, \max(8.0174, 4.5)]$$

$$= 8.0174 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.8259, 8.0174)$$

$$= 8.0174 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 47.1,	Allowable	: 133.2 MPa	Passed
Expansion	: 0.0,	Allowable	: 291.9 MPa	Passed
Occasional	: 6.3,	Allowable	: 177.2 MPa	Passed
Shear	: 32.1,	Allowable	: 93.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.797, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.114, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 4.988, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.384, Temp. Reduction = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2.21/10.20 = 0.217

Weld Size Calculations, Description: N5100

Intermediate Calc. for nozzle/shell welds Tmin 13.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (694.8866 - 427.6295 + 2 \cdot 27.0 \cdot 0.9709 \cdot (1.0 \cdot 13.0 - 7.9349)) \cdot 137)$
 $= 73093.47 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1651.735 + 97.0856 - 0.0 \cdot 0.97) \cdot 137$
 $= 239915.08 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1651.735 + 0.0 + 97.0856 + (681.5412)) \cdot 137$
 $= 333413.50 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1651.735 + 0.0 + 97.0856 + 0.0 + (681.5412)) \cdot 137$
 $= 333413.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 100.0 \cdot 10.0 \cdot 0.49 \cdot 133$
 $= 102514. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 46.5) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 133$
 $= 367732. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 100.0 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137$
 $= 159464. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (102514 + 367732) = 470246 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (102514 + 0 + 159464 + 0) = 261978 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (102514 + 159464 + 0) = 261978 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 470246 N	, must exceed W = 73093 N	or W1 = 239915 N
Path 2-2 = 261978 N	, must exceed W = 73093 N	or W2 = 333413 N
Path 3-3 = 261978 N	, must exceed W = 73093 N	or W3 = 333413 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa



Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 4.1290 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 320.1290 mm

Input Echo, WRC297 Item 1, Description: N5100 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	609.601	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	137.20 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	140.000	mm
Nozzle Thickness	30.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	133.20 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUsTained loads.

Design Internal Pressure	Dp	2.21	MPa
Radial Load	P	-7497.10	N
Circumferential Shear	Vc	4998.08	N
Longitudinal Shear	Vl	4998.08	N
Circumferential Moment	Mc	0.140E+07	N-mm
Longitudinal Moment	Ml	0.140E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (4.667) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 140.0 + 2 \cdot 1.65 \cdot \sqrt{(314.301(16.0 - 3.0))}$$

$$= 350.940 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	628.601 mm
Nozzle Outside Diameter	(d) =	140.000 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	27.000 mm
T / t	=	0.481
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.549$$



Nr/P = 0.080
 Mr/P = 0.169
 M0/P = 0.049
 N0/P = 0.109
 MrD/Mc = 0.295
 NrDL/Mc = 0.097
 M0d/Mc = 0.092
 N0DL/Mc = 0.060
 MrD/Ml = 0.140
 NrDL/Ml = 0.077
 M0D/Ml = 0.045
 N0DL/Ml = 0.129

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	51	53	51	53
Outplane Stress Summary	45	53	92	36

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-49	49	49	-49
Inplane Pressure Stress	25	25	25	25
Inplane Stress Summary	19	29	128	-61

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	3	3

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	47	55	128	98

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-32	32	32	-32



Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	25	25	25	25
Outplane Stress Summary	7	46	79	-11

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	45	-45	45	-45
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-104	104	104	-104
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	53	51	53
Inplane Stress Summary	-10	111	210	-87

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	19	111	210	87

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5
Hoop Stress Summary	0	3	15	18

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-6	-6	6	6
Axial Bending (ML)	-8	8	8	-8
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-4	-3	24	-7

Nozzle Stresses (MPa):



LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	1	1

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	7	9	25	26

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5
Hoop Stress Summary	4	7	11	14

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-20	20	20	-20
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-15	8	36	-19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	20	9	37	34

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	51	53	51	53	51	53	51	53

Circ. P1 (SUS)	-3	-3	11	11	-2	-2	8	8
Circ. Q (SUS)	-3	3	29	-29	-59	59	149	-149
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. P1 (SUS)	-1	-1	7	7	1	1	7	7
Long. Q (SUS)	-4	4	94	-94	-19	19	45	-45
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	51.0	53.0	51.0	53.0	51.0	53.0	51.0	53.0
Pm+P1 (SUS)	48.0	50.0	62.0	64.0	49.0	51.0	59.0	61.0
Pm+P1+Q (Total)	46.0	54.0	126.3	97.2	18.0	110.1	208.2	88.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	137.20	Passed
Pm+P1 (SUS)	64.03	205.80	Passed
Pm+P1+Q (TOTAL)	208.19	411.60	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	5	3	5	3	5	3	5
Circ. P1 (SUS)	-3	-3	11	11	1	1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. P1 (SUS)	-6	-6	6	6	-6	-6	6	6
Long. Q (SUS)	-1	1	15	-15	-13	13	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	3.0	5.0	3.0	5.0	3.0	5.0	3.0	5.0
Pm+P1 (SUS)	5.0	7.0	14.0	16.0	9.0	11.0	10.0	12.0
Pm+P1+Q (Total)	7.2	7.2	22.5	24.3	22.4	9.2	34.2	32.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	133.20	Passed
Pm+P1 (SUS)	16.00	199.80	Passed
Pm+P1+Q (TOTAL)	34.17	399.60	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

Input, Nozzle Desc: H3150

From: 70

Pressure for Reinforcement Calculations	P	2.2148	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	4380.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		3877.10	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	6.0000	mm

Flange Material		SA-105	
-----------------	--	--------	--

Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	24.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

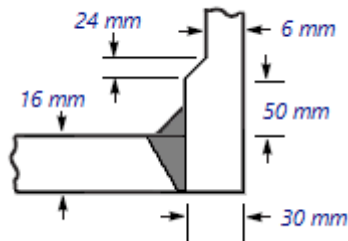
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: H3150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	150.000 mm.
Actual Thickness Used in Calculation	6.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.21 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.21)$
 $= 5.0174 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.6104 and the shell thk. is 16.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.21 \cdot 75.0 / (133 \cdot 1.0 + 0.4 \cdot 2.21)$
 $= 1.2388 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5490 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 24.0 / 0.5773$
 $= 41.5692 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	288.0000 mm
Parallel to Vessel Wall, opening length	d	144.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

Weld Strength Reduction Factor [fr1]:

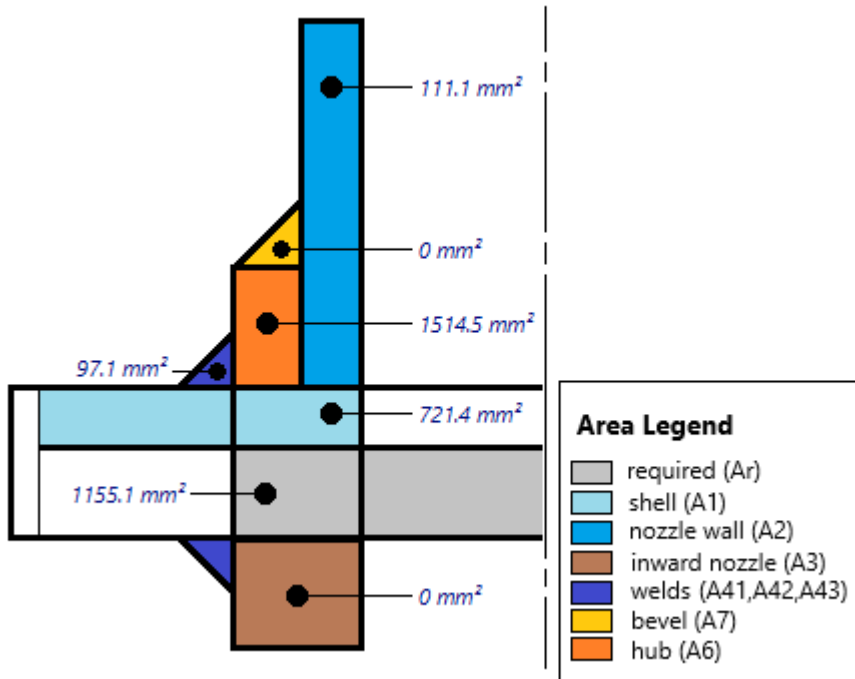
$= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
 = min(0.971, 1.0)
 = 0.971



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1155.109	343.394	NA
Area in Shell	A1	721.407	1179.603	NA
Area in Nozzle Wall	A2	111.140	154.672	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1514.536	1514.536	NA
TOTAL AREA AVAILABLE	Atot	2444.168	2945.897	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 = (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
 = (144.0•7.9349•1.0+2•27.0•7.9349•1.0•(1-0.97))
 = 1155.109 mm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 = d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
 = 144.0(1.0 • 13.0 - 1.0 • 7.935) - 2 • 27.0
 (1.0 • 13.0 - 1.0 • 7.9349) • (1 - 0.971)
 = 721.407 mm²

Area Available in Nozzle Projecting Outward [A2]:
 = (2 • tlnp)(tn - trn)fr2
 = (2 • 32.5)(3.0 - 1.24) 0.9709
 = 111.140 mm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:



$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(32.5, 200.0, 50.0)) \cdot (30.0 - 6.0) \cdot 0.971$$

$$= 1514.536 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.2388 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 4.2388 \text{ mm}$$

Available Nozzle Neck Thickness = 6.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 6.0$, $t_r = 1.196$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.399$, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 16.0$, $t_r = 4.988$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.384$, Temp. Reduction = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 2.21/5.10 = 0.434$$

Weld Size Calculations, Description: H3150

Intermediate Calc. for nozzle/shell Welds $T_{min} = 13.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) S_v)$$

$$= \max(0, (1155.1094 - 721.4069 + 2 \cdot 27.0 \cdot 0.9709 \cdot (1.0 \cdot 13.0 - 7.9349)) 137)$$

$$= 95927.59 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can / 0.707)^2 \cdot fr2) \cdot S_v$$



$$= (1625.6759 + 97.0856 - 0.0 \cdot 0.97) \cdot 137$$
$$= 236340.09 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$
$$= (1625.6759 + 0.0 + 97.0856 + (681.5412)) \cdot 137$$
$$= 329838.56 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$
$$= (1625.6759 + 0.0 + 97.0856 + 0.0 + (681.5412)) \cdot 137$$
$$= 329838.56 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$
$$= (3.1416/2.0) \cdot 150.0 \cdot 10.0 \cdot 0.49 \cdot 133$$
$$= 153771. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.1416 \cdot 73.5) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 133$$
$$= 581254. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$
$$= (3.1416/2.0) \cdot 150.0 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 137$$
$$= 239197. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (153771 + 581254) = 735025 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (153771 + 0 + 239197 + 0) = 392968 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$
$$= (153771 + 239197 + 0) = 392968 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 735025 N	, must exceed W = 95927 N	or W1 = 236340 N
Path 2-2 = 392967 N	, must exceed W = 95927 N	or W2 = 329838 N
Path 3-3 = 392967 N	, must exceed W = 95927 N	or W3 = 329838 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 MPa

The Drop for this Nozzle is : 9.3714 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 225.3714 mm

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Input, Nozzle Desc: K350

From: 110

Pressure for Reinforcement Calculations	P	2.2148	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	4300.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7202.20	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

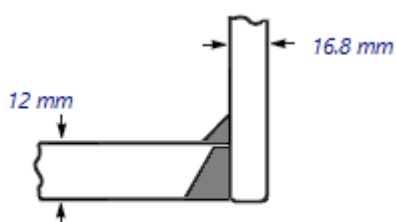
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K350

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 16.800 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.21 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.21)$
= 5.0174 mm

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.5937 and the shell thk. is 12.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.21 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 2.21)$
= 0.4770 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $133.2 / 137.2$)
= 0.971

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $133.2 / 137.2$)
= 0.971

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.971, 1.0)
= 0.971

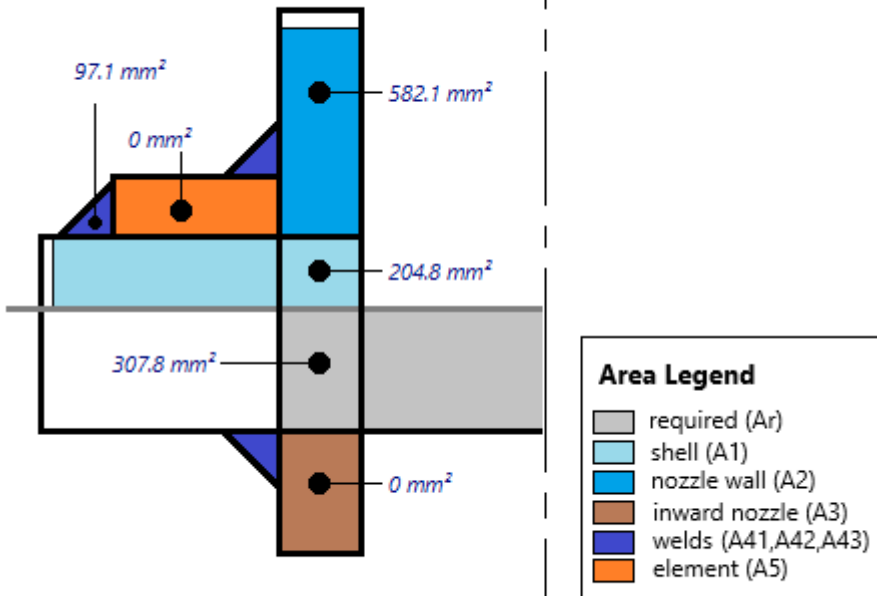


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	307.784	133.833	NA
Area in Shell	A1	204.773	243.770	NA
Area in Nozzle Wall	A2	582.063	585.756	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	883.921	926.612	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 5.3431 \cdot 1.0 + 2 \cdot 13.8 \cdot 5.3431 \cdot 1.0 \cdot (1 - 0.97))$
 $= 307.784 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 56.8(1.0 \cdot 9.0 - 1.0 \cdot 5.343) - 2 \cdot 13.8$
 $(1.0 \cdot 9.0 - 1.0 \cdot 5.3431) \cdot (1 - 0.971)$
 $= 204.773 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 22.5)(13.8 - 0.48)0.9709$
 $= 582.063 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$
 $= 97.086 \text{ mm}^2$



UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.4770$ mm
Wall Thickness per UG16(b), $tr_{16b} = 4.5000$ mm
Wall Thickness, shell/head, internal pressure $trb_1 = 8.0174$ mm
Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.0174$ mm
Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000$ mm
Wall Thickness per table UG-45 $tb_3 = 7.8000$ mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[7.8, \max(8.0174, 4.5)]$
 $= 7.8000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(3.477, 7.8)$
 $= 7.8000$ mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 4.988$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio $= t_r * E^* / (t_g - c) = 0.554$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $2.21/5.10 = 0.434$

Weld Size Calculations, Description: K350

Intermediate Calc. for nozzle/shell Welds $T_{min} = 9.0000$ mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A-A1+2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (307.7836 - 204.7726 + 2 \cdot 13.8 \cdot 0.9709 \cdot (1.0 \cdot 9.0 - 5.3431)) \cdot 137)$
 $= 27574.69$ N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A2+A5+A4-(W_i - Can/.707)^2 \cdot fr_2) \cdot Sv$
 $= (582.0632 + 0.0 + 97.0856 - 0.0 \cdot 0.97) \cdot 137$
 $= 93170.24$ N

Weld Load [W2]:

$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv$
 $= (582.0632 + 0.0 + 97.0856 + (241.1608)) \cdot 137$
 $= 126254.30$ N



Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S \\ &= (582.0632 + 0.0 + 97.0856 + 0.0 + (241.1608)) \cdot 137 \\ &= 126254.30 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 133 \\ &= 86522. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 35.3) \cdot (16.8 - 3.0) \cdot 0.7 \cdot 133 \\ &= 142682. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 137 \\ &= 121129. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (86522 + 142682) = 229204 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (86522 + 0 + 121129 + 0) = 207651 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (86522 + 121129 + 0) = 207651 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 229203 N	, must exceed W = 27574 N	or W1 = 93170 N
Path 2-2 = 207651 N	, must exceed W = 27574 N	or W2 = 126254 N
Path 3-3 = 207651 N	, must exceed W = 27574 N	or W3 = 126254 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.9355 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.9355 mm

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Input, Nozzle Desc: H2150

From: 110

Pressure for Reinforcement Calculations	P	2.2148	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	4300.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		8652.20	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	6.0000	mm

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	24.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

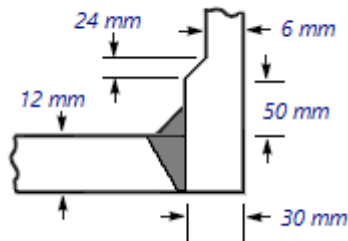
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: H2150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	150.000 mm.
Actual Thickness Used in Calculation	6.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.21 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.21)$
 $= 5.0174 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.5937 and the shell thk. is 12.0000 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.21 \cdot 75.0 / (133 \cdot 1.0 + 0.4 \cdot 2.21)$
 $= 1.2388 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5490 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 24.0 / 0.5773$
 $= 41.5692 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	288.0000 mm
Parallel to Vessel Wall, opening length	d	144.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm

Weld Strength Reduction Factor [fr1]:

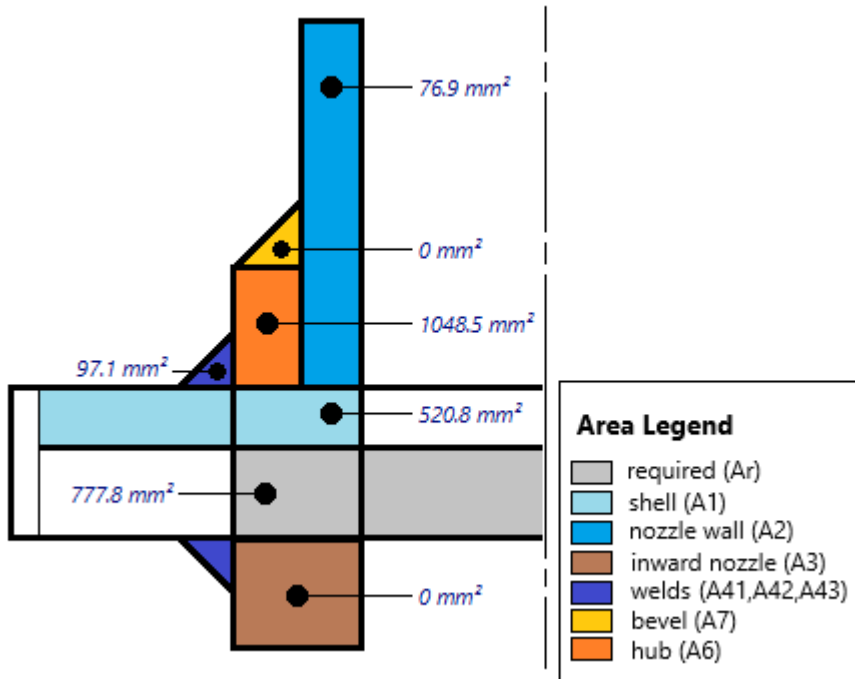
$= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
 = min(0.971, 1.0)
 = 0.971



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	777.810	338.213	NA
Area in Shell	A1	520.844	620.036	NA
Area in Nozzle Wall	A2	76.943	107.081	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1048.525	1048.525	NA
TOTAL AREA AVAILABLE	Atot	1743.397	1872.727	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 = (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
 = (144.0•5.3431•1.0+2•27.0•5.3431•1.0•(1-0.97))
 = 777.810 mm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 = d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)
 = 144.0(1.0 • 9.0 - 1.0 • 5.343) - 2 • 27.0
 (1.0 • 9.0 - 1.0 • 5.3431) • (1 - 0.971)
 = 520.844 mm²

Area Available in Nozzle Projecting Outward [A2]:
 = (2 • tlnp)(tn - trn)fr2
 = (2 • 22.5)(3.0 - 1.24) 0.9709
 = 76.943 mm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:



$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$$

$$= 97.086 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(22.5, 200.0, 50.0)) \cdot (30.0 - 6.0) \cdot 0.971$$

$$= 1048.525 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.2388 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 4.2388 \text{ mm}$$

Available Nozzle Neck Thickness = 6.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 6.0$, $t_r = 1.196$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.399$, Temp. Reduction = 51 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 4.988$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.554$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 2.21 / 5.10 = 0.434$$

Weld Size Calculations, Description: H2150

Intermediate Calc. for nozzle/shell Welds $T_{min} = 9.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) S_v)$$

$$= \max(0, (777.8096 - 520.8439 + 2 \cdot 27.0 \cdot 0.9709 \cdot (1.0 \cdot 9.0 - 5.3431)) 137)$$

$$= 61553.67 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can / 0.707)^2 \cdot fr2) \cdot S_v$$



$$= (1125.4679 + 97.0856 - 0.0 \cdot 0.97) \cdot 137$$
$$= 167718.19 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$
$$= (1125.4679 + 0.0 + 97.0856 + (471.8362)) \cdot 137$$
$$= 232447.88 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$
$$= (1125.4679 + 0.0 + 97.0856 + 0.0 + (471.8362)) \cdot 137$$
$$= 232447.88 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$
$$= (3.1416/2.0) \cdot 150.0 \cdot 10.0 \cdot 0.49 \cdot 133$$
$$= 153771. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.1416 \cdot 73.5) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 133$$
$$= 581254. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$
$$= (3.1416/2.0) \cdot 150.0 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 137$$
$$= 215277. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (153771 + 581254) = 735025 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (153771 + 0 + 215277 + 0) = 369048 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$
$$= (153771 + 215277 + 0) = 369048 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 735025 N	, must exceed W = 61553 N	or W1 = 167718 N
Path 2-2 = 369048 N	, must exceed W = 61553 N	or W2 = 232447 N
Path 3-3 = 369048 N	, must exceed W = 61553 N	or W3 = 232447 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

The Drop for this Nozzle is : 9.3714 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.3714 mm

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Input, Nozzle Desc: N180

From: 150

Pressure for Reinforcement Calculations	P	2.2148	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	4700.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		12450.30	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
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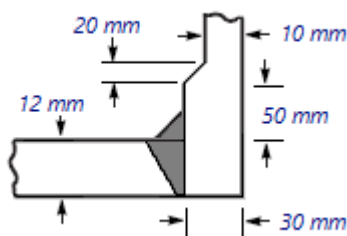
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: N180

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 80.000 mm.
Actual Thickness Used in Calculation 10.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.21 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.21)$
= 5.0174 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.21 \cdot 43.0 / (133 \cdot 1.0 - 0.6 \cdot 2.21)$
= 0.7222 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4327 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= 20.0/0.5773
= 34.6410 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

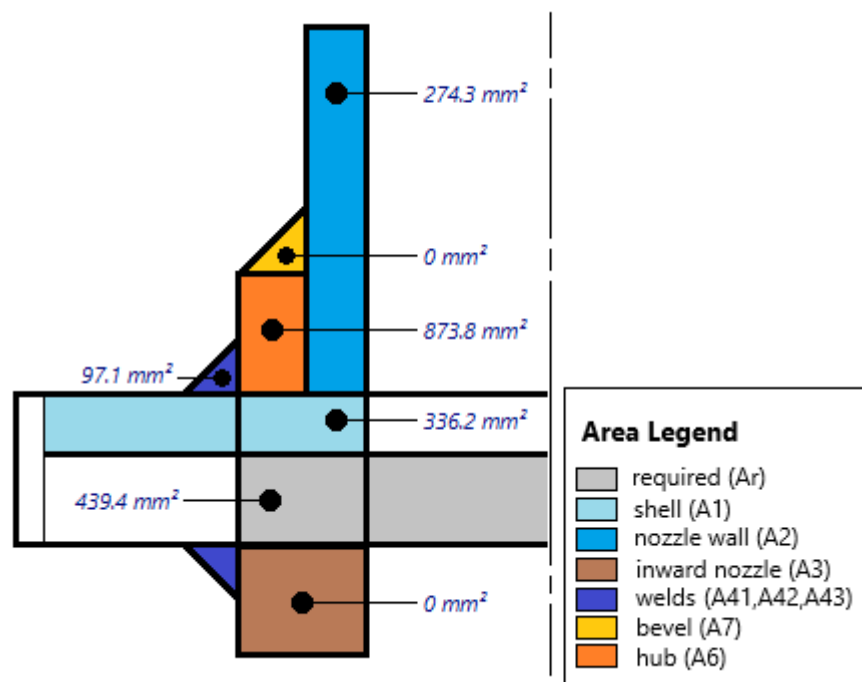
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.0000	mm
Parallel to Vessel Wall, opening length	d	86.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n/S_v)$
= $\min(1, 133.2/137.2)$
= 0.971

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n/S_v)$
= $\min(1, 133.2/137.2)$
= 0.971

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.971, 1.0)$
= 0.971



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	439.395	211.854	NA
Area in Shell	A1	336.233	351.356	NA
Area in Nozzle Wall	A2	274.268	286.915	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	873.771	873.771	NA
TOTAL AREA AVAILABLE	Atot	1581.358	1609.127	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (86.0 \cdot 5.0174 \cdot 1.0 + 2 \cdot 27.0 \cdot 5.0174 \cdot 1.0 \cdot (1 - 0.97))$
 $= 439.395 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 86.0(1.0 \cdot 9.0 - 1.0 \cdot 5.017) - 2 \cdot 27.0$
 $(1.0 \cdot 9.0 - 1.0 \cdot 5.0174) \cdot (1 - 0.971)$
 $= 336.233 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 22.5)(7.0 - 0.72) \cdot 0.9709$
 $= 274.268 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$
 $= 97.086 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(22.5, 200.0, 50.0)) \cdot (30.0 - 10.0) \cdot 0.971 \\ &= 873.771 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.7222 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.0174 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.0174 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[8.02, \max(8.0174, 4.5)] \\ &= 8.0174 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(3.7222, 8.0174) \\ &= 8.0174 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.8,	Allowable	: 133.2 MPa	Passed
Expansion	: 0.0,	Allowable	: 307.2 MPa	Passed
Occasional	: 6.3,	Allowable	: 177.2 MPa	Passed
Shear	: 21.7,	Allowable	: 93.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.697, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.1, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 4.988, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.554, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 2.21/10.20 = 0.217$$

Weld Size Calculations, Description: N180

Intermediate Calc. for nozzle/shell welds Tmin 9.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (439.3955 - 336.2331 + 2 • 27.0 • 0.9709 •
(1.0 • 9.0 - 5.0174))137)
= 42795.88 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (1148.0392 + 97.0856 - 0.0 • 0.97) • 137
= 170814.64 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1148.0392 + 0.0 + 97.0856 + (471.8362)) • 137
= 235544.36 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1148.0392 + 0.0 + 97.0856 + 0.0 + (471.8362)) • 137
= 235544.36 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 140.0 • 10.0 • 0.49 • 133
= 143520. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 56.5) • (30.0 - 3.0) • 0.7 • 133
= 446815. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 140.0 • (12.0 - 3.0) • 0.74 • 137
= 200925. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (143520 + 446815) = 590334 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (143520 + 0 + 200925 + 0) = 344445 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (143520 + 200925 + 0) = 344445 N

Summary of Failure Path Calculations:

Path 1-1 = 590334 N	, must exceed W = 42795 N	or W1 = 170814 N
Path 2-2 = 344444 N	, must exceed W = 42795 N	or W2 = 235544 N
Path 3-3 = 344444 N	, must exceed W = 42795 N	or W3 = 235544 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 8.1469 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1719 mm

Input Echo, WRC297 Item 1, Description: N180 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	609.601	mm	
Shell Thickness	12.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	137.20	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	80.000	mm	
Nozzle Thickness	30.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-105		
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	133.20	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2.21	MPa
Radial Load	P	-5997.68	N
Circumferential Shear	Vc	3998.47	N
Longitudinal Shear	Vl	3998.47	N
Circumferential Moment	Mc	896046.44	N-mm
Longitudinal Moment	Ml	896046.44	N-mm
Torsional Moment	Mt	0.128E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (4.667) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 140.0 + 2 \cdot 1.65 \cdot \sqrt{(312.301(12.0 - 3.0))}$$

$$= 314.953 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	624.601 mm
Nozzle Outside Diameter	(d) =	140.000 mm
Vessel Thickness used	(T) =	9.000 mm
Nozzle Thickness used	(t) =	27.000 mm
T / t	=	0.333
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.867$$

Nr/P	=	0.072
Mr/P	=	0.151
M0/P	=	0.044
N0/P	=	0.091
MrD/Mc	=	0.289
NrDL/Mc	=	0.093
M0d/Mc	=	0.090
N0DL/Mc	=	0.072
MrD/Ml	=	0.118
NrDL/Ml	=	0.080
M0D/Ml	=	0.037
N0DL/Ml	=	0.115

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	74	76	74	76
Outplane Stress Summary	74	72	127	55

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	67	-67	67	-67
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-55	55	55	-55
Inplane Pressure Stress	37	37	37	37
Inplane Stress Summary	47	25	172	-74

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	75	73	172	129

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19



Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	37	37	37	37

Outplane Stress Summary	15	61	111	-12
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	67	-67	67	-67
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-137	137	137	-137
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	74	76	74	76

Inplane Stress Summary	2	144	291	-114
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	6	6
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	15	145	291	115
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5

Hoop Stress Summary	0	3	18	21
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-4	4	4	-4
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	0	-3	15	-3
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	2	2	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	4	8	19	25

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5
Hoop Stress Summary	4	6	15	18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-12	12	12	-12
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-8	5	24	-12

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	2	2

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	13	7	25	30

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	74	76	74	76	74	76	74	76
Circ. Pl (SUS)	-3	-3	15	15	-2	-2	12	12
Circ. Q (SUS)	2	-2	36	-36	-70	70	204	-204
Long. Pm (SUS)	37	37	37	37	37	37	37	37
Long. Pl (SUS)	-1	-1	11	11	1	1	11	11
Long. Q (SUS)	12	-12	122	-122	-23	23	61	-61
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	74.0	76.0	74.0	76.0	74.0	76.0	74.0	76.0
Pm+Pl (SUS)	71.1	73.1	89.1	91.1	72.1	74.1	86.1	88.1
Pm+Pl+Q (Total)	74.4	71.8	170.1	129.1	15.3	144.0	290.2	116.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	76.00	137.20	Passed
Pm+Pl (SUS)	91.09	205.80	Passed
Pm+Pl+Q (TOTAL)	290.20	411.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	5	3	5	3	5	3	5
Circ. Pl (SUS)	-3	-3	15	15	1	1	11	11
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	1	-1	9	-9	-7	7	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	3.0	5.0	3.0	5.0	3.0	5.0	3.0	5.0
Pm+Pl (SUS)	2.0	4.0	18.0	20.0	6.0	8.0	14.0	16.0
Pm+Pl+Q (Total)	2.2	5.4	18.2	25.1	13.2	6.6	21.1	29.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	133.20	Passed
Pm+Pl (SUS)	20.00	199.80	Passed
Pm+Pl+Q (TOTAL)	29.07	399.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: H1150

From: 150

Pressure for Reinforcement Calculations	P	2.2148	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	4700.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		13985.30	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	6.0000	mm

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	29.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		35.0000	mm

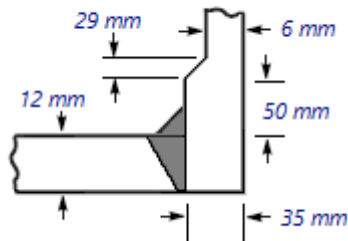
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: H1150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	150.000 mm.
Actual Thickness Used in Calculation	6.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.21 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.21)$
 $= 5.0174 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.21 \cdot 75.0 / (133 \cdot 1.0 + 0.4 \cdot 2.21)$
 $= 1.2388 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5490 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 29.0 / 0.5773$
 $= 50.2295 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	288.0000 mm
Parallel to Vessel Wall, opening length	d	144.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000 mm

Weld Strength Reduction Factor [fr1]:

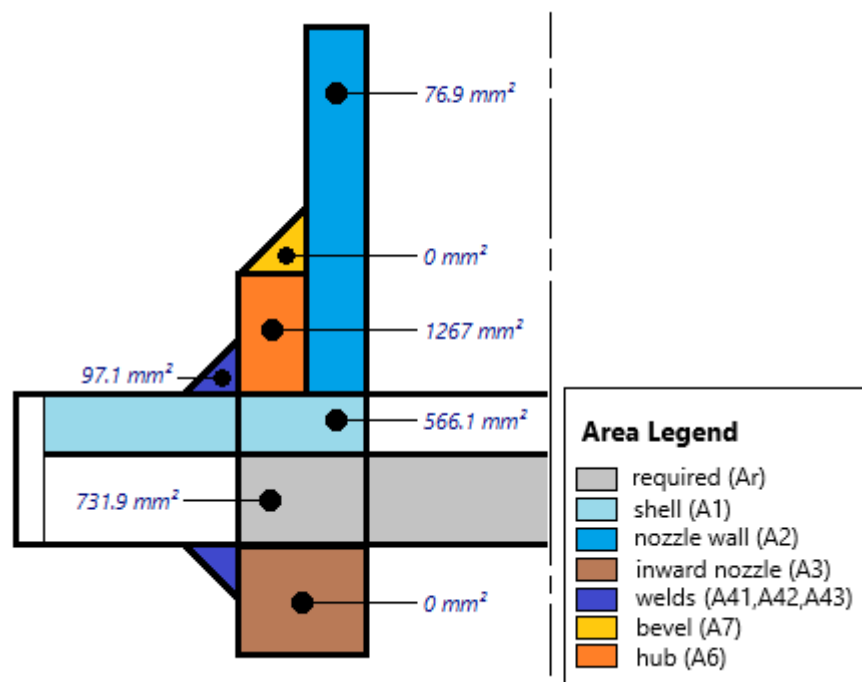
$= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.2)$
 $= 0.971$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.971, 1.0)$
 $= 0.971$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	731.869	352.871	NA
Area in Shell	A1	566.061	591.521	NA
Area in Nozzle Wall	A2	76.943	107.081	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1266.968	1266.968	NA
TOTAL AREA AVAILABLE	Atot	2007.058	2062.655	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (144.0 \cdot 5.0174 \cdot 1.0 + 2 \cdot 32.0 \cdot 5.0174 \cdot 1.0 \cdot (1 - 0.97))$
 $= 731.869 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 144.0(1.0 \cdot 9.0 - 1.0 \cdot 5.017) - 2 \cdot 32.0$
 $(1.0 \cdot 9.0 - 1.0 \cdot 5.0174) \cdot (1 - 0.971)$
 $= 566.061 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 22.5)(3.0 - 1.24) \cdot 0.9709$
 $= 76.943 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$
 $= 97.086 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2 \\ &= (2 \cdot \min(22.5, 200.0, 50.0)) \cdot (35.0 - 6.0) \cdot 0.971 \\ &= 1266.968 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 4.2388 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= t_a \\ &= 4.2388 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 6.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 6.0$, $t_r = 1.196$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.399$, Temp. Reduction = $51 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-48 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 4.988$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.554$, Temp. Reduction = $26 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-48 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -48 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $2.21/5.10 = 0.434$

Weld Size Calculations, Description: H1150

Intermediate Calc. for nozzle/shell Welds $T_{min} = 9.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (731.869 - 566.0613 + 2 \cdot 32.0 \cdot 0.9709 \cdot (1.0 \cdot 9.0 - 5.0174)) \cdot 137) \\ &= 56694.31 \text{ N} \end{aligned}$$

For hub type nozzles, A_2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_4 - (W_i - C_{an} / .707)^2 \cdot fr_2) \cdot S_v \\ &= (1343.9106 + 97.0856 - 0.0 \cdot 0.97) \cdot 137 \\ &= 197685.64 \text{ N} \end{aligned}$$

Weld Load [W2]:



$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (1343.9106 + 0.0 + 97.0856 + (559.2134)) \cdot 137 \\ &= 274402.31 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (1343.9106 + 0.0 + 97.0856 + 0.0 + (559.2134)) \cdot 137 \\ &= 274402.31 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 150.0 \cdot 10.0 \cdot 0.49 \cdot 133 \\ &= 153771. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 73.5) \cdot (35.0 - 3.0) \cdot 0.7 \cdot 133 \\ &= 688894. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 150.0 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 137 \\ &= 215277. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (153771 + 688894) = 842665 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (153771 + 0 + 215277 + 0) = 369048 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (153771 + 215277 + 0) = 369048 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 842665 N	, must exceed W = 56694 N	or W1 = 197685 N
Path 2-2 = 369048 N	, must exceed W = 56694 N	or W2 = 274402 N
Path 3-3 = 369048 N	, must exceed W = 56694 N	or W3 = 274402 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 9.3714 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.3714 mm

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Input, Nozzle Desc: N480

From: 190

Pressure for Reinforcement Calculations	P	2.2148	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	968.1400	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		17016.40	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
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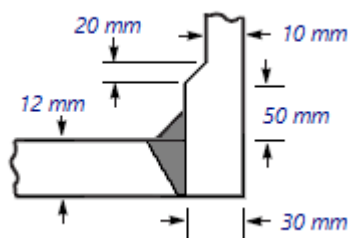
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: N480

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 80.000 mm.
Actual Thickness Used in Calculation 10.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.21 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.21)$
= 5.0174 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.21 \cdot 43.0 / (133 \cdot 1.0 - 0.6 \cdot 2.21)$
= 0.7222 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4327 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= 20.0/0.5773
= 34.6410 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

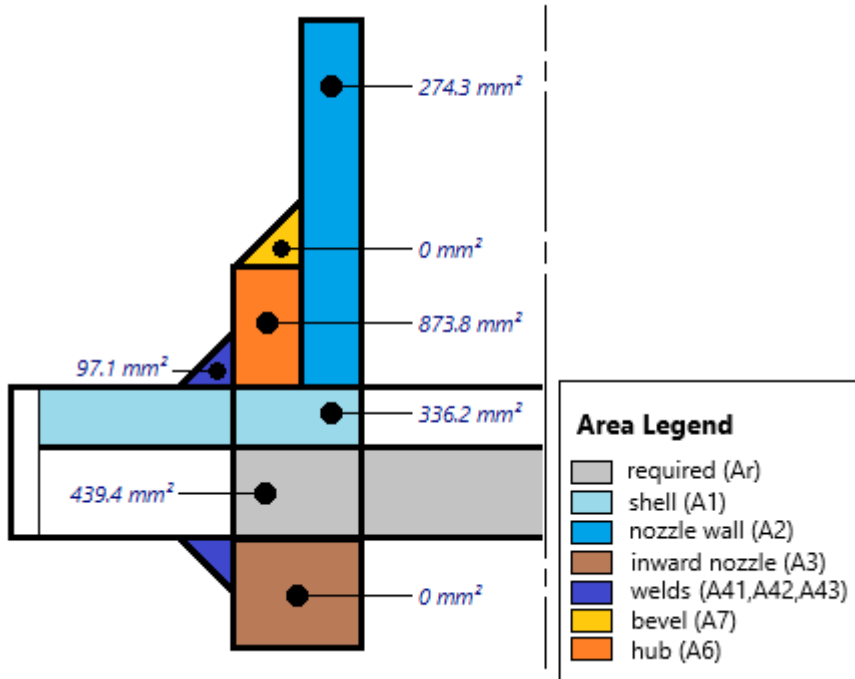
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.0000	mm
Parallel to Vessel Wall, opening length	d	86.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n/S_v)$
= $\min(1, 133.2/137.2)$
= 0.971

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n/S_v)$
= $\min(1, 133.2/137.2)$
= 0.971

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.971, 1.0)$
= 0.971



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	439.395	107.588	NA
Area in Shell	A1	336.233	552.395	NA
Area in Nozzle Wall	A2	274.268	286.915	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	873.771	873.771	NA
TOTAL AREA AVAILABLE	Atot	1581.358	1810.167	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (86.0 \cdot 5.0174 \cdot 1.0 + 2 \cdot 27.0 \cdot 5.0174 \cdot 1.0 \cdot (1 - 0.97))$
 $= 439.395 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 86.0(1.0 \cdot 9.0 - 1.0 \cdot 5.017) - 2 \cdot 27.0$
 $(1.0 \cdot 9.0 - 1.0 \cdot 5.0174) \cdot (1 - 0.971)$
 $= 336.233 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 22.5)(7.0 - 0.72) \cdot 0.9709$
 $= 274.268 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$
 $= 97.086 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(22.5, 200.0, 50.0)) \cdot (30.0 - 10.0) \cdot 0.971 \\ &= 873.771 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.7222 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.0174 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.0174 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.0200 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[8.02, \max(8.0174, 4.5)] \\ &= 8.0174 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(3.7222, 8.0174) \\ &= 8.0174 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.8,	Allowable	: 133.2 MPa	Passed
Expansion	: 0.0,	Allowable	: 307.2 MPa	Passed
Occasional	: 6.3,	Allowable	: 177.2 MPa	Passed
Shear	: 21.7,	Allowable	: 93.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.697, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.1, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 4.988, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.554, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 2.21/10.20 = 0.217$$

Weld Size Calculations, Description: N480

Intermediate Calc. for nozzle/shell welds Tmin 9.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
= max(0, (439.3955 - 336.2331 + 2 * 27.0 * 0.9709 *
(1.0 * 9.0 - 5.0174))137)
= 42795.88 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²*fr2)*Sv
= (1148.0392 + 97.0856 - 0.0 * 0.97) * 137
= 170814.64 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 * tn * t * fr1)) * Sv
= (1148.0392 + 0.0 + 97.0856 + (471.8362)) * 137
= 235544.36 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2*tn*t*fr1))*S
= (1148.0392 + 0.0 + 97.0856 + 0.0 + (471.8362)) * 137
= 235544.36 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) * Dlo * Wo * 0.49 * Snw
= (3.1416/2.0) * 140.0 * 10.0 * 0.49 * 133
= 143520. N

Shear, Nozzle Wall [Snw]:
= (π * (Dlr + Dlo)/4) * (Thk - Can) * 0.7 * Sn
= (3.1416 * 56.5) * (30.0 - 3.0) * 0.7 * 133
= 446815. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) * Dlo * (T - cas) * 0.74 * Sng
= (3.1416/2.0) * 140.0 * (12.0 - 3.0) * 0.74 * 137
= 200925. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (143520 + 446815) = 590334 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (143520 + 0 + 200925 + 0) = 344445 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (143520 + 200925 + 0) = 344445 N

Summary of Failure Path Calculations:

Path 1-1 = 590334 N	, must exceed W = 42795 N	or W1 = 170814 N
Path 2-2 = 344444 N	, must exceed W = 42795 N	or W2 = 235544 N
Path 3-3 = 344444 N	, must exceed W = 42795 N	or W3 = 235544 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 8.1469 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1469 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 86.00 mm

Input Echo, WRC297 Item 1,		Description: N480	:
Diameter Basis for Cylindrical Shell		ID	
Shell Internal Corrosion Allowance		3.0000	mm
Shell External Corrosion Allowance		0.0000	mm
Shell Diameter		609.601	mm
Shell Thickness		12.0000	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	137.20	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		3.0000	mm
Nozzle Diameter		80.000	mm
Nozzle Thickness		30.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	133.20	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2.21	MPa
Radial Load	P	-5997.68	N
Circumferential Shear	Vc	3998.47	N
Longitudinal Shear	Vl	3998.47	N
Circumferential Moment	Mc	896046.44	N-mm
Longitudinal Moment	Ml	896046.44	N-mm
Torsional Moment	Mt	0.128E+07	N-mm
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

*Warning - The ratio, Dn/Tn (4.667) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 140.0 + 2 \cdot 1.65 \cdot \sqrt{(312.301(12.0 - 3.0))} \\
 &= 314.953 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	624.601 mm
Nozzle Outside Diameter	(d) =	140.000 mm
Vessel Thickness used	(T) =	9.000 mm



Nozzle Thickness used (t) = 27.000 mm
 T / t = 0.333
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.867$$

Nr/P = 0.072
 Mr/P = 0.151
 M0/P = 0.044
 N0/P = 0.091
 MrD/Mc = 0.289
 NrDL/Mc = 0.093
 M0d/Mc = 0.090
 N0DL/Mc = 0.072
 MrD/M1 = 0.118
 NrDL/M1 = 0.080
 M0D/M1 = 0.037
 N0DL/M1 = 0.115

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	74	76	74	76
Outplane Stress Summary	74	72	127	55

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	67	-67	67	-67
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-55	55	55	-55
Inplane Pressure Stress	37	37	37	37
Inplane Stress Summary	47	25	172	-74

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	75	73	172	129

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
--	------------	------------	-------------	-------------



circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	37	37	37	37
Outplane Stress Summary	15	61	111	-12

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	67	-67	67	-67
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-137	137	137	-137
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	74	76	74	76
Inplane Stress Summary	2	144	291	-114

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	15	145	291	115

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5
Hoop Stress Summary	0	3	18	21

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3



Axial Bending (ML)	-4	4	4	-4
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	0	-3	15	-3

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	2	2	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	4	8	19	25

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5
Hoop Stress Summary	4	6	15	18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-12	12	12	-12
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-8	5	24	-12

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	2	2

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	13	7	25	30

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	74	76	74	76	74	76	74	76
Circ. Pl (SUS)	-3	-3	15	15	-2	-2	12	12
Circ. Q (SUS)	2	-2	36	-36	-70	70	204	-204
Long. Pm (SUS)	37	37	37	37	37	37	37	37
Long. Pl (SUS)	-1	-1	11	11	1	1	11	11
Long. Q (SUS)	12	-12	122	-122	-23	23	61	-61
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	74.0	76.0	74.0	76.0	74.0	76.0	74.0	76.0
Pm+Pl (SUS)	71.1	73.1	89.1	91.1	72.1	74.1	86.1	88.1
Pm+Pl+Q (Total)	74.4	71.8	170.1	129.1	15.3	144.0	290.2	116.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	76.00	137.20	Passed
Pm+Pl (SUS)	91.09	205.80	Passed
Pm+Pl+Q (TOTAL)	290.20	411.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	5	3	5	3	5	3	5
Circ. Pl (SUS)	-3	-3	15	15	1	1	11	11
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	1	-1	9	-9	-7	7	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	3.0	5.0	3.0	5.0	3.0	5.0	3.0	5.0
Pm+Pl (SUS)	2.0	4.0	18.0	20.0	6.0	8.0	14.0	16.0
Pm+Pl+Q (Total)	2.2	5.4	18.2	25.1	13.2	6.6	21.1	29.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	133.20	Passed
Pm+Pl (SUS)	20.00	199.80	Passed
Pm+Pl+Q (TOTAL)	29.07	399.60	Passed



*Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.*

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Input, Nozzle Desc: K550

From: 190

Pressure for Reinforcement Calculations	P	2.2148	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	609.60	mm
Design Length of Section	L	968.1400	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		17016.40	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

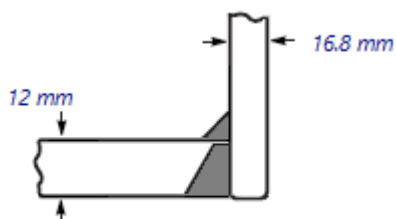
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K550

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 16.800 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.21 \cdot 307.8005 / (137 \cdot 1.0 - 0.6 \cdot 2.21)$
= 5.0174 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.21 \cdot 28.4 / (133 \cdot 1.0 - 0.6 \cdot 2.21)$
= 0.4770 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $133.2 / 137.2$)
= 0.971

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $133.2 / 137.2$)
= 0.971

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(0.971, 1.0)
= 0.971

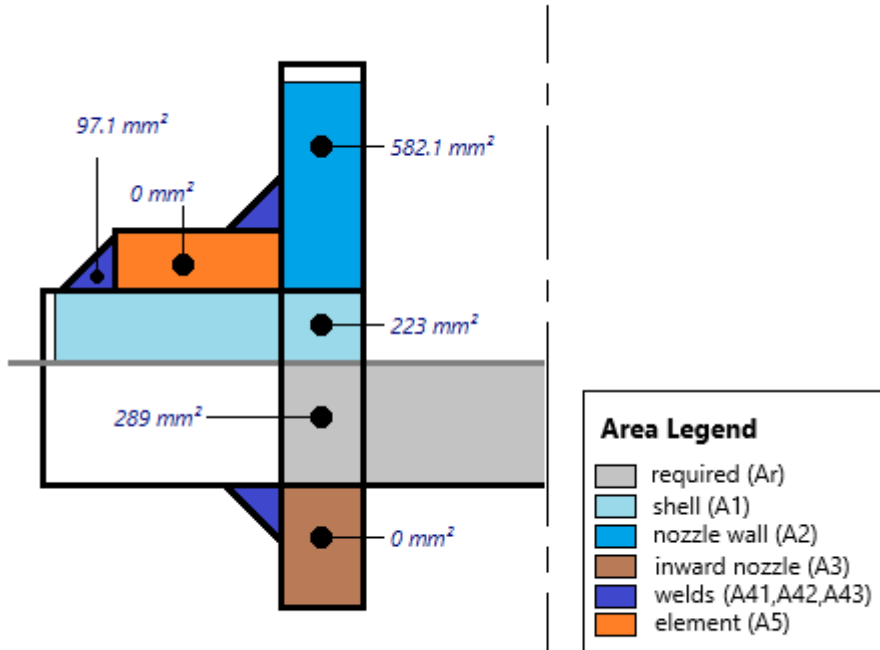


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	289.026	70.769	NA
Area in Shell	A1	223.006	366.375	NA
Area in Nozzle Wall	A2	582.063	585.756	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	97.086	97.086	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	902.155	1049.217	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (56.8 \cdot 5.0174 \cdot 1.0 + 2 \cdot 13.8 \cdot 5.0174 \cdot 1.0 \cdot (1 - 0.97))$
 $= 289.026 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 56.8(1.0 \cdot 9.0 - 1.0 \cdot 5.017) - 2 \cdot 13.8$
 $(1.0 \cdot 9.0 - 1.0 \cdot 5.0174) \cdot (1 - 0.971)$
 $= 223.006 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 22.5)(13.8 - 0.48)0.9709$
 $= 582.063 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9709 + (0.0)^2 \cdot 0.9709$
 $= 97.086 \text{ mm}^2$



UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.4770$ mm
 Wall Thickness per UG16(b), $tr_{16b} = 4.5000$ mm
 Wall Thickness, shell/head, internal pressure $trb_1 = 8.0174$ mm
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 8.0174$ mm
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000$ mm
 Wall Thickness per table UG-45 $tb_3 = 7.8000$ mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[7.8, \max(8.0174, 4.5)]$
 $= 7.8000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(3.477, 7.8)$
 $= 7.8000$ mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 4.988$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio $= t_r * E^* / (t_g - c) = 0.554$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $2.21/5.10 = 0.434$

Weld Size Calculations, Description: K550

Intermediate Calc. for nozzle/shell Welds $T_{min} = 9.0000$ mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A-A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (289.0261 - 223.0063 + 2 \cdot 13.8 \cdot 0.9709 \cdot (1.0 \cdot 9.0 - 5.0174)) \cdot 137)$
 $= 23696.98$ N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2) \cdot S_v$
 $= (582.0632 + 0.0 + 97.0856 - 0.0 \cdot 0.97) \cdot 137$
 $= 93170.24$ N

Weld Load [W2]:

$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$
 $= (582.0632 + 0.0 + 97.0856 + (241.1608)) \cdot 137$
 $= 126254.30$ N



Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S \\ &= (582.0632 + 0.0 + 97.0856 + 0.0 + (241.1608)) \cdot 137 \\ &= 126254.30 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 84.4 \cdot 10.0 \cdot 0.49 \cdot 133 \\ &= 86522. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 35.3) \cdot (16.8 - 3.0) \cdot 0.7 \cdot 133 \\ &= 142682. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 84.4 \cdot (12.0 - 3.0) \cdot 0.74 \cdot 137 \\ &= 121129. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (86522 + 142682) = 229204 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (86522 + 0 + 121129 + 0) = 207651 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (86522 + 121129 + 0) = 207651 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 229203 N	, must exceed W = 23696 N	or W1 = 93170 N
Path 2-2 = 207651 N	, must exceed W = 23696 N	or W2 = 126254 N
Path 3-3 = 207651 N	, must exceed W = 23696 N	or W3 = 126254 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 2.9355 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.9355 mm

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Input, Nozzle Desc: N2100

From: 210

Pressure for Reinforcement Calculations	P	2.2148	MPa
Temperature for Internal Pressure	Temp	270	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	200	°C

Parent Material		SA-266 2	
Parent Allowable Stress at Temperature	Sv	133.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Outside Diameter of Bolted Blind Flange	D	914.40	mm
Head Finished (Minimum) Thickness	t	88.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	133.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		100.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		120	

Flange Material		SA-105	
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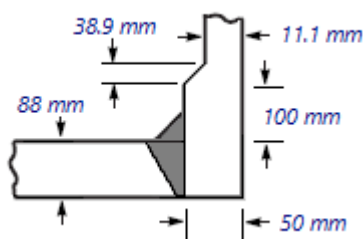
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	38.8748	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		66.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 92.050 mm.
Actual Thickness Used in Calculation 11.125 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.21 \cdot 49.02 / (133 \cdot 1.0 - 0.6 \cdot 2.21)$
= 0.8234 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5560 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 38.875/0.5773
= 67.3332 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

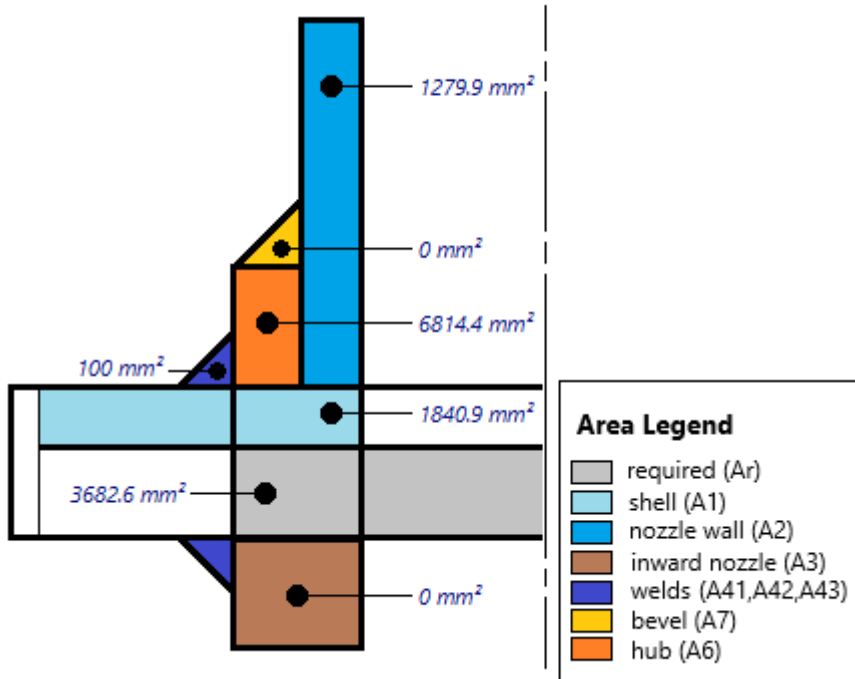
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	284.3000	mm
Parallel to Vessel Wall	Rn+tn+t	142.1500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	87.6462	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 133.2/133.2)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 133.2/133.2)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1., 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3682.575	3683.275	NA
Area in Shell	A1	1840.868	1838.211	NA
Area in Nozzle Wall	A2	1279.945	1326.821	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	99.999	99.999	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	6814.413	6814.413	NA
TOTAL AREA AVAILABLE	Atot	10035.225	10079.443	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 \cdot d \cdot t + t \cdot \tan(1 - fr1)$ per UG-39(b)(1)
 $= 0.5 \cdot 98.0496 \cdot 75.1161 + 75.1161 \cdot 47.0(1 - 1.)$
 $= 3682.575 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot \tan(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 186.25(1.0 \cdot 85.0 - 1.0 \cdot 75.116) - 2 \cdot 47.0$
 $(1.0 \cdot 85.0 - 1.0 \cdot 75.1161) \cdot (1 - 1.)$
 $= 1840.868 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot \tan p)(\tan - \tan)fr2$
 $= (2 \cdot 87.65)(8.13 - 0.82)1.$
 $= 1279.945 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$
 $= 99.999 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(87.6, 300.0, 100.0)) \cdot (50.0 - 11.1) \cdot 1. \\ &= 6814.413 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 3.8234 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 78.1161 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 78.1161 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb_3, \max(tb_1, tb_2)] \\ &= \min[8.258, \max(78.1161, 4.5)] \\ &= 8.2578 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(3.8234, 8.2578) \\ &= 8.2578 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 9.735$, $t_r = 0.794$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.118$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 50.0$, $t_r = 0.794$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.017$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	17 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $2.21/5.10 = 0.434$

Weld Size Calculations, Description: N2100

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) S_v) \\ &= \max(0, (3682.5754 - 1840.8679 + 2 \cdot 47.0 \cdot 1. \cdot \\ &\quad (1.0 \cdot 85.0 - 75.1161)) 133) \\ &= 369039.44 \text{ N} \end{aligned}$$



For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
$$= (A2 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$
$$= (8094.3574 + 100.0 - 0.0 \cdot 1.) \cdot 133$$
$$= 1091402.25 \text{ N}$$

Weld Load [W2]:
$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$
$$= (8094.3574 + 0.0 + 100.0 + (7989.9507)) \cdot 133$$
$$= 2155579.75 \text{ N}$$

Weld Load [W3]:
$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$
$$= (8094.3574 + 0.0 + 100.0 + 0.0 + (7989.9507)) \cdot 133$$
$$= 2155579.75 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$
$$= (3.1416/2.0) \cdot 192.0496 \cdot 10.0 \cdot 0.49 \cdot 133$$
$$= 196878. \text{ N}$$

Shear, Nozzle Wall [Snw]:
$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.1416 \cdot 72.5248) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 133$$
$$= 998389. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:
$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$
$$= (3.1416/2.0) \cdot 192.0496 \cdot (66.0 - 0.0) \cdot 0.74 \cdot 133$$
$$= 1962361. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
$$= (SONW + SNW) = (196878 + 998389) = 1195266 \text{ N}$$

Failure Path [Path 2-2]:
$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (196878 + 0 + 1962361 + 0) = 2159239 \text{ N}$$

Failure Path [Path 3-3]:
$$= (Sonw + Tngw + Sinw)$$
$$= (196878 + 1962361 + 0) = 2159239 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 1195266 N	, must exceed W = 369039 N	or W1 = 1091402 N
Path 2-2 = 2159238 N	, must exceed W = 369039 N	or W2 = 2155579 N
Path 3-3 = 2159238 N	, must exceed W = 369039 N	or W3 = 2155579 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Cut Length for this Nozzle is, Drop + Ho + H + T : 388.0508 mm

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

- ** Warning: Wind/Earthquake Deflection Exceeded the Specified Allowable!**
- ** Warning: PWHT was required for at least 1 Shell/Head in this vessel!**
- ** Warning: PWHT was required for at least 1 Nozzle/Pad in this vessel!**

Please review all reports carefully!

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